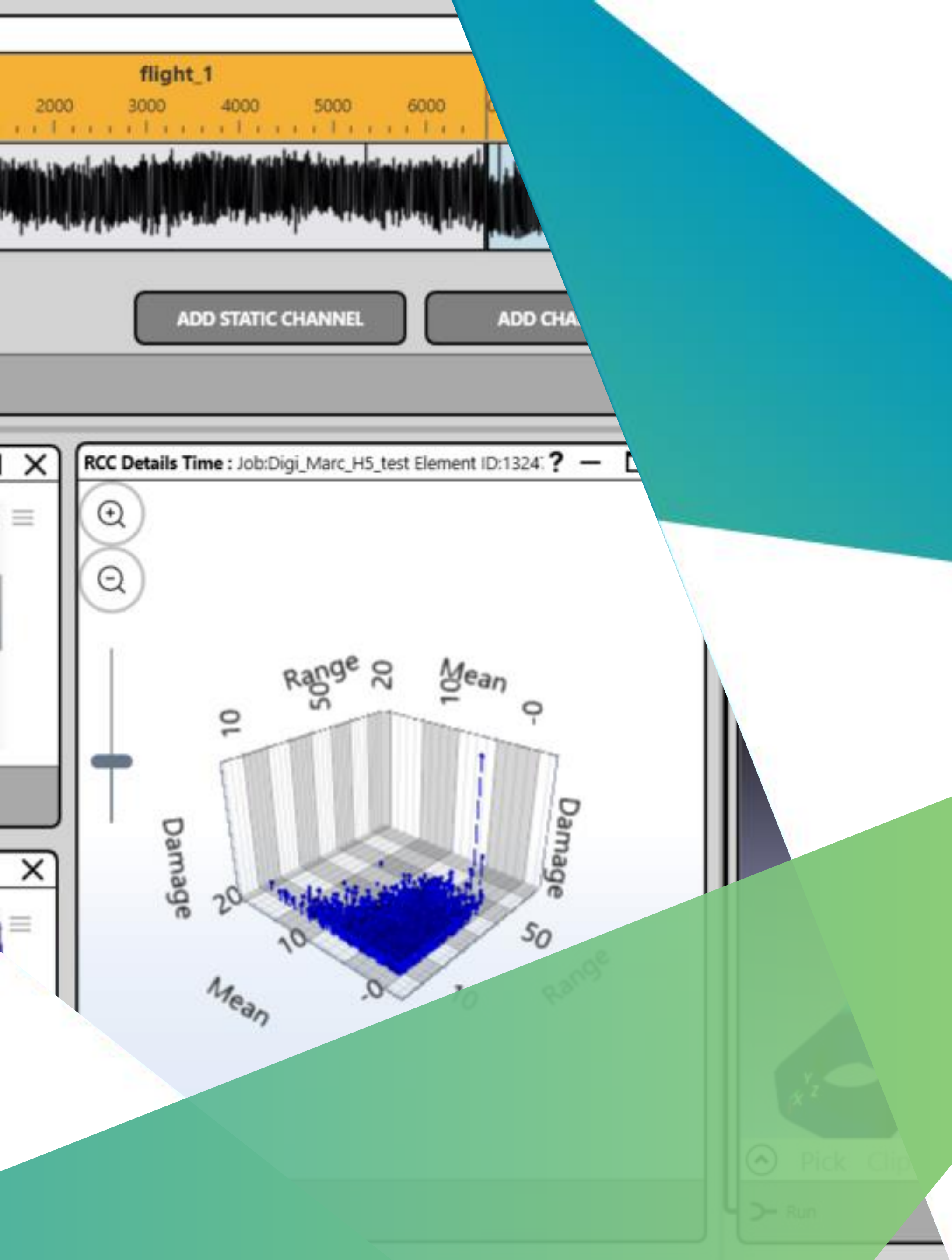


HEXAGON

CAEfatigue Software

Customer Success Stories & Use Case examples

"This CFV job ran in 4 minutes, before it took 6 hours to do this analysis".

Collins Aerospace

Customer Success Stories

CAEfatigue

HEXAGON

Customer Success Stories

Advanced Mixed Loading required for aerospace and automotive applications

Vibration Fatigue Benchmarks

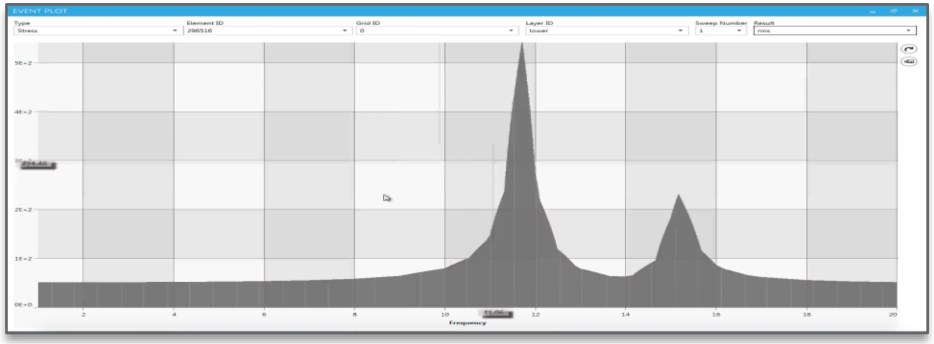
Collins Aerospace (ex UTAS)

Collins Aerospace (a United Technologies subsidiary) is one of the world's largest suppliers of aerospace and defense products. Collins is the supporting industry to the major aircraft manufacturing industry and aerospace engineering industry. Collins is the world's leading independent full-service supplier of nacelles, pylons, thrust reversers, and other structural aircraft components for commercial, business and military aircraft, and helicopters. Major commercial aircraft manufacturers such Boeing and Airbus have the facilities to assemble the whole structure and components of the aircraft, but they don't manufacture all the necessary systems and technologies, they just bring out those things from the subsidiary companies such as Collins Aerospace.

Collins Aerospace is engaged in designing, manufacturing and servicing systems and components for 2 main sectors,

- The Aircraft Systems consists of Actuation & Propeller Systems, Aerostructures, Air Management Systems, Interiors, and Landing Systems (formerly Landing Gear and Wheels and Brakes).

- The Power, Control & Sensing Systems consists of Electric Systems, Engine Systems, ISR & Space Systems, and Sensors & Integrated Systems.



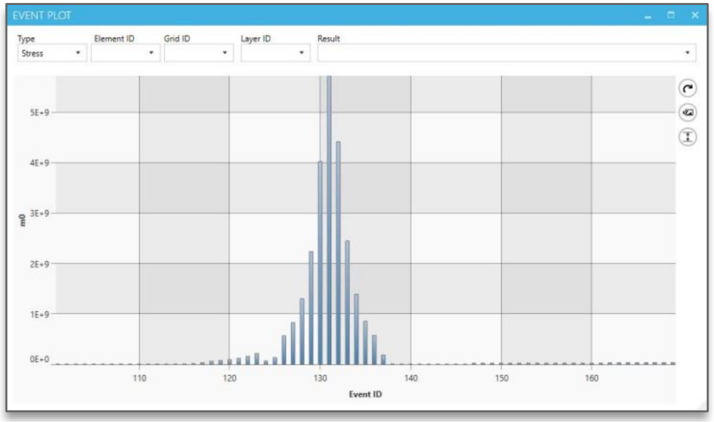
HEXAGON

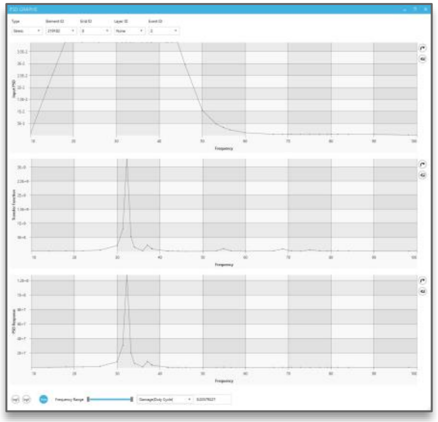
*Key technologies covered: **Base Shake** (single input) analysis and **Mixed Loading**, i.e. random with deterministic.*

Vibration Fatigue Benchmarks

Collins Aerospace (ex UTAS)

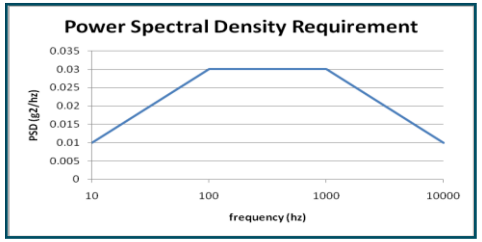
Advanced Mixed Loading required for aerospace and automotive applications





Random Vibration

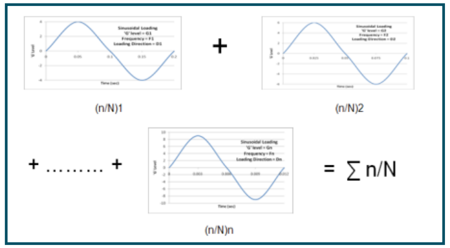
* Each direction of loading has an independent:
    - PSD level
    - Clipping level
    - Length of testing



Consecutive Sines ('windmilling condition')

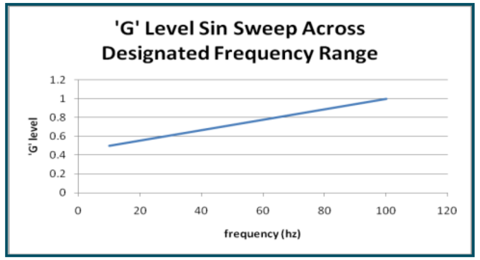
(with or without a consecutive random vibration loading condition)

* Each direction of loading has sinusoids with independent:
    - Magnitude
    - Frequency
    - Length of time



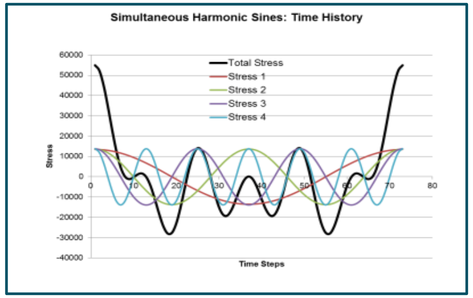
Sine Sweep

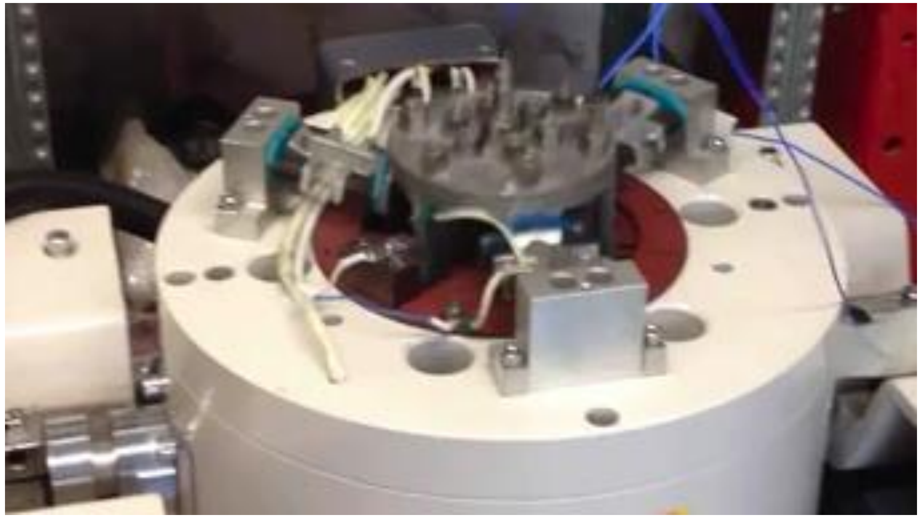
* Each direction of loading has an independent:
    - G loading
    - Sweep rate
    - Number of passes
    - Sweep range



Simultaneous Sines (with or without random background)

* Each direction of loading has an independent:
    - Length of Test
* Each direction of loading has sinusoids with independent:
    - Magnitude
    - Frequency
* If required, each direction of random vibration loading has independent:
    - PSD level, Clipping level, Length of Testing





*Key technologies covered: **Base Shake** (single input) analysis and **Mixed Loading**, i.e. random with deterministic.*

Collins Aerospace
An RTX Business

Vibration Fatigue with Advanced Mixed Loading for aerospace applications

“This vibration job ran in 4 minutes, before it took 6 hours to do this analysis”.

Collins Aerospace

Challenge

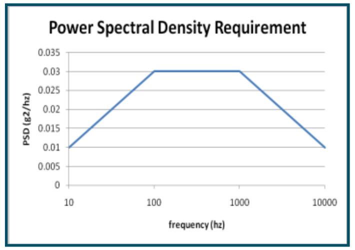
- Designing, manufacturing and servicing complex systems and components for Aircrafts, Power, Control, Sensing, ISR & Space Systems
- Demanding Virtual vibration testing under different specifications
- Multiple loading environments, from consecutive or simultaneous sines to Power Spectral Density, sine sweep and random on random

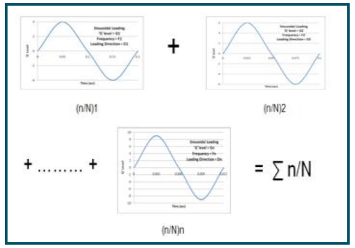
Solution

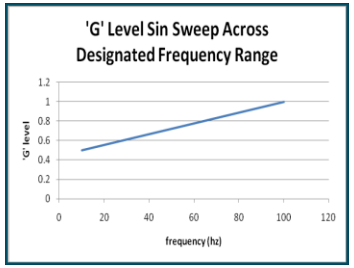
- Hexagon collaborated intensively with Collins Aerospace to prove consistency of results between the main in-house codes
- Upon successful testing, CAEfatigue was recommended to the worldwide Collins community.
- CAEfatigue control files are implemented into standard workflows deployed across the organization

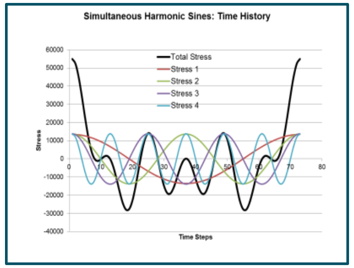
Benefits

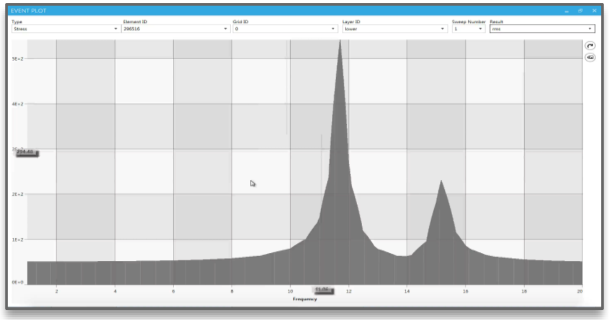
- Performances, completeness and ease of use,
- Consistency across multiple applications and departments
- Remove the need to maintain in-house code
- Reduce and improve physical testing.









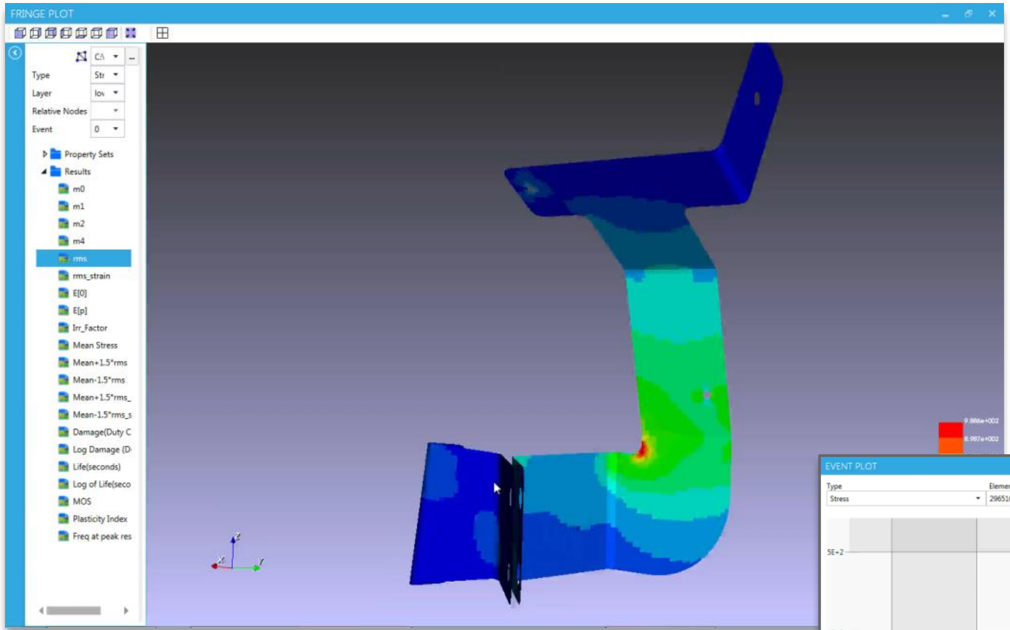


HEXAGON

Navistar

Truck Cabs and Trailers

Advanced Sine on Random required for aerospace and automotive applications



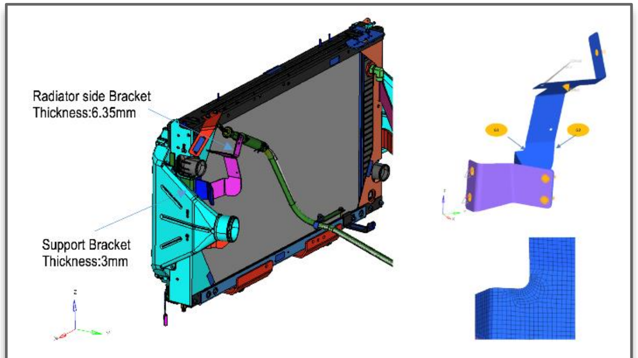
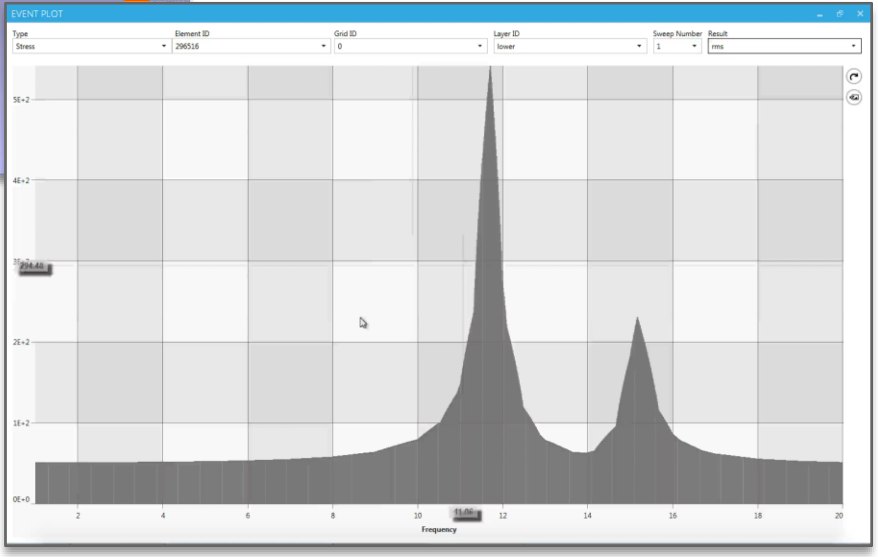
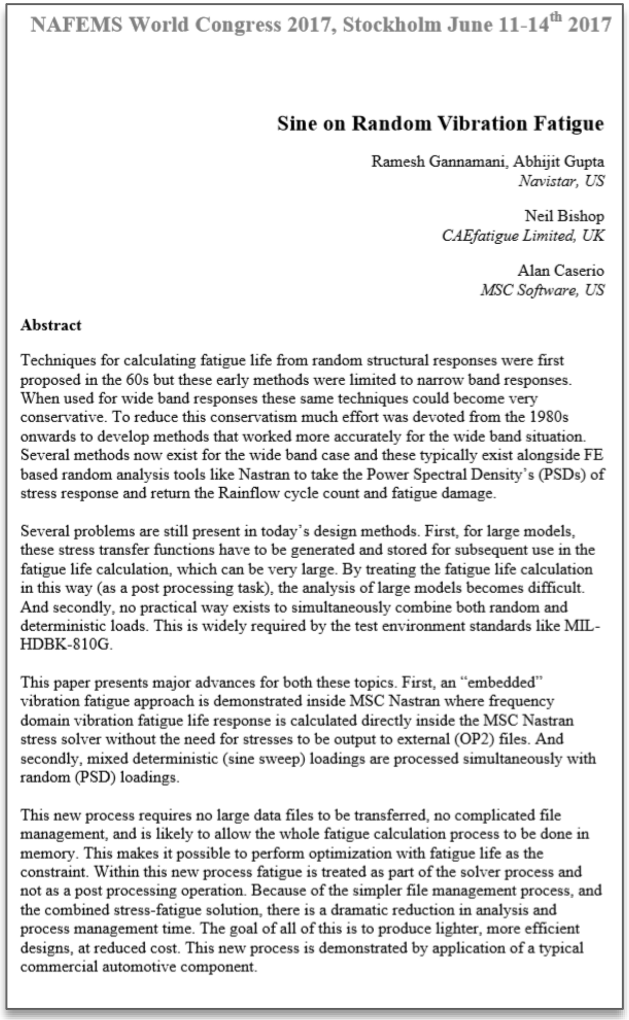


Figure 1. Fill tube bracket layout





NAFEMS World Congress 2017, Stockholm June 11-14th 2017

Sine on Random Vibration Fatigue

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Navistar, US

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CAEfatigue Limited, UK

Alan Caserio
MSC Software, US

Abstract

Techniques for calculating fatigue life from random structural responses were first proposed in the 60s but these early methods were limited to narrow band responses. When used for wide band responses these same techniques could become very conservative. To reduce this conservatism much effort was devoted from the 1980s onwards to develop methods that worked more accurately for the wide band situation. Several methods now exist for the wide band case and these typically exist alongside FE based random analysis tools like Nastran to take the Power Spectral Density's (PSDs) of stress response and return the Rainflow cycle count and fatigue damage.

Several problems are still present in today's design methods. First, for large models, these stress transfer functions have to be generated and stored for subsequent use in the fatigue life calculation, which can be very large. By treating the fatigue life calculation in this way (as a post processing task), the analysis of large models becomes difficult. And secondly, no practical way exists to simultaneously combine both random and deterministic loads. This is widely required by the test environment standards like MIL-HDBK-810G.

This paper presents major advances for both these topics. First, an "embedded" vibration fatigue approach is demonstrated inside MSC Nastran where frequency domain vibration fatigue life response is calculated directly inside the MSC Nastran stress solver without the need for stresses to be output to external (OP2) files. And secondly, mixed deterministic (sine sweep) loadings are processed simultaneously with random (PSD) loadings.

This new process requires no large data files to be transferred, no complicated file management, and is likely to allow the whole fatigue calculation process to be done in memory. This makes it possible to perform optimization with fatigue life as the constraint. Within this new process fatigue is treated as part of the solver process and not as a post processing operation. Because of the simpler file management process, and the combined stress-fatigue solution, there is a dramatic reduction in analysis and process management time. The goal of all of this is to produce lighter, more efficient designs, at reduced cost. This new process is demonstrated by application of a typical commercial automotive component.

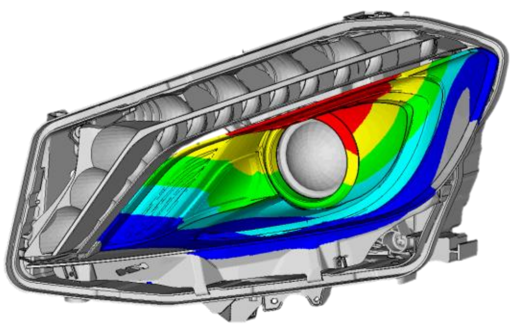
HEXAGON

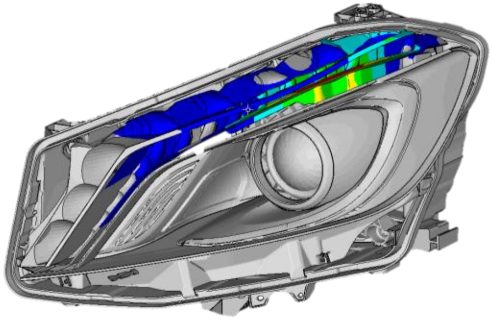
*Key technologies covered: **Base Shake** (single input) analysis and **Mixed Loading**, i.e. random with deterministic.*

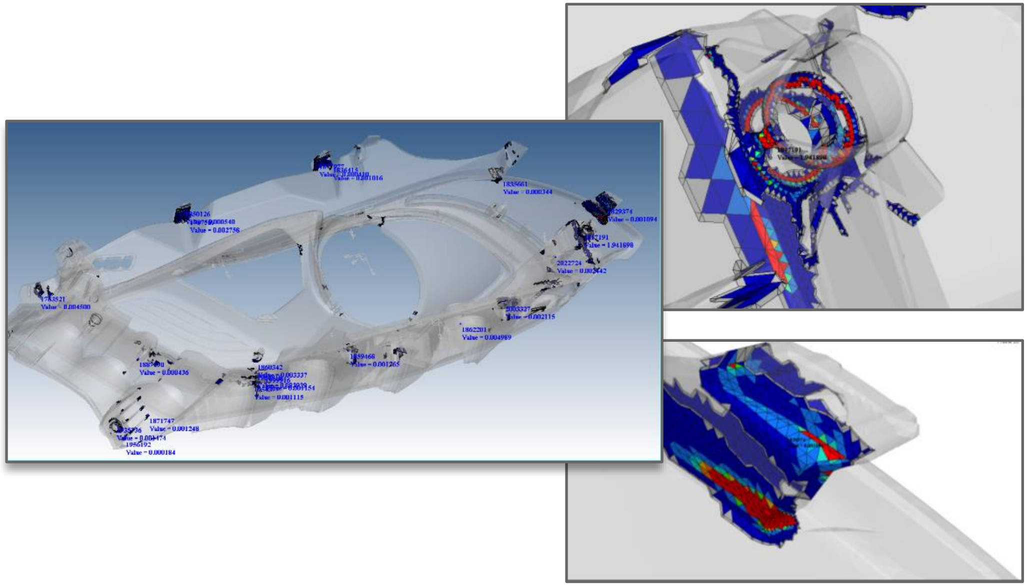
Hella

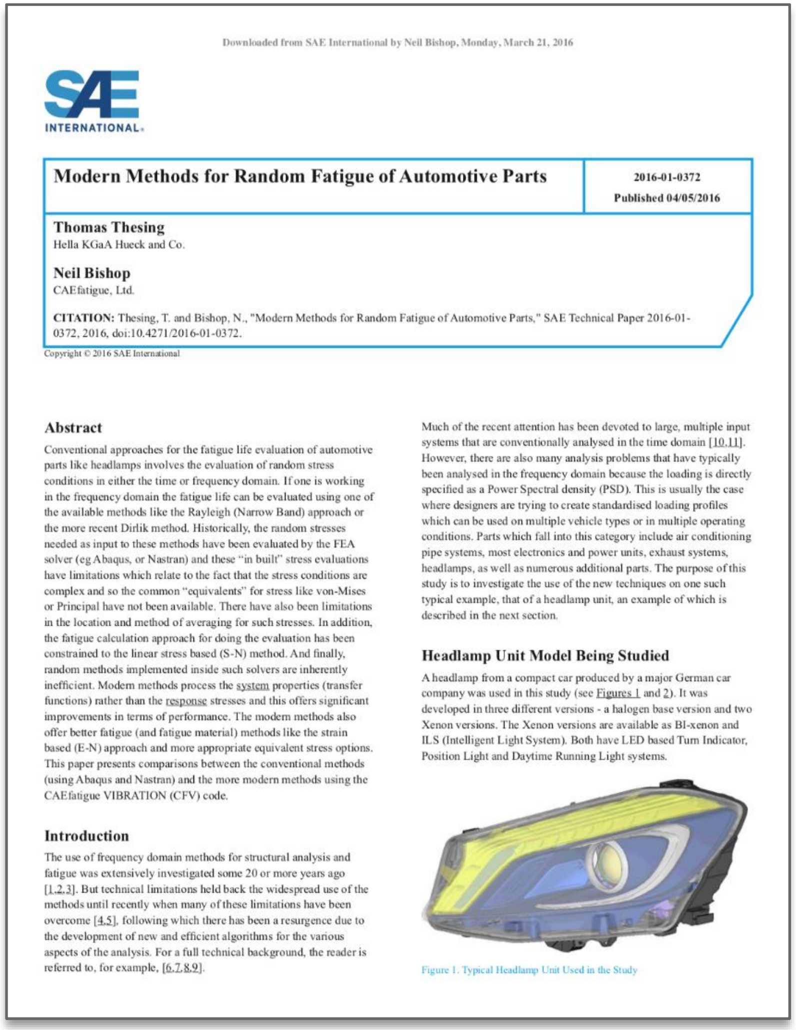
Headlamps and Components

Frequency Domain fatigue analysis of automotive headlamps









SAE INTERNATIONAL

Modern Methods for Random Fatigue of Automotive Parts

2016-01-0372
Published 04/05/2016

Thomas Thesing
Hella KGaA Hueck and Co.

Neil Bishop
CAEfatigue, Ltd.

CITATION: Thesing, T. and Bishop, N., "Modern Methods for Random Fatigue of Automotive Parts," SAE Technical Paper 2016-01-0372, 2016, doi:10.4271/2016-01-0372.

Abstract

Conventional approaches for the fatigue life evaluation of automotive parts like headlamps involves the evaluation of random stress conditions in either the time or frequency domain. If one is working in the frequency domain the fatigue life can be evaluated using one of the available methods like the Rayleigh (Narrow Band) approach or the more recent Dirlik method. Historically, the random stresses needed as input to these methods have been evaluated by the FEA solver (eg Abaqus, or Nastran) and these "in built" stress evaluations have limitations which relate to the fact that the stress conditions are complex and so the common "equivalents" for stress like von-Mises or Principal have not been available. There have also been limitations in the location and method of averaging for such stresses. In addition, the fatigue calculation approach for doing the evaluation has been constrained to the linear stress based (S-N) method. And finally, random methods implemented inside such solvers are inherently inefficient. Modern methods process the system properties (transfer functions) rather than the response stresses and this offers significant improvements in terms of performance. The modern methods also offer better fatigue (and fatigue material) methods like the strain based (E-N) approach and more appropriate equivalent stress options. This paper presents comparisons between the conventional methods (using Abaqus and Nastran) and the more modern methods using the CAEfatigue VIBRATION (CFV) code.

Introduction

The use of frequency domain methods for structural analysis and fatigue was extensively investigated some 20 or more years ago [1,2,3]. But technical limitations held back the widespread use of the methods until recently when many of these limitations have been overcome [4,5], following which there has been a resurgence due to the development of new and efficient algorithms for the various aspects of the analysis. For a full technical background, the reader is referred to, for example, [6,7,8,9].

Much of the recent attention has been devoted to large, multiple input systems that are conventionally analysed in the time domain [10,11]. However, there are also many analysis problems that have typically been analysed in the frequency domain because the loading is directly specified as a Power Spectral density (PSD). This is usually the case where designers are trying to create standardised loading profiles which can be used on multiple vehicle types or in multiple operating conditions. Parts which fall into this category include air conditioning pipe systems, most electronics and power units, exhaust systems, headlamps, as well as numerous additional parts. The purpose of this study is to investigate the use of the new techniques on one such typical example, that of a headlamp unit, an example of which is described in the next section.

Headlamp Unit Model Being Studied

A headlamp from a compact car produced by a major German car company was used in this study (see Figures 1 and 2). It was developed in three different versions - a halogen base version and two Xenon versions. The Xenon versions are available as BI-xenon and ILS (Intelligent Light System). Both have LED based Turn Indicator, Position Light and Daytime Running Light systems.

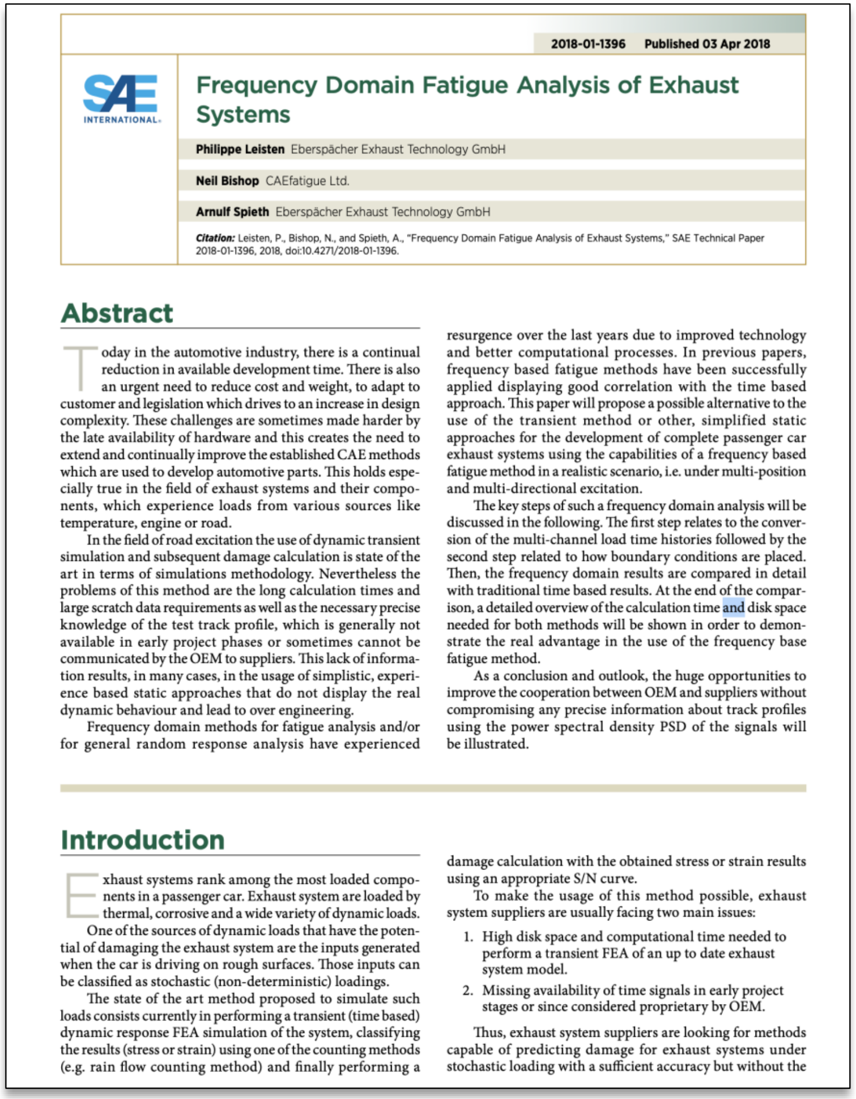
Figure 1. Typical Headlamp Unit Used in the Study

Key technologies covered: ***Base Shake*** *(single input) analysis using random input PSDs of loading.*

Frequency Domain fatigue analysis of exhausts

Eberspaecher

Exhausts



2018-01-1396 Published 03 Apr 2018

Frequency Domain Fatigue Analysis of Exhaust Systems

Philippe Leisten Eberspächer Exhaust Technology GmbH

Neil Bishop CAEfatigue Ltd.

Arnulf Spieth Eberspächer Exhaust Technology GmbH

Citation: Leisten, P., Bishop, N., and Spieth, A., "Frequency Domain Fatigue Analysis of Exhaust Systems," SAE Technical Paper 2018-01-1396, 2018, doi:10.4271/2018-01-1396.

Abstract

Today in the automotive industry, there is a continual reduction in available development time. There is also an urgent need to reduce cost and weight, to adapt to customer and legislation which drives to an increase in design complexity. These challenges are sometimes made harder by the late availability of hardware and this creates the need to extend and continually improve the established CAE methods which are used to develop automotive parts. This holds especially true in the field of exhaust systems and their components, which experience loads from various sources like temperature, engine or road.

In the field of road excitation the use of dynamic transient simulation and subsequent damage calculation is state of the art in terms of simulations methodology. Nevertheless the problems of this method are the long calculation times and large scratch data requirements as well as the necessary precise knowledge of the test track profile, which is generally not available in early project phases or sometimes cannot be communicated by the OEM to suppliers. This lack of information results, in many cases, in the usage of simplistic, experience based static approaches that do not display the real dynamic behaviour and lead to over engineering.

Frequency domain methods for fatigue analysis and/or for general random response analysis have experienced resurgence over the last years due to improved technology and better computational processes. In previous papers, frequency based fatigue methods have been successfully applied displaying good correlation with the time based approach. This paper will propose a possible alternative to the use of the transient method or other, simplified static approaches for the development of complete passenger car exhaust systems using the capabilities of a frequency based fatigue method in a realistic scenario, i.e. under multi-position and multi-directional excitation.

The key steps of such a frequency domain analysis will be discussed in the following. The first step relates to the conversion of the multi-channel load time histories followed by the second step related to how boundary conditions are placed. Then, the frequency domain results are compared in detail with traditional time based results. At the end of the comparison, a detailed overview of the calculation time and disk space needed for both methods will be shown in order to demonstrate the real advantage in the use of the frequency base fatigue method.

As a conclusion and outlook, the huge opportunities to improve the cooperation between OEM and suppliers without compromising any precise information about track profiles using the power spectral density PSD of the signals will be illustrated.

Introduction

Exhaust systems rank among the most loaded components in a passenger car. Exhaust system are loaded by thermal, corrosive and a wide variety of dynamic loads.

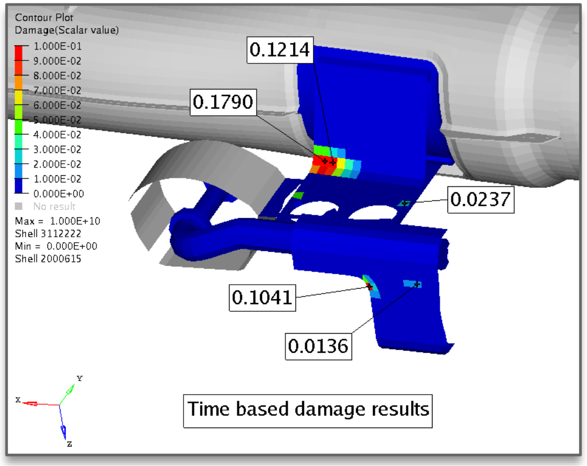
One of the sources of dynamic loads that have the potential of damaging the exhaust system are the inputs generated when the car is driving on rough surfaces. Those inputs can be classified as stochastic (non-deterministic) loadings.

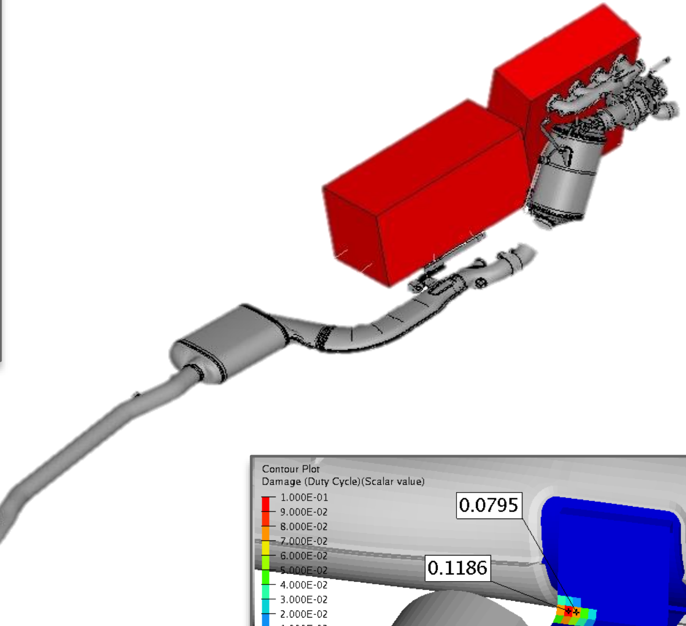
The state of the art method proposed to simulate such loads consists currently in performing a transient (time based) dynamic response FEA simulation of the system, classifying the results (stress or strain) using one of the counting methods (e.g. rain flow counting method) and finally performing a damage calculation with the obtained stress or strain results using an appropriate S/N curve.

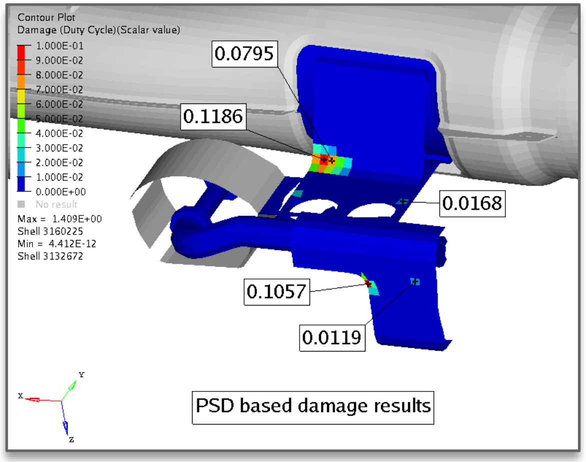
To make the usage of this method possible, exhaust system suppliers are usually facing two main issues:

1. High disk space and computational time needed to perform a transient FEA of an up to date exhaust system model.
2. Missing availability of time signals in early project stages or since considered proprietary by OEM.

Thus, exhaust system suppliers are looking for methods capable of predicting damage for exhaust systems under stochastic loading with a sufficient accuracy but without the







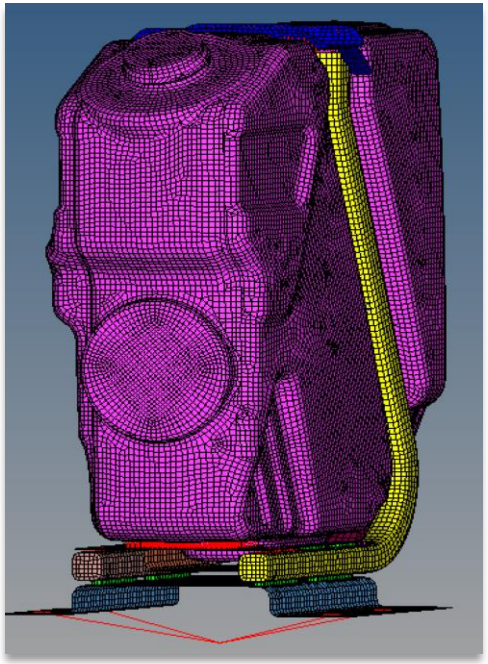
HEXAGON

*Key technologies covered: **Multi-input** analysis using random input PSDs of loading.*

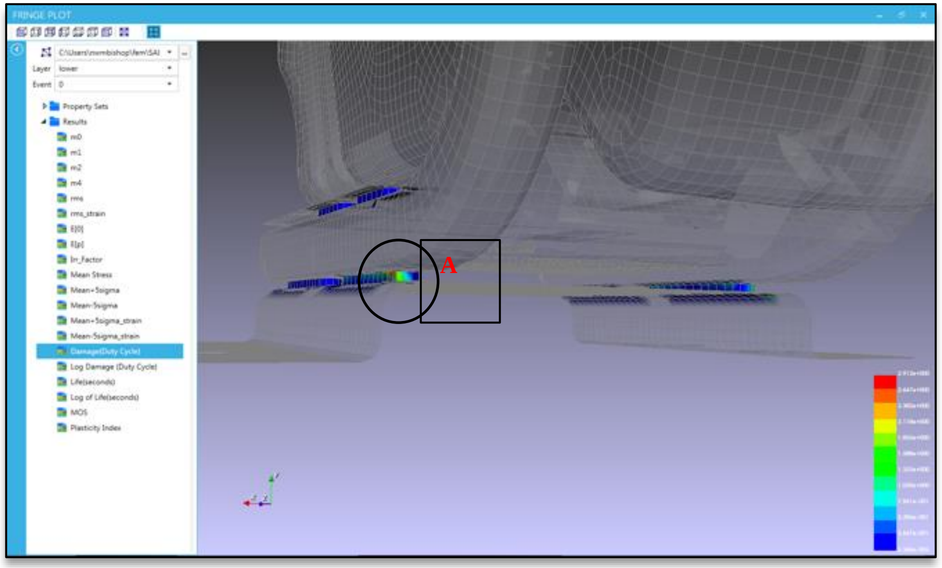
Ford

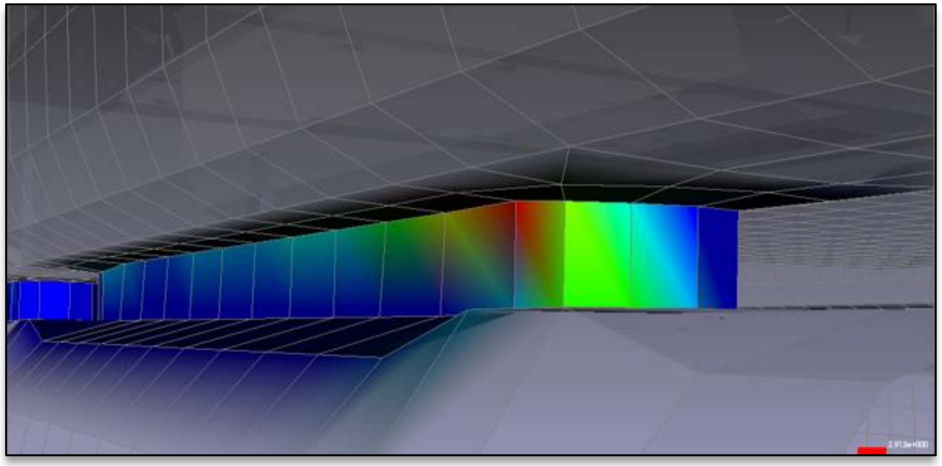
Automotive OEM

Frequency Domain fatigue analysis of seam welds from time domain loads



Fuel tank model used in study





SAE INTERNATIONAL

Frequency FE-Based Weld Fatigue Life Prediction of Dynamic Systems

2017-01-0355

Published 03/28/2017

Edinilson Alves Costa
Ford Motor Company

Neil Bishop
CAEfatigue Ltd

Valdir Cardoso
Altair Engineering

CITATION: Costa, E., Bishop, N., and Cardoso, V., "Frequency FE-Based Weld Fatigue Life Prediction of Dynamic Systems," SAE Technical Paper 2017-01-0355, 2017, doi:10.4271/2017-01-0355.

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Abstract

In most aspects of mechanical design related to a motor vehicle there are two ways to treat dynamic fatigue problems. These are the time domain and the frequency domain approaches. Time domain approaches are the most common and most widely used especially in the automotive industries and accordingly it is the method of choice for the fatigue calculation of welded structures. In previous papers the frequency approach has been successful applied showing a good correlation with the life and damage estimated using a time based approach; in this paper the same comparative process has been applied but now extended specifically to welded structures. Both the frequency domain approach and time domain approach are used for numerically predicting the fatigue life of the seam welds of a thin sheet powertrain installation bracketry of a commercial truck submitted to variable amplitude loading. Predicted results are then compared with bench tests results, and their accuracy are rated. Ultimately the key question addressed within this study is whether the frequency domain approach can be a better alternative when compared to the conventional time domain approaches, when used to determine weld fatigue durability. It is demonstrated that when the fatigue damage is efficiently evaluated using the frequency approach (PSDs), it permits a better understand of the importance of the dynamic characteristic of the actual system, isolating the mode contributions to the total damage, guiding how much a test can be accelerated or even how this damage can be minimized by changing the normal modes of this system.

Introduction

In most aspects of mechanical design related to a motor vehicle there are two ways to treat dynamic fatigue problems, that is: time domain vs. frequency domain approach. Time domain approaches are the most common and most widely used especially in the automotive industries and accordingly it is the method of choice for the fatigue calculation of welded structures [1].

Both solutions in time domain or frequency domain are mathematically the same. However computational and technological issues have held back the implementation of fatigue calculations in the frequency domain until now. In a previous paper the frequency approach has been successful applied showing a good correlation with the life and damage estimated using the time based approach [1,2,3,4]; in this paper the same comparative process has been applied but now extended specifically to welded structures.

Both the frequency domain approach and the time domain approach are used for numerically predicting the fatigue life of seam welds of a thin sheet powertrain installation bracketry of a commercial truck subjected to variable amplitude loading. Predicted results are then compared with bench tests, and their accuracy is rated. Ultimately the key question addressed within this study is whether the frequency domain approach can be a better alternative when compared to the conventional time domain approaches, when used to determine weld fatigue durability.

Methods for Calculating Fatigue Life in Welds

For thin-sheet structures the fatigue strength is mainly given by the strength of the joints [5]. Accordingly, FE-based methods which can accurately compute the stress state of the joints are inherently able to support good fatigue predictions. However, when it comes to welded joints the actual stress state in a weld is difficult or, in practice, maybe impossible to determine [5]. The heat from the welding process causes local tensile residual stresses at the weld toe, changing material properties near the weld, and cause geometric distortions which lead to additional bending stresses. In addition, the local geometry along the weld toe varies along its length [6].

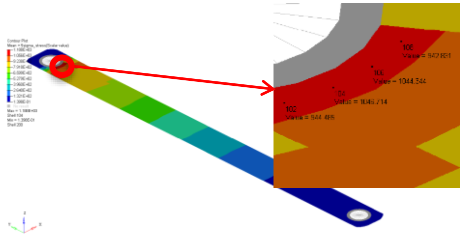
HEXAGON

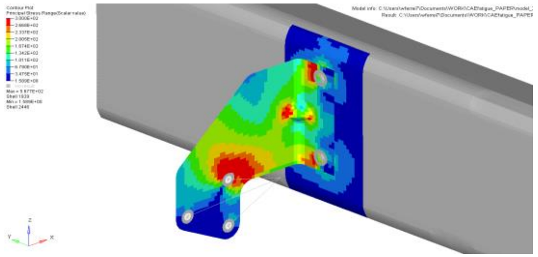
Key technologies covered: ***Multi-input*** *analysis of* ***Joints*** *(seam welds) using* ***Loads Conditioning*** *to produce the random input PSDs.*

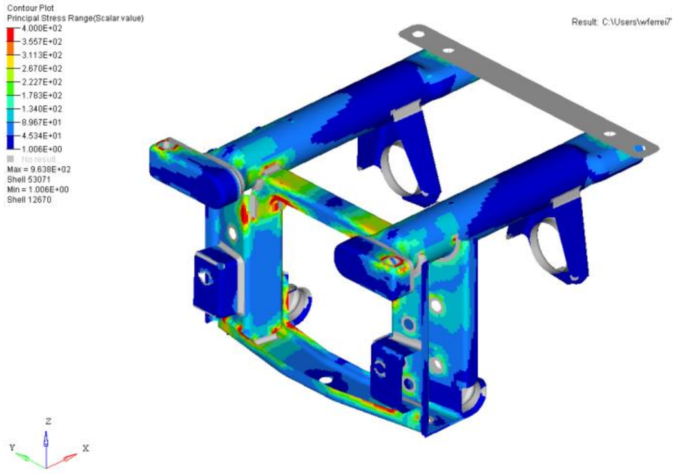
Ford

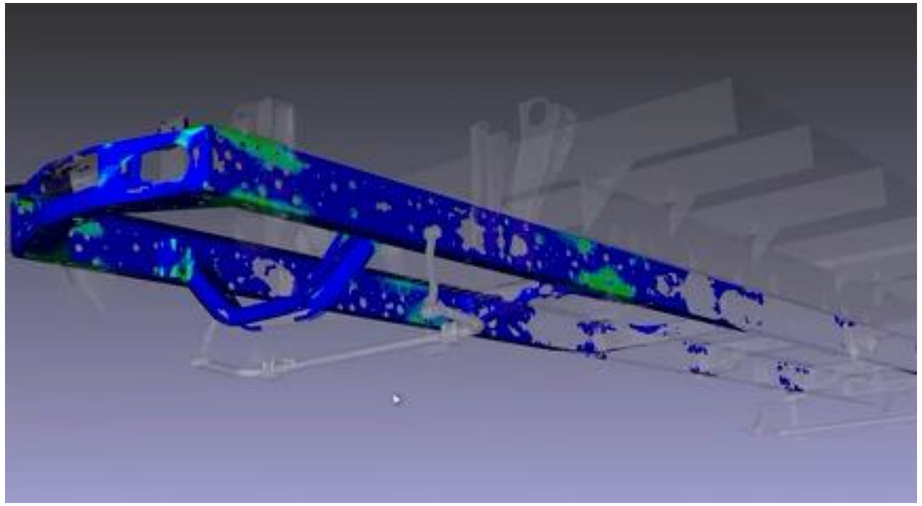
Automotive OEM

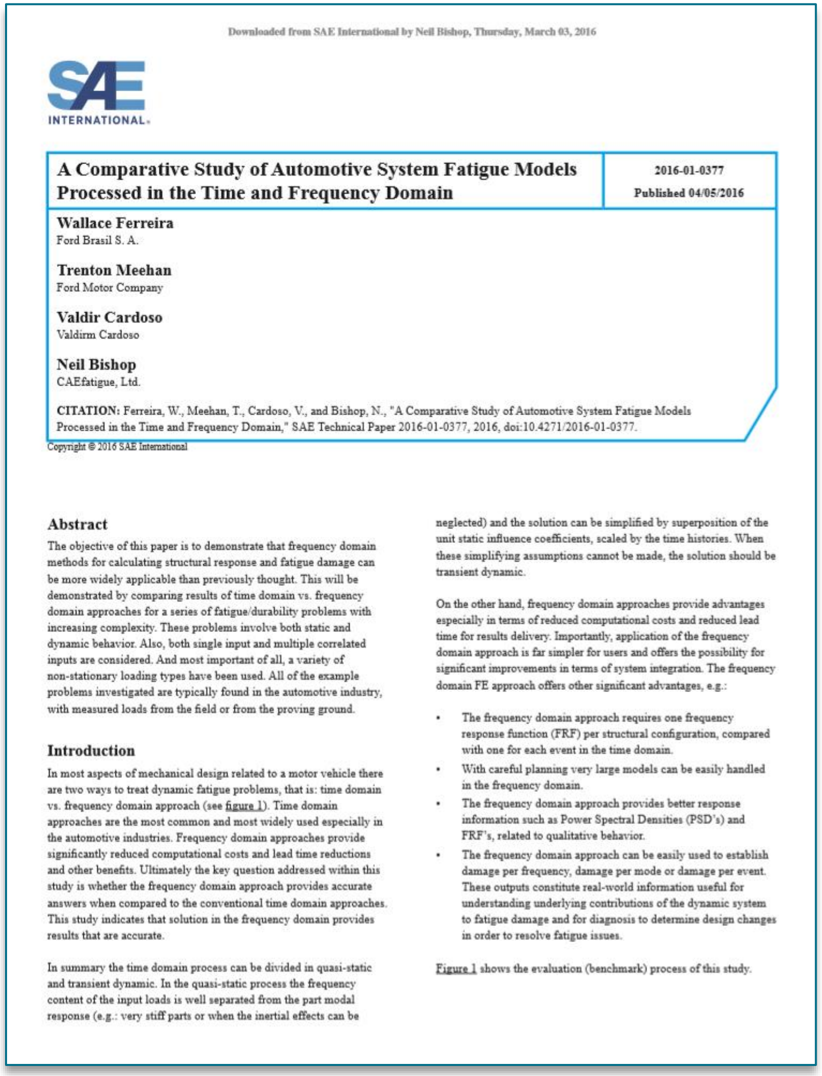
Frequency Domain compared to Time Domain for 5 automotive subsystems











SAE INTERNATIONAL

A Comparative Study of Automotive System Fatigue Models Processed in the Time and Frequency Domain

2016-01-0377
Published 04/05/2016

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Neil Bishop
CAEfatigue, Ltd.

CITATION: Ferreira, W., Meehan, T., Cardoso, V., and Bishop, N., "A Comparative Study of Automotive System Fatigue Models Processed in the Time and Frequency Domain," SAE Technical Paper 2016-01-0377, 2016, doi:10.4271/2016-01-0377.

Abstract

The objective of this paper is to demonstrate that frequency domain methods for calculating structural response and fatigue damage can be more widely applicable than previously thought. This will be demonstrated by comparing results of time domain vs. frequency domain approaches for a series of fatigue/durability problems with increasing complexity. These problems involve both static and dynamic behavior. Also, both single input and multiple correlated inputs are considered. And most important of all, a variety of non-stationary loading types have been used. All of the example problems investigated are typically found in the automotive industry, with measured loads from the field or from the proving ground.

Introduction

In most aspects of mechanical design related to a motor vehicle there are two ways to treat dynamic fatigue problems, that is: time domain vs. frequency domain approach (see figure 1). Time domain approaches are the most common and most widely used especially in the automotive industries. Frequency domain approaches provide significantly reduced computational costs and lead time reductions and other benefits. Ultimately the key question addressed within this study is whether the frequency domain approach provides accurate answers when compared to the conventional time domain approaches. This study indicates that solution in the frequency domain provides results that are accurate.

In summary the time domain process can be divided in quasi-static and transient dynamic. In the quasi-static process the frequency content of the input loads is well separated from the part modal response (e.g.: very stiff parts or when the inertial effects can be neglected) and the solution can be simplified by superposition of the unit static influence coefficients, scaled by the time histories. When these simplifying assumptions cannot be made, the solution should be transient dynamic.

On the other hand, frequency domain approaches provide advantages especially in terms of reduced computational costs and reduced lead time for results delivery. Importantly, application of the frequency domain approach is far simpler for users and offers the possibility for significant improvements in terms of system integration. The frequency domain FE approach offers other significant advantages, e.g.:

- The frequency domain approach requires one frequency response function (FRF) per structural configuration, compared with one for each event in the time domain.
- With careful planning very large models can be easily handled in the frequency domain.
- The frequency domain approach provides better response information such as Power Spectral Densities (PSD's) and FRF's, related to qualitative behavior.
- The frequency domain approach can be easily used to establish damage per frequency, damage per mode or damage per event. These outputs constitute real-world information useful for understanding underlying contributions of the dynamic system to fatigue damage and for diagnosis to determine design changes in order to resolve fatigue issues.

Figure 1 shows the evaluation (benchmark) process of this study.

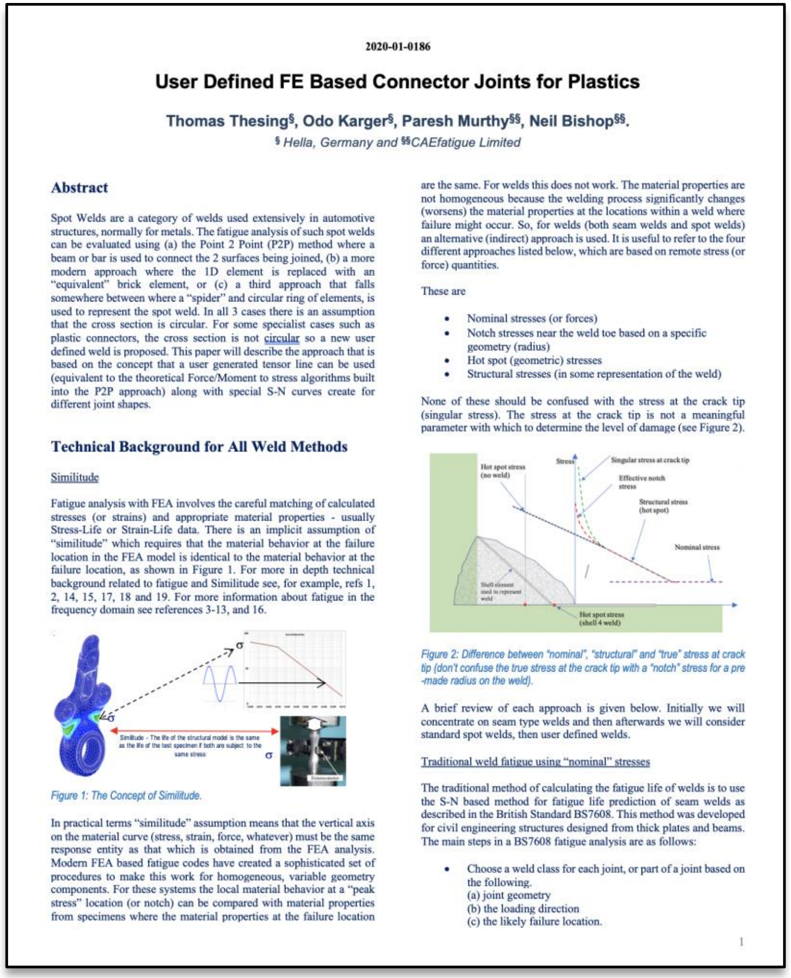
HEXAGON

*Key technologies covered: **Multi-input** analysis using **Loads Conditioning** to produce the random input PSDs.*

Hella

Headlamps and Components

Advanced User Connector feature used for plastic joints in automotive headlamps



2020-01-0186

User Defined FE Based Connector Joints for Plastics

Thomas Thesing§, Odo Karger§, Paresh Murthy§§, Neil Bishop§§.
§ Hella, Germany and §§CAEfatigue Limited

Abstract

Spot Welds are a category of welds used extensively in automotive structures, normally for metals. The fatigue analysis of such spot welds can be evaluated using (a) the Point 2 Point (P2P) method where a beam or bar is used to connect the 2 surfaces being joined, (b) a more modern approach where the 1D element is replaced with an "equivalent" brick element, or (c) a third approach that falls somewhere between where a "spider" and circular ring of elements, is used to represent the spot weld. In all 3 cases there is an assumption that the cross section is circular. For some specialist cases such as plastic connectors, the cross section is not circular so a new user defined weld is proposed. This paper will describe the approach that is based on the concept that a user generated tensor line can be used (equivalent to the theoretical Force/Moment to stress algorithms built into the P2P approach) along with special S-N curves create for different joint shapes.

Technical Background for All Weld Methods

Similitude

Fatigue analysis with FEA involves the careful matching of calculated stresses (or strains) and appropriate material properties - usually Stress-Life or Strain-Life data. There is an implicit assumption of "similitude" which requires that the material behavior at the failure location in the FEA model is identical to the material behavior at the failure location, as shown in Figure 1. For more in depth technical background related to fatigue and Similitude see, for example, refs 1, 2, 14, 15, 17, 18 and 19. For more information about fatigue in the frequency domain see references 3-13, and 16.

Similitude - The life of the structural model is the same as the life of the test specimen if both are subject to the same stress

Figure 1: The Concept of Similitude.

In practical terms "similitude" assumption means that the vertical axis on the material curve (stress, strain, force, whatever) must be the same response entity as that which is obtained from the FEA analysis. Modern FEA based fatigue codes have created a sophisticated set of procedures to make this work for homogeneous, variable geometry components. For these systems the local material behavior at a "peak stress" location (or notch) can be compared with material properties from specimens where the material properties at the failure location are the same. For welds this does not work. The material properties are not homogeneous because the welding process significantly changes (worsens) the material properties at the locations within a weld where failure might occur. So, for welds (both seam welds and spot welds) an alternative (indirect) approach is used. It is useful to refer to the four different approaches listed below, which are based on remote stress (or force) quantities.

These are

- Nominal stresses (or forces)
- Notch stresses near the weld toe based on a specific geometry (radius)
- Hot spot (geometric) stresses
- Structural stresses (in some representation of the weld)

None of these should be confused with the stress at the crack tip (singular stress). The stress at the crack tip is not a meaningful parameter with which to determine the level of damage (see Figure 2).

Hot spot stress (no weld)
Stress
Singular stress at crack tip
Effective notch stress
Structural stress (hot spot)
Nominal stress
Hot spot stress (shell 4 weld)

Figure 2: Difference between "nominal", "structural" and "true" stress at crack tip (don't confuse the true stress at the crack tip with a "notch" stress for a pre-made radius on the weld).

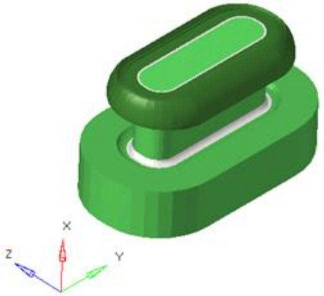
A brief review of each approach is given below. Initially we will concentrate on seam type welds and then afterwards we will consider standard spot welds, then user defined welds.

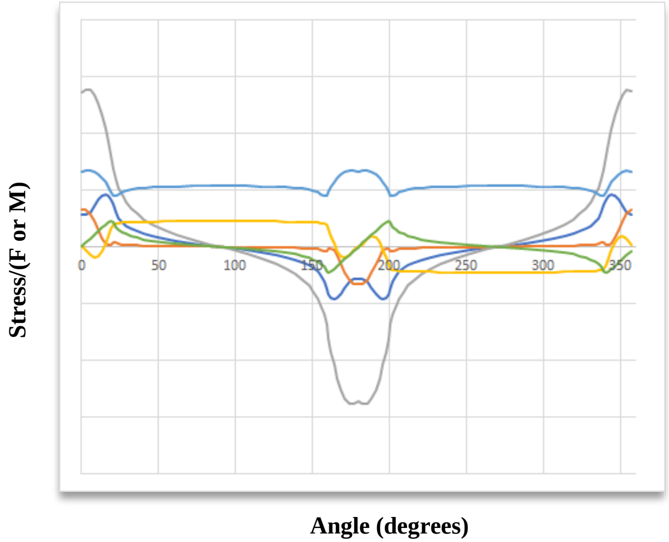
Traditional weld fatigue using "nominal" stresses

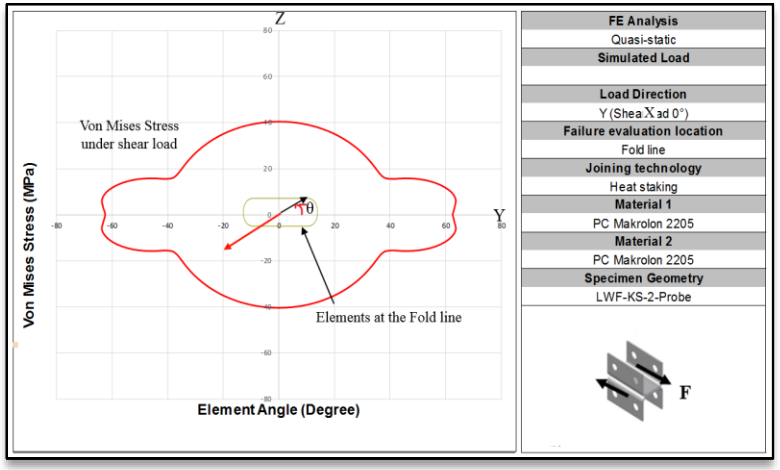
The traditional method of calculating the fatigue life of welds is to use the S-N based method for fatigue life prediction of seam welds as described in the British Standard BS7608. This method was developed for civil engineering structures designed from thick plates and beams. The main steps in a BS7608 fatigue analysis are as follows:

- Choose a weld class for each joint, or part of a joint based on the following.
  (a) joint geometry
  (b) the loading direction
  (c) the likely failure location.

1







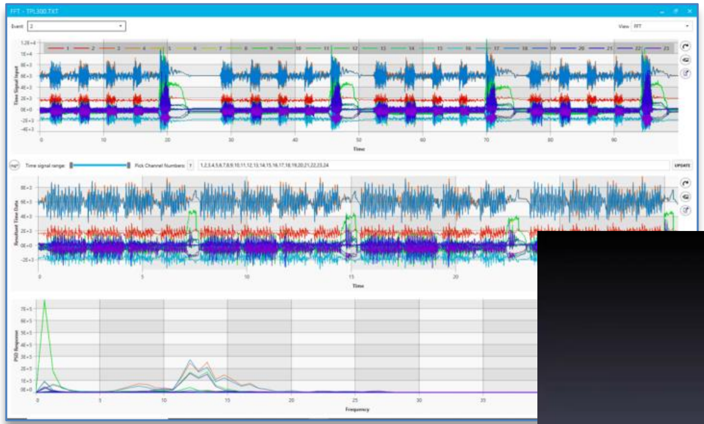
Key technologies covered: ***Base Shake*** *(single input) analysis of user defined* ***Joints*** *(plastic welds) using random input PSDs of loading.*

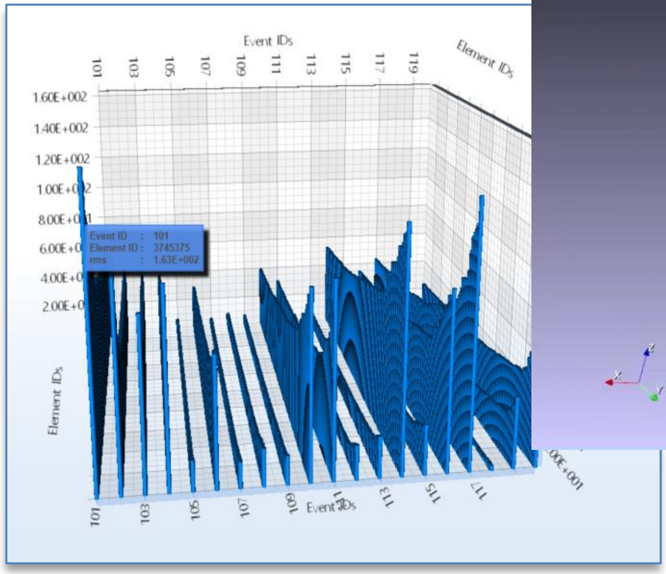
Customer Success Stories

Ford

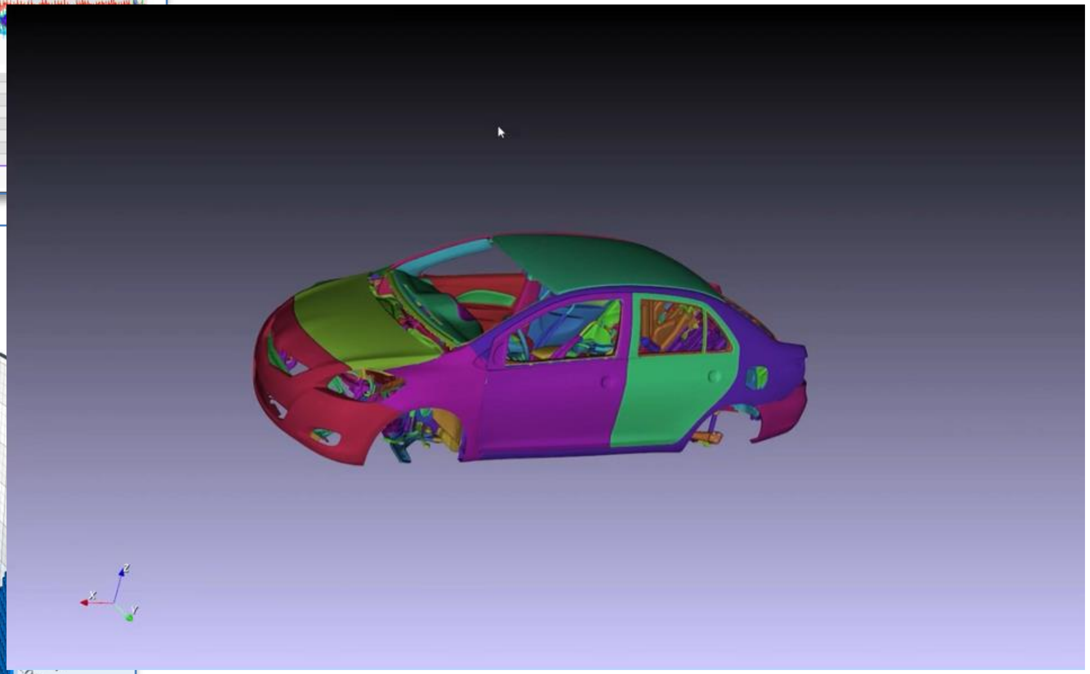
Automotive OEM

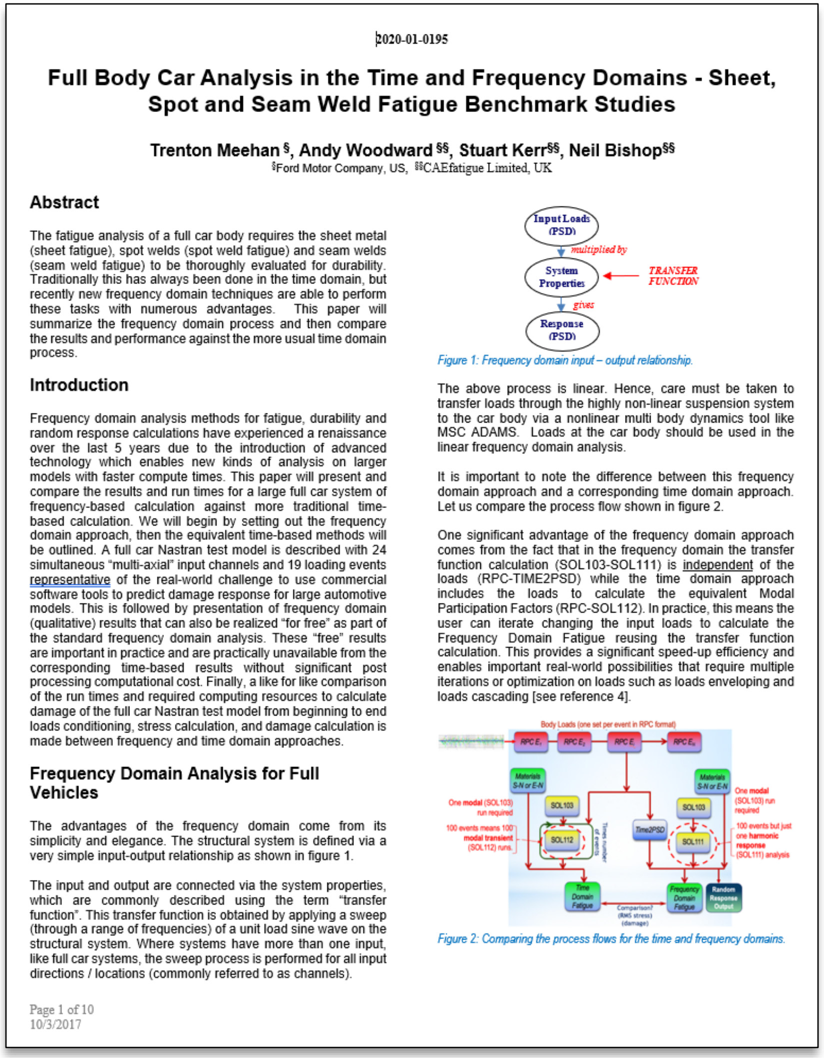
Frequency Domain analysis compared to Time Domain analysis for a full car





"An important aspect … is that a careful preprocessing of the loads must be performed before applying the frequency domain …"





2020-01-0195

Full Body Car Analysis in the Time and Frequency Domains - Sheet, Spot and Seam Weld Fatigue Benchmark Studies

Trenton Meehan §, Andy Woodward §§, Stuart Kerr§§, Neil Bishop§§
§Ford Motor Company, US, §§CAEfatigue Limited, UK

Abstract

The fatigue analysis of a full car body requires the sheet metal (sheet fatigue), spot welds (spot weld fatigue) and seam welds (seam weld fatigue) to be thoroughly evaluated for durability. Traditionally this has always been done in the time domain, but recently new frequency domain techniques are able to perform these tasks with numerous advantages. This paper will summarize the frequency domain process and then compare the results and performance against the more usual time domain process.

Introduction

Frequency domain analysis methods for fatigue, durability and random response calculations have experienced a renaissance over the last 5 years due to the introduction of advanced technology which enables new kinds of analysis on larger models with faster compute times. This paper will present and compare the results and run times for a large full car system of frequency-based calculation against more traditional time-based calculation. We will begin by setting out the frequency domain approach, then the equivalent time-based methods will be outlined. A full car Nastran test model is described with 24 simultaneous "multi-axial" input channels and 19 loading events representative of the real-world challenge to use commercial software tools to predict damage response for large automotive models. This is followed by presentation of frequency domain (qualitative) results that can also be realized "for free" as part of the standard frequency domain analysis. These "free" results are important in practice and are practically unavailable from the corresponding time-based results without significant post processing computational cost. Finally, a like for like comparison of the run times and required computing resources to calculate damage of the full car Nastran test model from beginning to end loads conditioning, stress calculation, and damage calculation is made between frequency and time domain approaches.

Frequency Domain Analysis for Full Vehicles

The advantages of the frequency domain come from its simplicity and elegance. The structural system is defined via a very simple input-output relationship as shown in figure 1.

The input and output are connected via the system properties, which are commonly described using the term "transfer function". This transfer function is obtained by applying a sweep (through a range of frequencies) of a unit load sine wave on the structural system. Where systems have more than one input, like full car systems, the sweep process is performed for all input directions / locations (commonly referred to as channels).

Figure 1: Frequency domain input – output relationship.

The above process is linear. Hence, care must be taken to transfer loads through the highly non-linear suspension system to the car body via a nonlinear multi body dynamics tool like MSC ADAMS. Loads at the car body should be used in the linear frequency domain analysis.

It is important to note the difference between this frequency domain approach and a corresponding time domain approach. Let us compare the process flow shown in figure 2.

One significant advantage of the frequency domain approach comes from the fact that in the frequency domain the transfer function calculation (SOL103-SOL111) is independent of the loads (RPC-TIME2PSD) while the time domain approach includes the loads to calculate the equivalent Modal Participation Factors (RPC-SOL112). In practice, this means the user can iterate changing the input loads to calculate the Frequency Domain Fatigue reusing the transfer function calculation. This provides a significant speed-up efficiency and enables important real-world possibilities that require multiple iterations or optimization on loads such as loads enveloping and loads cascading [see reference 4].

Figure 2: Comparing the process flows for the time and frequency domains.

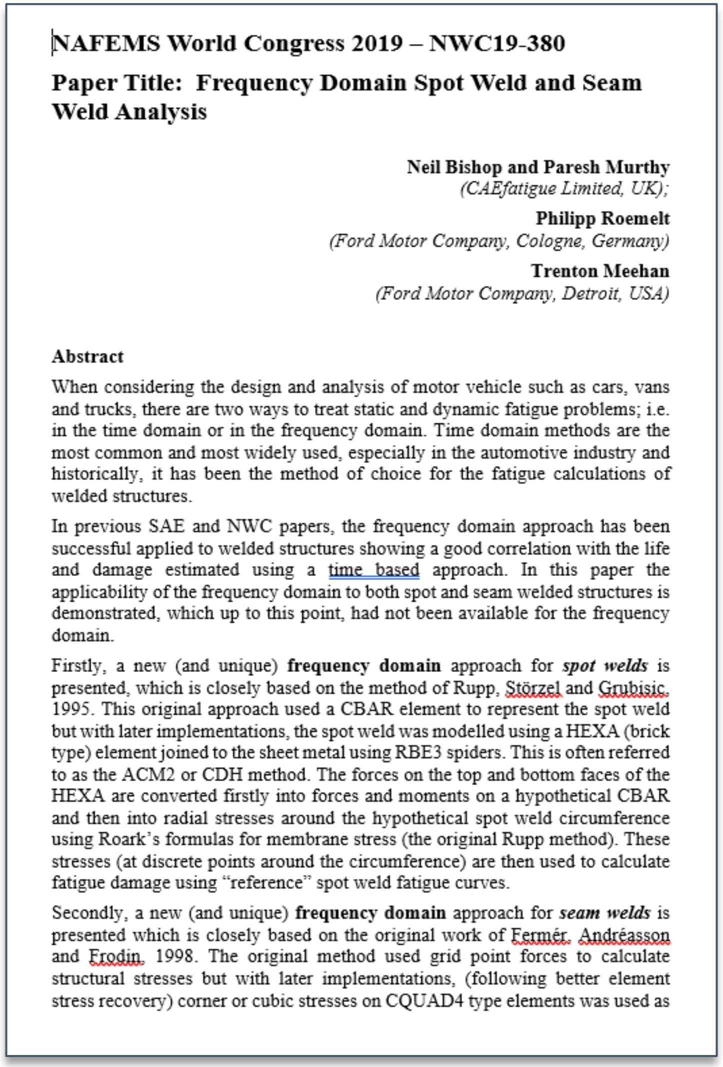
Page 1 of 10
10/3/2017

Key technologies covered: ***Multi-input*** analysis of ***Joints*** (spot and seam welds) using ***Loads Conditioning*** to generate the input PSDs of loading. Advanced ***Random*** output results were compared to a ***Time Domain*** Analysis with ***Parallel Processing***.

Ford

Automotive OEM

Advanced Frequency Domain Spot Weld and Seam Weld for automotive applications



NAFEMS World Congress 2019 – NWC19-380

Paper Title: Frequency Domain Spot Weld and Seam Weld Analysis

Neil Bishop and Paresh Murthy
(CAEfatigue Limited, UK);
Philipp Roemelt
(Ford Motor Company, Cologne, Germany)
Trenton Meehan
(Ford Motor Company, Detroit, USA)

Abstract

When considering the design and analysis of motor vehicle such as cars, vans and trucks, there are two ways to treat static and dynamic fatigue problems; i.e. in the time domain or in the frequency domain. Time domain methods are the most common and most widely used, especially in the automotive industry and historically, it has been the method of choice for the fatigue calculations of welded structures.

In previous SAE and NWC papers, the frequency domain approach has been successful applied to welded structures showing a good correlation with the life and damage estimated using a time based approach. In this paper the applicability of the frequency domain to both spot and seam welded structures is demonstrated, which up to this point, had not been available for the frequency domain.

Firstly, a new (and unique) **frequency domain** approach for ***spot welds*** is presented, which is closely based on the method of Rupp, Störzel and Grubisic, 1995. This original approach used a CBAR element to represent the spot weld but with later implementations, the spot weld was modelled using a HEXA (brick type) element joined to the sheet metal using RBE3 spiders. This is often referred to as the ACM2 or CDH method. The forces on the top and bottom faces of the HEXA are converted firstly into forces and moments on a hypothetical CBAR and then into radial stresses around the hypothetical spot weld circumference using Roark's formulas for membrane stress (the original Rupp method). These stresses (at discrete points around the circumference) are then used to calculate fatigue damage using "reference" spot weld fatigue curves.

Secondly, a new (and unique) **frequency domain** approach for ***seam welds*** is presented which is closely based on the original work of Fermér, Andréasson and Frodin, 1998. The original method used grid point forces to calculate structural stresses but with later implementations, (following better element stress recovery) corner or cubic stresses on CQUAD4 type elements was used as

Spot Weld

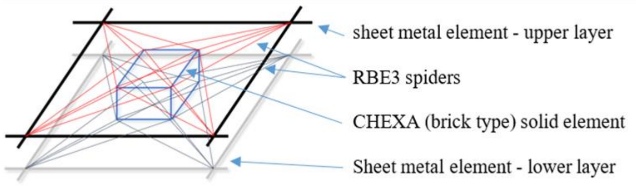


Figure 1: CHEXA element (brick type) jointed to sheet metal by RBE3 spiders.

Seam Weld

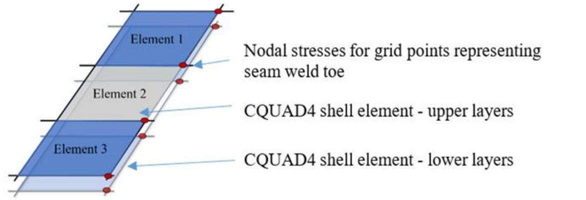
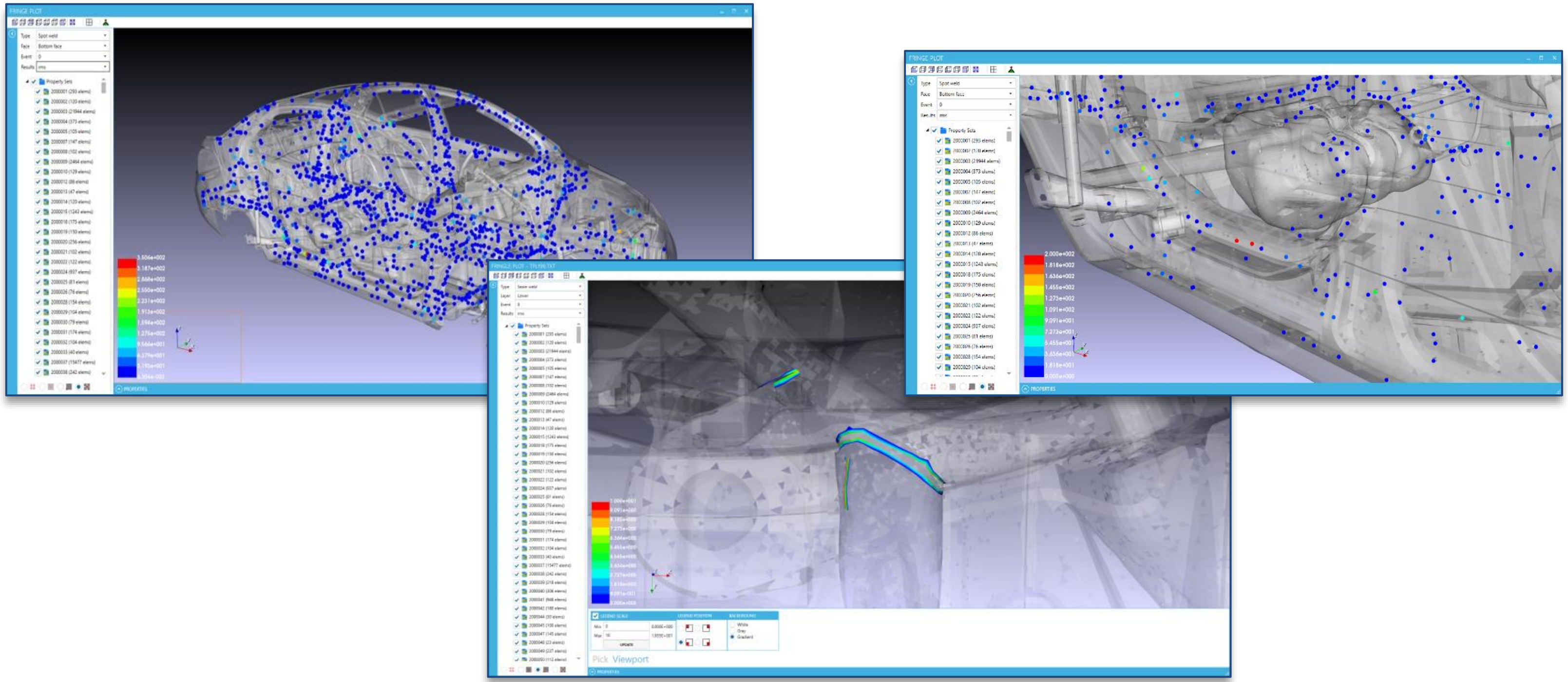


Figure 2: Nodal Streses required along sheet metal grid line representing the seam weld toe.



HEXAGON

Key technologies covered: ***Multi-input*** *analysis of* ***Joints*** *(spot and seam welds) using* ***Loads Conditioning*** *to generate the input PSDs of loading that produce advanced* ***Random*** *output results.*

Virtual vibration fatigue testing of an automotive EV battery pack

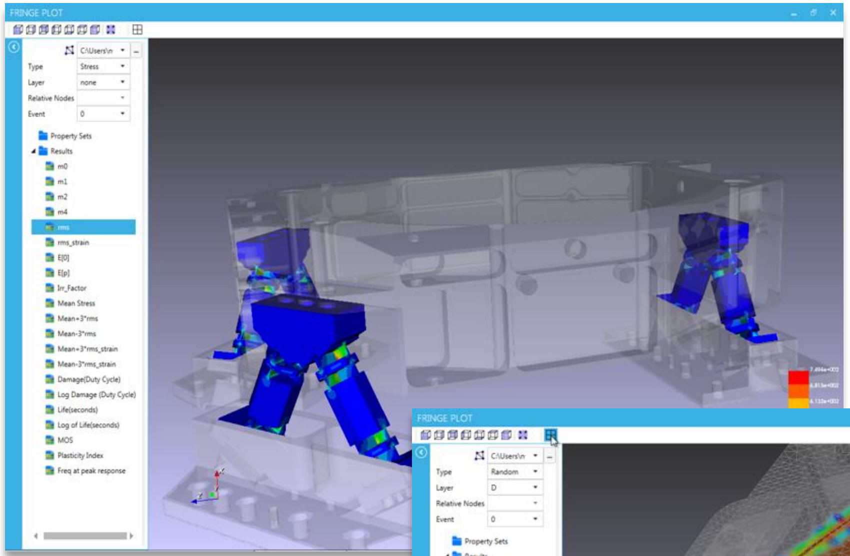
automotive CAE Grand Challenge 2021, Marco Veltri

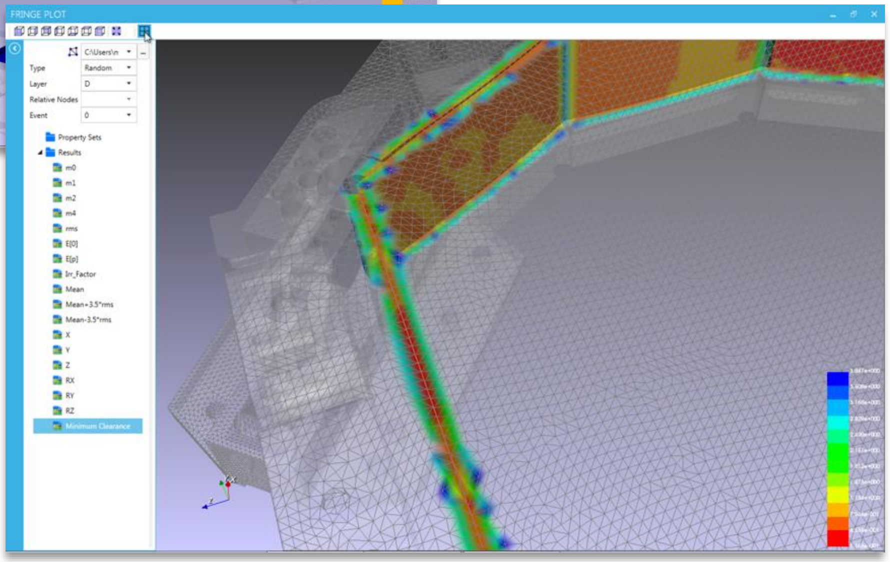
Customer Success Stories

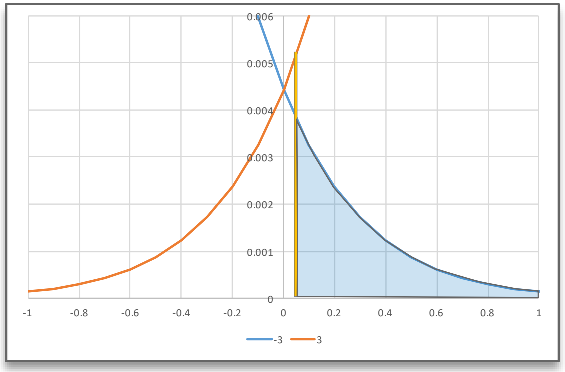
Advanced Collision Detection to predict part collisions in a satellite system

OHB

Satellites







NAFEMS World Congress 2017, Stockholm June 11-14th 2017

Simultaneous Durability Assessment and Relative Random Analysis Under Base Shake Loading Conditions

Abstract

For many aerospace syste
to rule out the possibility
shake vibration conditions

Advanced frequency dom
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efficient analysis tools. In
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applied. The most advanc
algorithms (eg, Strain-Lif
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The check for "collisions"
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However, new technologi
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the CAEfatigue VIBRATI

Using this new approach,
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Success Story

Altair | Partner Alliance

HyperWorks and the Altair Partner Alliance Streamline Development Process of Optical Satellite Components at OHB System AG

OHB

Key Highlights

Industry
Space Vehicles

Challenge
Satellite program development

Altair Solution
CAEfatigue VIBRATION, Altair HyperMesh, Altair OptiStruct

Benefits
- Drastically reduced random analysis time from days to hours
- More refined prototypes early on
- Access to all necessary software in one environment

Overview and Customer Profile

With so many subsystems and components, spacecrafts are subject to various loads during their lifetime, all of which have to be considered in the design process to avoid functional errors or failure of the system. While the launch of a spacecraft generates the highest loads, there are also other events that can be critical for parts of the structure. To ensure the structure's ability to survive as well as the system's functionality, every component has to be analyzed in detail. Engineers are able to predict some of the loads as a function of time (deterministic loads), but they can only estimate others statistically (random loads).

For these random loads, the engineers normally employ a random environment introducing broadband and high frequency vibration into the system, which comes from the engine, acoustic loads or aerodynamic turbulences. Besides other virtual tests prior to a physical prototype, the components are analyzed for random response to get a deeper insight on the resulting forces and stresses. To reduce development time and costs, it is a standard practice to use sophisticated CAE tools to avoid design changes when problems occur later in the development process (i.e. in the physical testing phase of prototypes).

OHB System AG is one of the leading space companies in Europe and specializes in high-tech solutions for space, science and industry. In two recent projects, the engineers used the CAE tool CAEfatigue VIBRATION (CFV), as well as Altair's HyperWorks software suite, to streamline the development process of optical instruments used in satellites, such as the MTG (European Space Agency (ESA) weather satellite) or ENMAP (German Aerospace Center (DLR) satellite). By applying CAE tools such as CFV, they were able to cut down the time needed for random response analysis of those components from several days to a few hours.

HyperWorks is a division of Altair

Keywords: HyperWorks, CAEfatigue VIBRATION, APA, CAE, Satellites, Space Vehicles

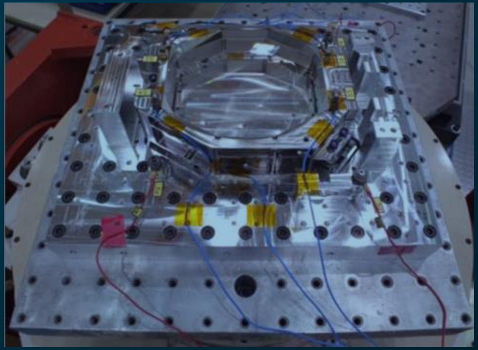
Key technologies covered: ***Base Shake*** *(single input) analysis together with advanced* ***Random*** *output to do* ***Collision Detection*** *between parts.*

Virtual qualification for a new satellite optical unit

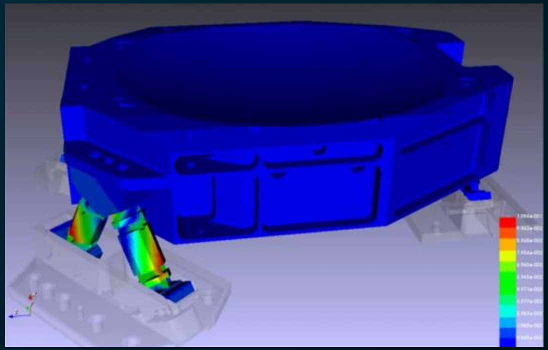
Industry Pain Points

Qualification test for highly sensitive optical units (by-pod mirror)

- Severe vibration
- Determination of allowable random acceleration and stress
- Avoid contact between Sensitive optical components (mirrors)



Why Hexagon?



High-performance simultaneous random vibration and fatigue analysis

- Proven consistency with industry, standard stress algorithms
- Determination of parts clearance as well as probability of collision
- Unique visual random and fatigue process, intuitive and repeatable

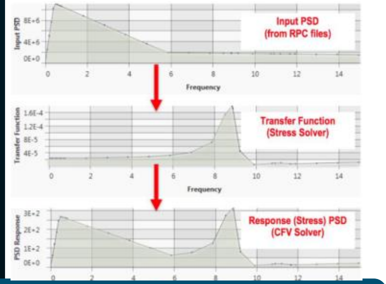
Report

Simultaneous Durability Assessment and Relative Random Analysis Under Base Shake Loading Conditions

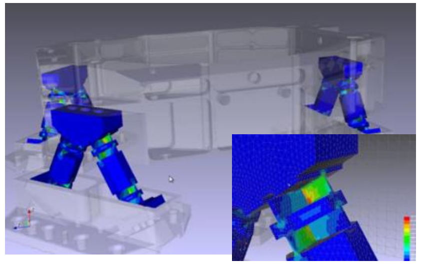
NAFEMS World Congress 2017, Stockholm June 11-14th 2017

Full article

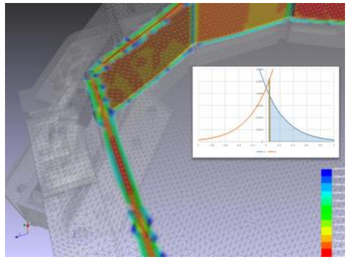
Solution Elements



Visual Random Response for vibration analysis



Stress and vibration fatigue analysis



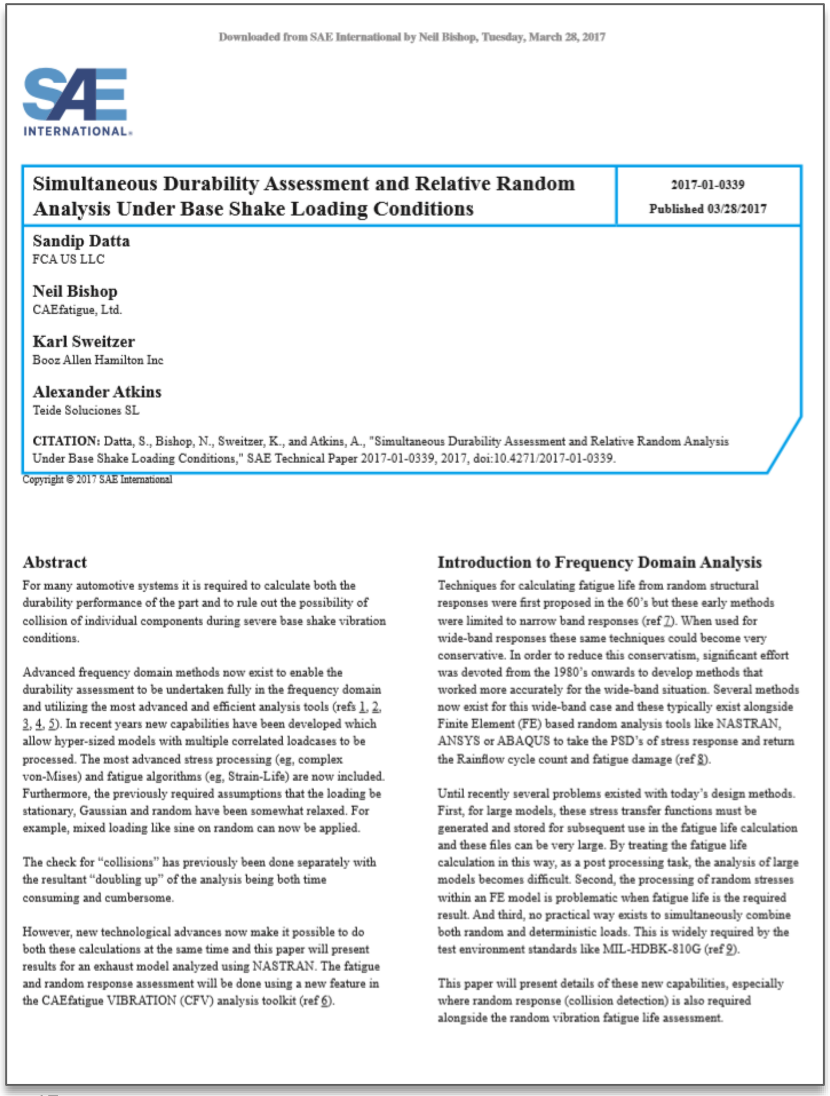
Parts clearance and collision detection

Customer Success Stories

FCA

Automotive OEM

Advanced Collision Detection to predict part collisions in an automotive system



SAE INTERNATIONAL

Simultaneous Durability Assessment and Relative Random Analysis Under Base Shake Loading Conditions

2017-01-0339

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Abstract

For many automotive systems it is required to calculate both the durability performance of the part and to rule out the possibility of collision of individual components during severe base shake vibration conditions.

Advanced frequency domain methods now exist to enable the durability assessment to be undertaken fully in the frequency domain and utilizing the most advanced and efficient analysis tools (refs 1, 2, 3, 4, 5). In recent years new capabilities have been developed which allow hyper-sized models with multiple correlated loadcases to be processed. The most advanced stress processing (eg, complex von-Mises) and fatigue algorithms (eg, Strain-Life) are now included. Furthermore, the previously required assumptions that the loading be stationary, Gaussian and random have been somewhat relaxed. For example, mixed loading like sine on random can now be applied.

The check for "collisions" has previously been done separately with the resultant "doubling up" of the analysis being both time consuming and cumbersome.

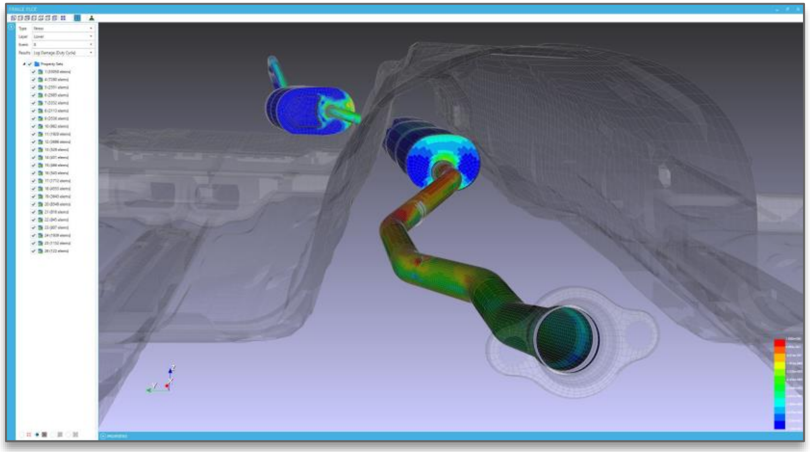
However, new technological advances now make it possible to do both these calculations at the same time and this paper will present results for an exhaust model analyzed using NASTRAN. The fatigue and random response assessment will be done using a new feature in the CAEfatigue VIBRATION (CFV) analysis toolkit (ref 6).

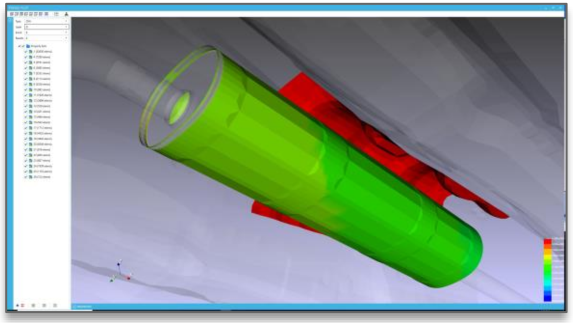
Introduction to Frequency Domain Analysis

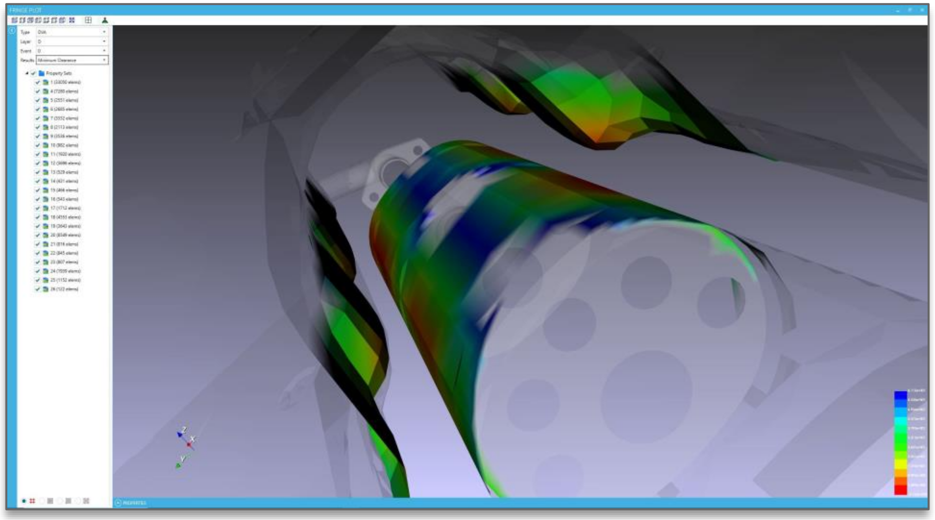
Techniques for calculating fatigue life from random structural responses were first proposed in the 60's but these early methods were limited to narrow band responses (ref 7). When used for wide-band responses these same techniques could become very conservative. In order to reduce this conservatism, significant effort was devoted from the 1980's onwards to develop methods that worked more accurately for the wide-band situation. Several methods now exist for this wide-band case and these typically exist alongside Finite Element (FE) based random analysis tools like NASTRAN, ANSYS or ABAQUS to take the PSD's of stress response and return the Rainflow cycle count and fatigue damage (ref 8).

Until recently several problems existed with today's design methods. First, for large models, these stress transfer functions must be generated and stored for subsequent use in the fatigue life calculation and these files can be very large. By treating the fatigue life calculation in this way, as a post processing task, the analysis of large models becomes difficult. Second, the processing of random stresses within an FE model is problematic when fatigue life is the required result. And third, no practical way exists to simultaneously combine both random and deterministic loads. This is widely required by the test environment standards like MIL-HDBK-810G (ref 9).

This paper will present details of these new capabilities, especially where random response (collision detection) is also required alongside the random vibration fatigue life assessment.







HEXAGON

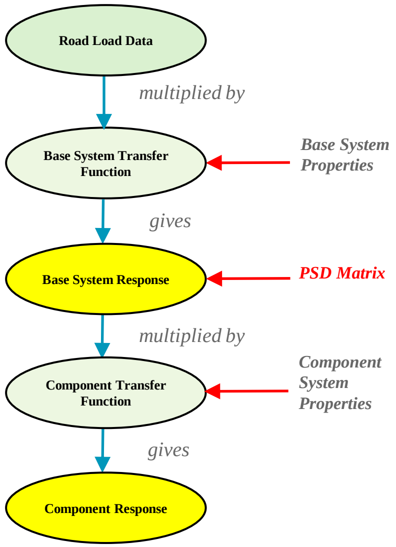
*Key technologies covered: **Base Shake** (single input) analysis together with advanced **Random** output to do **Collision Detection** between parts.*

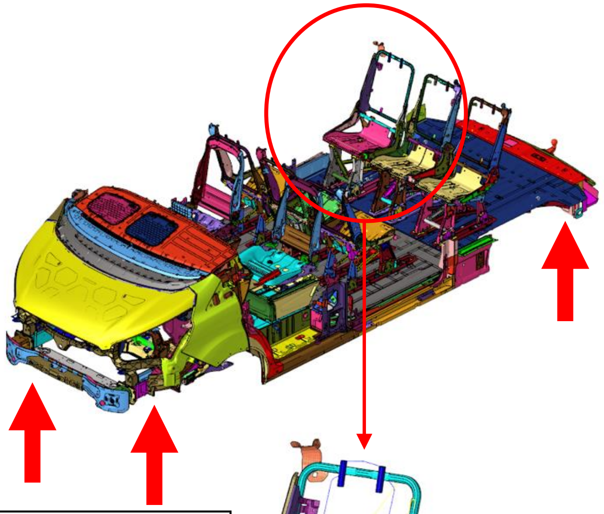
Customer Success Stories

Ford

Automotive OEM

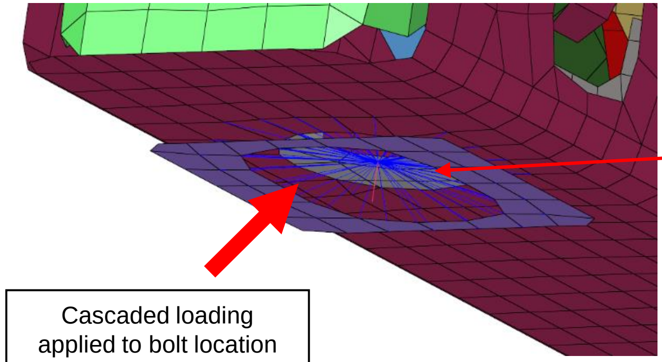
Advanced Loads Cascading to generate accurate loading for component design





Test track data collected for chassis loading

Chassis loading cascaded to seat bolt locations



Cascaded loading applied to bolt location

2020-01-0762

Loads Cascading for Full Vehicle Component Design

Philipp Roemelt §, Christoph Hallet§,
§Ford, Germany, §§CAE

Abstract

Frequency domain methods of analysis are now being used for the evaluation of fatigue for large vehicle systems and these methods offer advantages over equivalent time domain approaches in a number of ways, including analysis efficiency and the usefulness of derived results. One big potential advantage is to be able to do localized sub-component analysis using "cascaded" loads. Such sub-components can be analysed with refined parameters such as more sophisticated damping or a different frequency range. Local parts can also be re-analysed at a different phase in the design program. This paper will demonstrate the approach and show examples of the method.

Introduction

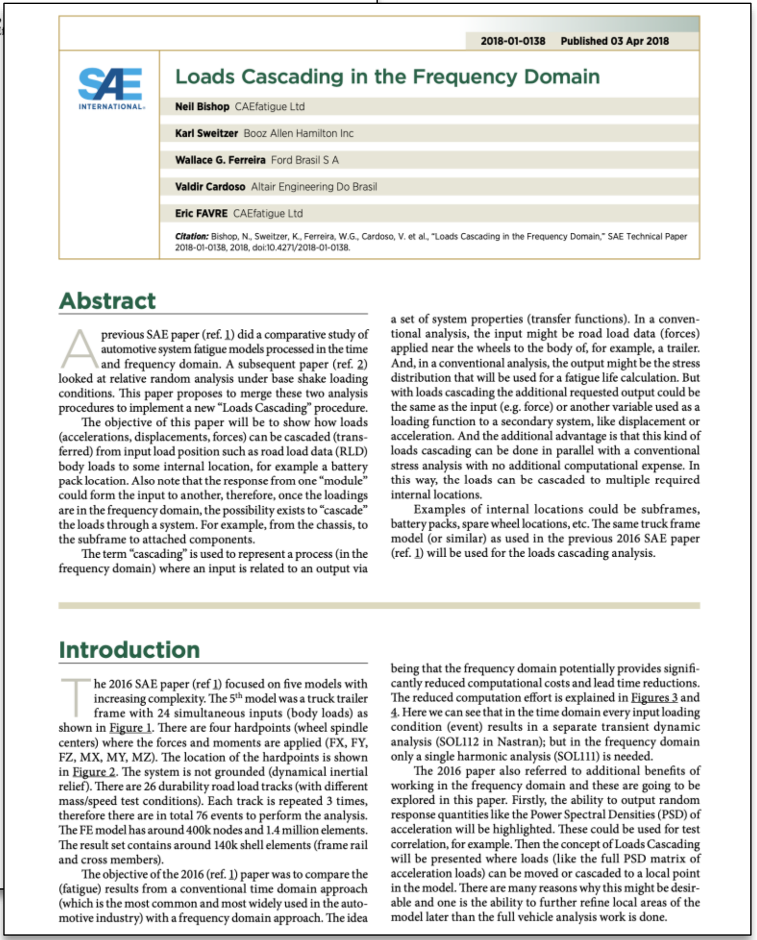
Over their lifetime cars, vans and trucks are subjected to random loading during normal usage due to road load input. This load input can lead to fatigue cracking, which makes fatigue analyses during the design process a must. This loading data must be derived first either by using prototype vehicles on a proving ground, or by using analytical Multi Body Dynamics (MBD) models [24].

Fatigue analyses can be conducted in combination with different type of finite element models, quasi-static, transient or frequency response, depending on the nature of the system or component that is analysed. If the fatigue behaviour is driven by the load magnitude only, i.e. the highest frequency of loading is 1/3 (or less) of the lowest eigenfrequency of the system, then a quasi-static analysis is performed [23]. For this unit loads are applied at the load application points and the resulting stress tensors are scaled and superimposed for all load conditions. If the fatigue behaviour is also driven by the load frequency, i.e. the system responds dynamically to the frequency of the load input, either a transient or frequency response analysis need to be conducted [23].

After the full stress tensor – time or response history is calculated a representative stress is calculated. A number of different stresses are available, e.g. von-Mises, principle stress, critical plane etc., and the choice usually depends on the type of problem that is analysed [23]. Once the representative stress history is known a cycle counting technique is used to group the representative stress according to magnitude, mean stress and number of occurrences. Each cycle group is then corrected for mean stress and then used to calculate a damage parameter like Smith-Watson-Topper or Morrow. Finally the individual damages are accumulated using Palmgren-Miner's Law to calculate a total damage value.

Fatigue analysis in the frequency domain has been widely used by the automotive industry for a number of system and subsystem analysis [1-3]. The capabilities of these type of analyses have recently been extended with the addition of spot and seam weld fatigue [4] and collision detection [5]. An additional feature now available is the cascading of loads from the original load attachment points to subsystem interface points [6]. This allows for more efficient

Page 1 of 7



2018-01-0138 Published 03 Apr 2018

SAE INTERNATIONAL

Loads Cascading in the Frequency Domain

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Citation: Bishop, N., Sweitzer, K., Ferreira, W.G., Cardoso, V. et al., "Loads Cascading in the Frequency Domain," SAE Technical Paper 2018-01-0138, 2018, doi:10.4271/2018-01-0138.

Abstract

A previous SAE paper (ref. 1) did a comparative study of automotive system fatigue models processed in the time and frequency domain. A subsequent paper (ref. 2) looked at relative random analysis under base shake loading conditions. This paper proposes to merge these two analysis procedures to implement a new "Loads Cascading" procedure.

The objective of this paper will be to show how loads (accelerations, displacements, forces) can be cascaded (transferred) from input load position such as road load data (RLD) body loads to some internal location, for example a battery pack location. Also note that the response from one "module" could form the input to another, therefore, once the loadings are in the frequency domain, the possibility exists to "cascade" the loads through a system. For example, from the chassis, to the subframe to attached components.

The term "cascading" is used to represent a process (in the frequency domain) where an input is related to an output via a set of system properties (transfer functions). In a conventional analysis, the input might be road load data (forces) applied near the wheels to the body of, for example, a trailer. And, in a conventional analysis, the output might be the stress distribution that will be used for a fatigue life calculation. But with loads cascading the additional requested output could be the same as the input (e.g. force) or another variable used as a loading function to a secondary system, like displacement or acceleration. And the additional advantage is that this kind of loads cascading can be done in parallel with a conventional stress analysis with no additional computational expense. In this way, the loads can be cascaded to multiple required internal locations.

Examples of internal locations could be subframes, battery packs, spare wheel locations, etc. The same truck frame model (or similar) as used in the previous 2016 SAE paper (ref. 1) will be used for the loads cascading analysis.

Introduction

The 2016 SAE paper (ref 1) focused on five models with increasing complexity. The 5th model was a truck trailer frame with 24 simultaneous inputs (body loads) as shown in Figure 1. There are four hardpoints (wheel spindle centers) where the forces and moments are applied (FX, FY, FZ, MX, MY, MZ). The location of the hardpoints is shown in Figure 2. The system is not grounded (dynamical inertial relief). There are 26 durability road load tracks (with different mass/speed test conditions). Each track is repeated 3 times, therefore there are in total 76 events to perform the analysis. The FE model has around 400k nodes and 1.4 million elements. The result set contains around 140k shell elements (frame rail and cross members).

The objective of the 2016 (ref. 1) paper was to compare the (fatigue) results from a conventional time domain approach (which is the most common and most widely used in the automotive industry) with a frequency domain approach. The idea being that the frequency domain potentially provides significantly reduced computational costs and lead time reductions. The reduced computation effort is explained in Figures 3 and 4. Here we can see that in the time domain every input loading condition (event) results in a separate transient dynamic analysis (SOL112 in Nastran); but in the frequency domain only a single harmonic analysis (SOL111) is needed.

The 2016 paper also referred to additional benefits of working in the frequency domain and these are going to be explored in this paper. Firstly, the ability to output random response quantities like the Power Spectral Densities (PSD) of acceleration will be highlighted. These could be used for test correlation, for example. Then the concept of Loads Cascading will be presented where loads (like the full PSD matrix of acceleration loads) can be moved or cascaded to a local point in the model. There are many reasons why this might be desirable and one is the ability to further refine local areas of the model later than the full vehicle analysis work is done.

HEXAGON

*Key technologies covered: **Multi-input** analysis using **Loads Conditioning** and advanced **Random** results plus **Loads Cascading** to generate additional input loads at seat connection points.*

Navistar

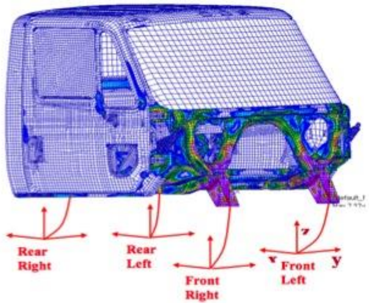
Truck Cabs and Trailers

Frequency Domain Analysis of Truck Cabs
(has now been extended with Surrogate Load Analysis, Loads Cascading and Collision Detection)

Test Track Duty Cycles

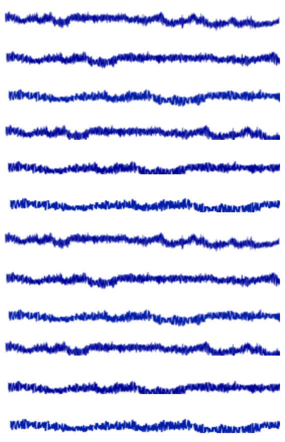
Event	Name	Description	Time
1	ea10	Empty Abbreviated Belgian Blocks at 10 mph	152.0
2	egvl	Empty Gravel Road	42.0
3	er20	Empty Railroad Crossing at 20 mph	18.0
4	er30	Empty Railroad crossing at 30 mph	18.0
5	lp10	Loaded Primary Belgian Blocks at 10 mph	480.0
6	lp12	Loaded Primary Belgian Blocks at 12 mph	1682.0
7	lp14	Loaded Primary Belgian Blocks at 14 mph	378.0
8	lr20	Loaded Railroad Crossing at 20 mph	18.0
9	lr30	Loaded Railroad Crossing at 30 mph	18.0
10	ls20	Loaded Supplemental Course at 20 mph	68.0

Example:
We want to apply the time signals for Event 6 as a PSD matrix to the 4 mounting points on the cab. To do this we **convert the signals into direct and cross PSDs** and do the **fatigue analysis** in the **Frequency Domain**.



Loading Channels for Event 6

time signal 1
time signal 2
time signal 3
time signal 4
time signal 5
time signal 6
time signal 7
time signal 8
time signal 9
time signal 10
time signal 11
time signal 12



Time Signals converted to Direct and Cross PSDs

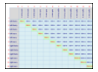
$$S(f) = \sum_{a=1}^{n}\sum_{b=1}^{n} H_a^*(f).H_b(f).G_{ab}(f)$$

Resulting PSD MATRIX (PSDM) for Event 6

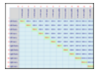
J \ K		1 left front x	2 left front y	3 left front z	4 left rear x	5 left rear y	6 left rear z	7 right front x	8 right front y	9 right front z	10 right rear x	11 right rear y	12 right rear z
1	left front x	PSD1-1	CSD1-2	CSD1-3	CSD1-4	CSD1-5	CSD1-6	CSD1-7	CSD1-8	CSD1-9	CSD1-10	CSD1-11	CSD1-12
2	left front y		PSD2-2	CSD2-3	CSD2-4	CSD2-5	CSD2-6	CSD2-7	CSD2-8	CSD2-9	CSD2-10	CSD2-11	CSD2-12
3	left front z			PSD3-3	CSD3-4	CSD3-5	CSD3-6	CSD3-7	CSD3-8	CSD3-9	CSD3-10	CSD3-11	CSD3-12
4	left rear x				PSD4-4	CSD4-5	CSD4-6	CSD4-7	CSD4-8	CSD4-9	CSD4-10	CSD4-11	CSD4-12
5	left rear y					PSD5-5	CSD5-6	CSD5-7	CSD5-8	CSD5-9	CSD5-10	CSD5-11	CSD5-12
6	left rear z						PSD6-6	CSD6-7	CSD6-8	CSD6-9	CSD6-10	CSD6-11	CSD6-12
7	right front x							PSD7-7	CSD7-8	CSD7-9	CSD7-10	CSD7-11	CSD7-12
8	right front y								PSD8-8	CSD8-9	CSD8-10	CSD8-11	CSD8-12
9	right front z									PSD9-9	CSD9-10	CSD9-11	CSD9-12
10	right rear x										PSD10-10	CSD10-11	CSD10-12
11	right rear y											PSD11-11	CSD11-12
12	right rear z												PSD12-12

Test Track Duty Cycles converted to PSDM files

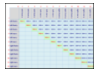
PSD Matrix Event 1
152 seconds



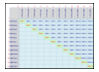
PSD Matrix Event 2
42 seconds



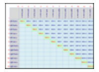
PSD Matrix Event 3
18 seconds



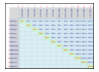
PSD Matrix Event 4
18 seconds



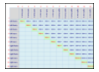
PSD Matrix Event 5
480 seconds



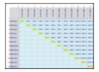
PSD Matrix Event 6
1682 seconds



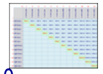
PSD Matrix Event 7
378 seconds



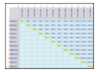
PSD Matrix Event 8
18 seconds



PSD Matrix Event 9
18 seconds



PSD Matrix Event 10
68 seconds



SAE INTERNATIONAL

Time vs Frequency Domain Analysis for Large Automotive Systems

2015-01-0535
Published 04/14/2015

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Booz Allen Hamilton Inc.

Stuart C. Kerr
CAEfatigue, Ltd.

CITATION: Bishop, N., Murthy, P., Sweitzer, K., and Kerr, S., "Time vs Frequency Domain Analysis For Large Automotive Systems," SAE Technical Paper 2015-01-0535, 2015, doi:10.4271/2015-01-0535.

Abstract

It has been recognised since the 1960's that the frequency domain method for structural analysis offers superior qualitative information about structural response (refs 1, 2, 3, 4); But computational and technological issues have held back the implementation for fatigue calculation until now. Recent technological developments (see refs 5, 6, 7, 8, 9) have now enabled the practical implementation of the frequency domain approach and this paper will focus on the accuracy of the approach when compared with the traditional time based (transient dynamic) approach.

Introduction

The general aim of this paper is to compare the time and frequency domain approaches for estimating response and fatigue damage for large automotive systems. The frequency domain approach has only recently become practically applicable because of a number of technical advances which are listed below.

- Extension from single to multiple load inputs.
- Correlation between inputs now properly dealt with (both complex terms).
- Both a modified von-Mises and/or Principal stress approach have been incorporated.
- The fatigue analysis has been extended to the Strain-Life (E-N) approach.
- Mixed random and deterministic loads allowed.
- Novel running sum moment procedure that enables the handling of very large transfer function.
- Both a Nastran modal (SOL111) and direct frequency response (SOL108) analysis allowed.
- New developments allow Stress-Life (S-N), E-N and random response only (including peak and rms elastic-plastic strain) to be calculated in one run.

We can therefore now assume that the frequency domain approach includes all the necessary technical ingredients needed for a full body fatigue assessment. An important specific aim of this paper is therefore to assess the level of accuracy associated with the frequency domain approach.

Performance Advantages

The potential performance advantages of working in the frequency domain are well known, as listed below,

- The frequency domain approach requires one frequency response function (FRF) per structural configuration, compared with one for each event in the time domain.
- With careful planning very large models can be easily handled in the frequency domain.
- The frequency domain approach provides better response information such as Power Spectral Densities (PSD's) and FRF's, related to qualitative behavior.
- The frequency domain approach can be easily used to establish damage per frequency, damage per mode or damage per event. This can then be used as a very useful diagnostic technique.
- Application of the frequency domain approach is far simpler for users and offers the possibility for significant improvements in terms of system integration.

Accuracy

Even though the frequency domain approach may offer potential performance advantages it is still necessary to determine that accuracy can be maintained and this paper will look at this issue in some detail. This detailed assessment will show that the time and frequency domain approaches give identical results (within statistical scatter limits), for both random response and fatigue life results.

HEXAGON

Key technologies covered: ***Multi Input*** *analysis,* ***Loads Cascading*** *(random) and* ***Surrogate Loads****.*

Customer Success Stories

Time Domain Analysis of Truck Cabs

Navistar

Truck Cabs and Trailers

Test Track Duty Cycles

Event	Name	Description	Time
1	ea10	Empty Abbreviated Belgian Blocks at 10 mph	152.0
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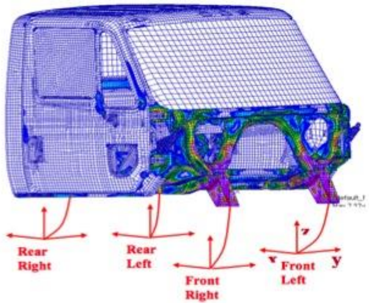
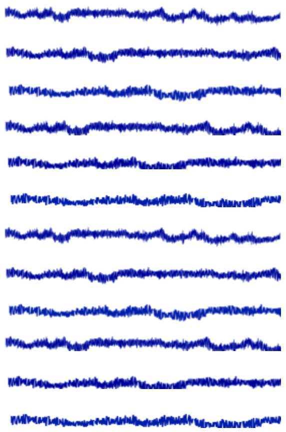
8	lr20	Loaded Railroad Crossing at 20 mph	18.0
9	lr30	Loaded Railroad Crossing at 30 mph	18.0
10	ls20	Loaded Supplemental Course at 20 mph	68.0

Example:
We want to apply the time signals for Event 6 to the 4 mounting points on the cab. The stress response is calculated using **LINEAR SUPERPOSITION** and the fatigue calculation is done in the **Time Domain.**



Loading Channels for Event 6

time signal 1
time signal 2
time signal 3
time signal 4
time signal 5
time signal 6
time signal 7
time signal 8
time signal 9
time signal 10
time signal 11
time signal 12



Nastran is used to calculate stress fields caused by unit loads on each of the 12 channels (Nastran subcases). These stress fields are then combined (for each Event) using linear superposition to obtain the stress time histories. These are then used for the fatigue calculation.

Damage Event 1
Damage Event 2
Damage Event 3
Damage Event 4
Damage Event 5
Damage Event 6
Damage Event 7
Damage Event 8
Damage Event 9
Damage Event 10

SAE INTERNATIONAL

Solver Embedded Fatigue

2014-01-0904
Published 04/01/2014

Timothy Palmer
MSC Software Corp.

Neil Bishop
CAEfatigue Ltd.

CITATION: Palmer, T. and Bishop, N., "Solver Embedded Fatigue," SAE Technical Paper 2014-01-0904, 2014, doi:10.4271/2014-01-0904.

Abstract

This paper presents a fundamental conceptual change to the traditional CAE based fatigue analysis process. Traditional approaches take the responses from a stress solver and these are then transferred into a secondary fatigue analysis step. In this way fatigue is, and always has been, treated as a post processing step. The new conceptual change described in this paper involves combining the two separate tasks into one (stress and fatigue together). This results in a simple, elegant and more powerful Durability Management concept. This new process requires no large data files to be transferred, no complicated file management and it is likely that whole fatigue calculation process can be done in memory. This makes it possible to perform optimization with fatigue life as the constraint. It also facilitates full body fatigue life calculations, including dynamic behavior, for much larger models than was previously possible. Within this new process, fatigue is treated as part of the solve process and not as a post processing operation.

Because of the simpler file management process, and the combined stress-fatigue solution, there will be dramatic reductions in analysis time and process management time. It should be anticipated that where a job takes a week or more to process, it should now be possible in a day or two. There will be a reduction in the time and resources needed for collaboration between different job functions and between different locations because of the simplified analysis definition format (everything in the bulk data file). The combination of optimization with fatigue will result in lighter, more efficient designs. The ability to handle larger models within a dynamic fatigue life calculation will provide greater design confidence.

A solver embedded fatigue process is easier to set up and fully transparent so fatigue analysis will become more accessible to the general analyst. Fatigue life calculations will no longer be the domain of the fatigue expert.

This paper provides a detailed overview of the solver embedded fatigue concept. This new process is demonstrated by application to a typical commercial automotive truck cab model.

Introduction

CAE fatigue methods involve the amalgamation of three separate topics - Fatigue, CAE and Dynamics, each with their own historical development and challenges. Methods for fatigue design became important with the birth of the industrialised society around 170 years ago and at around the same time August Wohler (ref 1) presented the basic relationship for fatigue life as a function of applied load. Then, around 100 years ago, Ewins and Humphrey (ref 2) identified what really causes fatigue, which is crack growth, but it was another 50 years before Paris, in the early 1960's, quantified the fundamental relationship for crack growth rate (ref 3). Around the same time computer based stress analysis tools, such as the **NA**SA **STR**ucture **AN**alysis program called Nastran, were being introduced which enable stresses to be evaluated over full model geometries. It was then only a matter of time before the connection was made between this new ability to generate stresses over the entirety of a structural model and a fatigue calculation.

They all have one undamental thing in common. In every case fatigue follows the stress calculation as a secondary post processing step as seen in Figure 2.

In fact this is true not only for all CAE based fatigue calculations but also for test based fatigue. Fatigue has always been considered as a post processing step following the stress calculation or stress measurement. This becomes very important when considering the data flow needed to undertake a fatigue design for a typical large structural design as shown in Figure 3.

HEXAGON

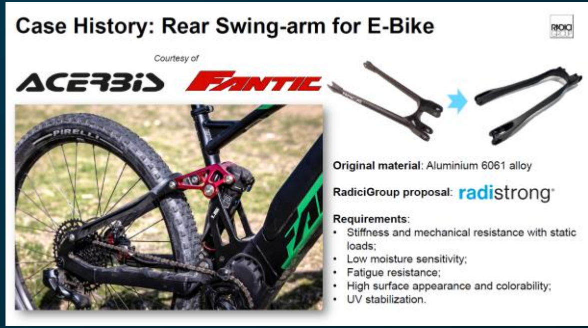
*Key technologies covered: **Time Domain** (multi input) analysis.*

Radici Group

Polymers

RADICI GROUP
HIGH PERFORMANCE POLYMERS

Metal Replacement with high performance polymers



Challenge

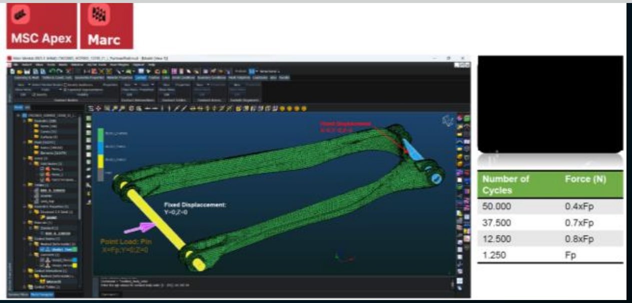
Substitute one or more parts currently made in metal with a part having the same Functionality but made with Engineering Plastics

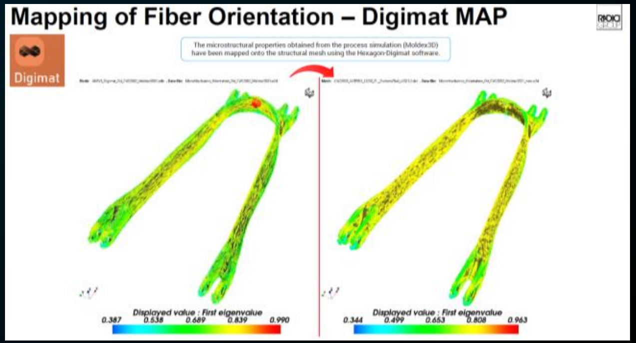
Solution

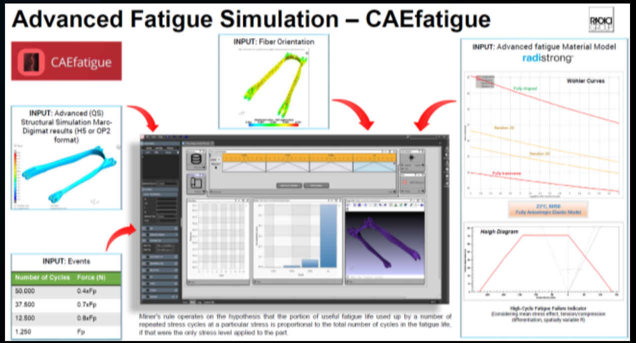
- Material Model generation (EP) Digimat
- Structural simulation MSC Apex and Marc
- Fiber Orientation mapping –Digimat-MAP
- Advanced Fatigue Simulation–CAEfatigue

Benefits

- **successful** performance prediction
- 10% weight reduction compared to metal version
- Sustainable design







Full public presentation here https://go.mi.hexagon.com/l/980223/2023-06-23/362mb7

HEXAGON

Husqvarna Group

Mechanical Systems

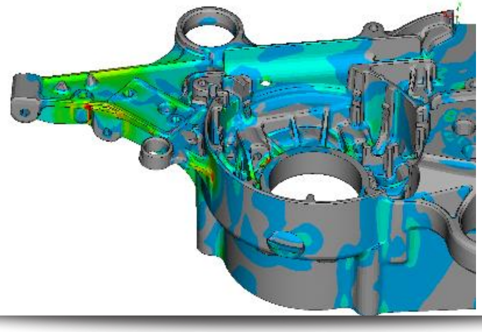
Husqvarna Group: Leveraging MSC Nastran E… increase result precision

Husqvarna Group

The Challenge

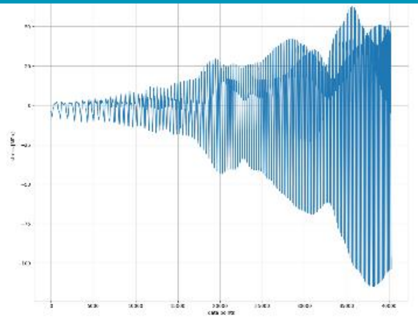
- Fatigue in any part of the product is caused by vibrations from the two-stroke engine, which drives for example a chainsaw or a handheld power cutter.

MSC Simulation Solution



- Husqvarna engineers decided to employ MSC Nastran Embedded Fatigue for virtual prototyping of a model to assess where the material could fail and to strengthen the part structural integrity accordingly
- NEF has useful functionality to automatically generate S-N curves.

Value Realized



- Using MSC Nastran Embedded Fatigue, Husquvarna can analyze the whole load cycle, not only the immediate surrounding of load peaks. The process has proven to be robust to changes in the excitation. Data transfer takes much less time.

"First there was no clearly defined workflow for durability analysis. This changed when MSC‘s Nastran Embedded Fatigue became available. After a year of intensive evaluation, we decided to use MSC Nastran Embedded Fatigue in productive operation for vibration analysis."

–Marcus Fälth, CAE Technology Specialist, Research and Development Department, Husqvarna AB

Read the article

Husqvarna Group

Mechanical Systems

Increase Result Precision with MSC Nastran Em

Husqvarna Group

"First there was no clearly defined workflow for durability analysis. This changed when MSC's Nastran Embedded Fatigue became available. After a year of intensive evaluation, we decided to use MSC Nastran Embedded Fatigue in productive operation for vibration analysis.

– Marcus Fälth, CAE Technology Specialist, Research and Development Department, Husqvarna AB

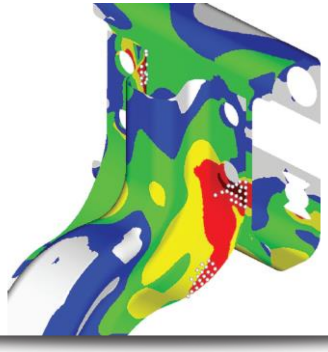
[Read the article]

HEXAGON

MSC NASTRAN Speeds Up Fatigue Lifecycle Predictions by 3X for Navistar- Tech Mahindra

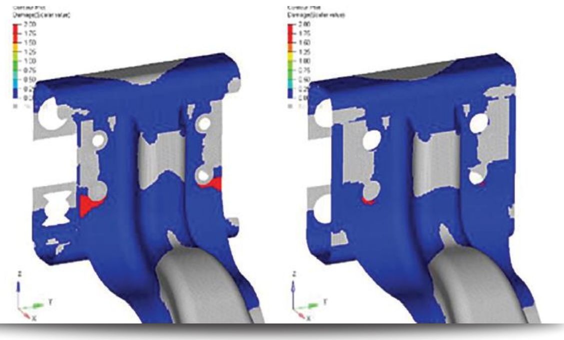
Mahindra

The Challenge



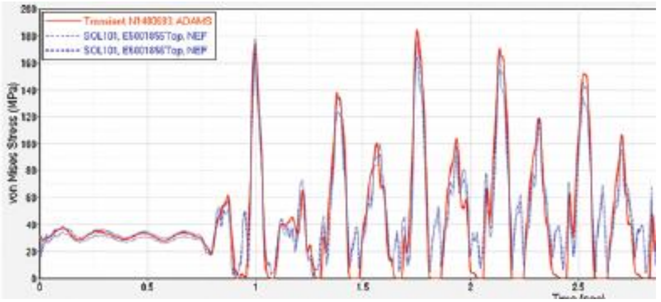
- Predicting Fatigue Life accurately in early stage of design enhances product life, reduces testing and prototype costs and ensures greater speed to market
- Physical tests are costly and time consuming

MSC Simulation Solution



- Nastran Embedded Fatigue provides a process that uses the entire time-history of loads, calculates fatigue life, and can display results of the entire component.
- Entire Adams time history of forces and moments are easily fed into NEF

Value Realized



- NEF enables identification of high locations more effectively
- 3X faster overall process than previous method.
- Fewer and smaller intermediate files makes bookkeeping simpler.

"Simplified process using MSC Nastran Embedded Fatigue enables highlighting high stress locations more effectively and has significant advantage in terms of time, ease of use as well as bookkeeping over traditional fatigue workflow."
-Pravin Kulkarni and Chinmay Pawaskar,
Tech Mahindra Ltd.

MSC NASTRAN Speeds Up Fatigue Lifecycle Predictions by 3X for Navistar- Tech Mahindra

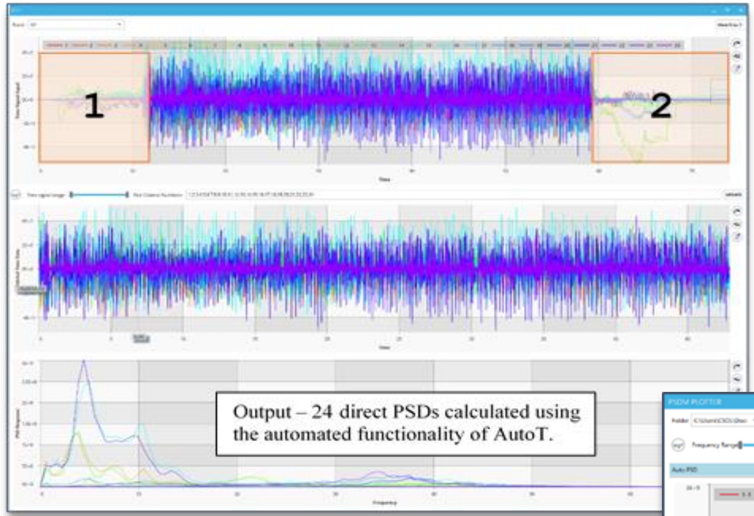
“Simplified process using MSC Nastran Embedded Fatigue enables highlighting high stress locations more effectively and has significant advantage in terms of time, ease of use as well as bookkeeping over traditional fatigue workflow.”

– *Pravin Kulkarni and Chinmay Pawaskar, Tech Mahindra Ltd.*

FCA

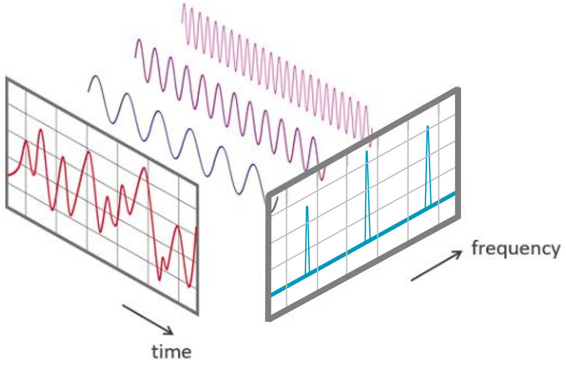
Automotive OEM

Advanced Loads Conditioning to convert time domain data to frequency domain



Creation of PSD Matrix

Direct PSD 1-1	Cross PSD 1-2	Cross PSD 1-3	Cross PSD 1-4
Cross PSD 1-2	Direct PSD 2-2	Cross PSD 2-3	Cross PSD 2-4
Cross PSD 1-3	Cross PSD 2-3	Direct PSD 3-3	Cross PSD 3-4
Cross PSD 1-4	Cross PSD 2-4	Cross PSD 3-4	Direct PSD 4-4



NAFEMS World Congress 2019 – NWC19-378

Paper Title: Loads Conditioning for Frequency Domain Analysis

Stuart Kerr and Neil Bishop
(CAEfatigue Limited, UK);
Sandip Datta
(Fiat Chrysler Corporation, USA).

Abstract

The accurate conversion of time-based data into an equivalent frequency domain form has become very important because of the increased use of frequency domain analysis for random response and durability calculations. Typically, for road vehicles, the loading conditions are measured at the wheel center and then cascaded into body loads using multi body dynamics systems (a partially virtual model). Or, in a more sophisticated system, the road profiles could be combined with a wheel/tire model to transfer the loads through the wheel system and onto the body of the vehicle (a fully virtual model). In both simulations, the cascaded loads on the body are in the time domain and usually in the form of multiple channel time signals. The correlation between the multiple channels can be very important to the final outputs for random response and fatigue.

If a frequency domain analysis is required, this time data must be converted into the frequency domain. This involves well understood "Fourier" conversion techniques in the form of Fourier Series or Fourier Transforms. However, a typical analyst faces 3 significant barriers when performing this conversion process.

Firstly, the frequency domain approach itself requires adherence to limiting assumptions. The data being processed must be,

- Stationary
- Gaussian
- Random

Typical users struggle to quantify these assumptions.

Secondly, the legacy use of Fourier Transforms (such as the Fast Fourier Transform – FFT) is difficult for a typical User because of several variables which must be set (such as FFT window shape, FFT window length, etc.). Setting these (and other variables) requires prior experience, which is beyond a typical User. Furthermore, the variables usually have to be set one by one for each load set (event) and this can be impractical.

HEXAGON

*Key technologies covered: **Loads Conditioning** and conversion of time signals into a PSD matrix.*

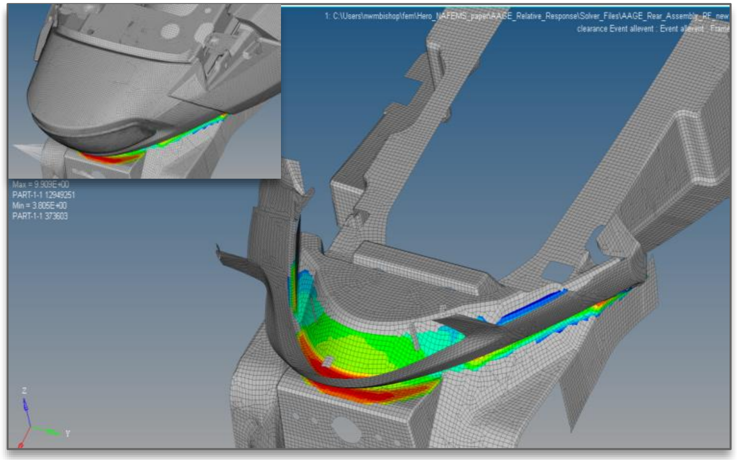
Customer Success Stories

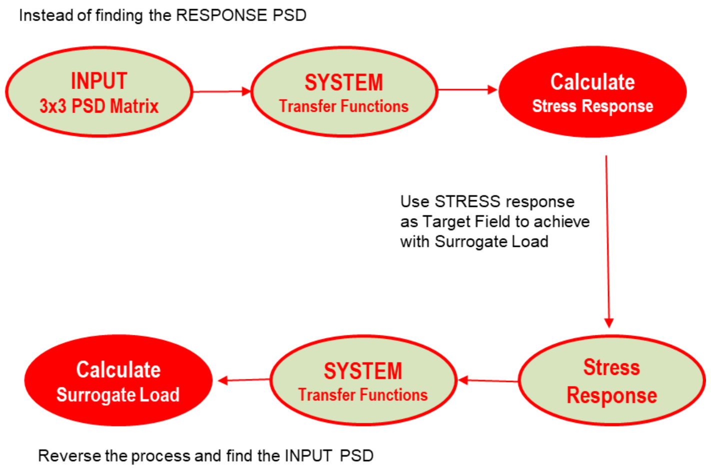
HeroMotoCorp

3-Wheeler OEM

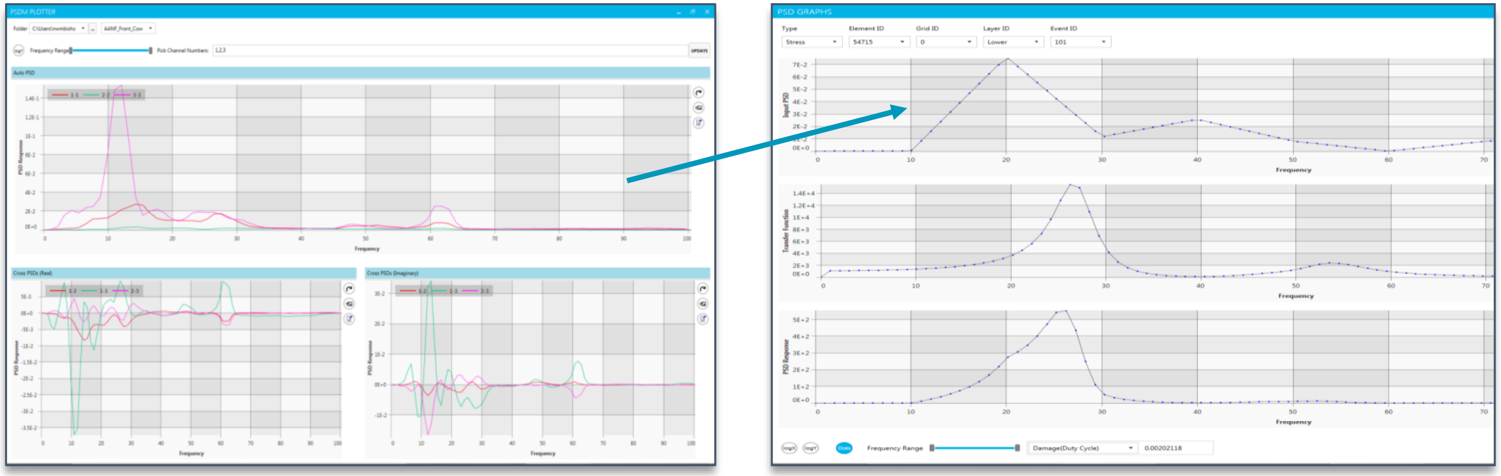
Advanced Collision Detection and Surrogate Loading used for motorcycles

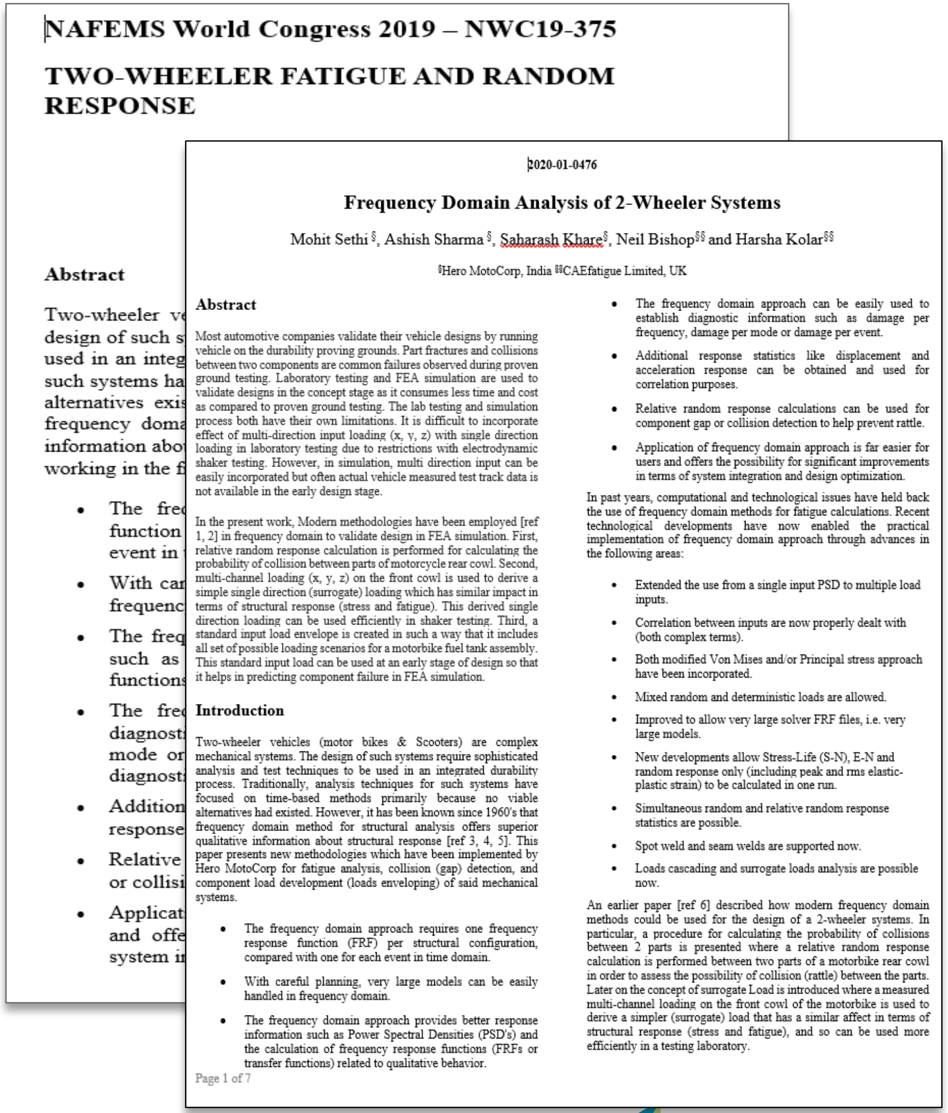
Collision Detection between seat and fender





Simplify multi-input loading into single input PSD





NAFEMS World Congress 2019 – NWC19-375

TWO-WHEELER FATIGUE AND RANDOM RESPONSE

Abstract

2020-01-0476

Frequency Domain Analysis of 2-Wheeler Systems

Mohit Sethi §, Ashish Sharma §, Saharash Khare§, Neil Bishop§§ and Harsha Kolar§§

§Hero MotoCorp, India §§CAEfatigue Limited, UK

Abstract

Most automotive companies validate their vehicle designs by running vehicle on the durability proving grounds. Part fractures and collisions between two components are common failures observed during proven ground testing. Laboratory testing and FEA simulation are used to validate designs in the concept stage as it consumes less time and cost as compared to proven ground testing. The lab testing and simulation process both have their own limitations. It is difficult to incorporate effect of multi-direction input loading (x, y, z) with single direction loading in laboratory testing due to restrictions with electrodynamic shaker testing. However, in simulation, multi direction input can be easily incorporated but often actual vehicle measured test track data is not available in the early design stage.

In the present work, Modern methodologies have been employed [ref 1, 2] in frequency domain to validate design in FEA simulation. First, relative random response calculation is performed for calculating the probability of collision between parts of motorcycle rear cowl. Second, multi-channel loading (x, y, z) on the front cowl is used to derive a simple single direction (surrogate) loading which has similar impact in terms of structural response (stress and fatigue). This derived single direction loading can be used efficiently in shaker testing. Third, a standard input load envelope is created in such a way that it includes all set of possible loading scenarios for a motorbike fuel tank assembly. This standard input load can be used at an early stage of design so that it helps in predicting component failure in FEA simulation.

Introduction

Two-wheeler vehicles (motor bikes & Scooters) are complex mechanical systems. The design of such systems require sophisticated analysis and test techniques to be used in an integrated durability process. Traditionally, analysis techniques for such systems have focused on time-based methods primarily because no viable alternatives had existed. However, it has been known since 1960's that frequency domain method for structural analysis offers superior qualitative information about structural response [ref 3, 4, 5]. This paper presents new methodologies which have been implemented by Hero MotoCorp for fatigue analysis, collision (gap) detection, and component load development (loads enveloping) of said mechanical systems.

- The frequency domain approach requires one frequency response function (FRF) per structural configuration, compared with one for each event in time domain.
- With careful planning, very large models can be easily handled in frequency domain.
- The frequency domain approach provides better response information such as Power Spectral Densities (PSD's) and the calculation of frequency response functions (FRFs or transfer functions) related to qualitative behavior.
- The frequency domain approach can be easily used to establish diagnostic information such as damage per frequency, damage per mode or damage per event.
- Additional response statistics like displacement and acceleration response can be obtained and used for correlation purposes.
- Relative random response calculations can be used for component gap or collision detection to help prevent rattle.
- Application of frequency domain approach is far easier for users and offers the possibility for significant improvements in terms of system integration and design optimization.

In past years, computational and technological issues have held back the use of frequency domain methods for fatigue calculations. Recent technological developments have now enabled the practical implementation of frequency domain approach through advances in the following areas:

- Extended the use from a single input PSD to multiple load inputs.
- Correlation between inputs are now properly dealt with (both complex terms).
- Both modified Von Mises and/or Principal stress approach have been incorporated.
- Mixed random and deterministic loads are allowed.
- Improved to allow very large solver FRF files, i.e. very large models.
- New developments allow Stress-Life (S-N), E-N and random response only (including peak and rms elastic-plastic strain) to be calculated in one run.
- Simultaneous random and relative random response statistics are possible.
- Spot weld and seam welds are supported now.
- Loads cascading and surrogate loads analysis are possible now.

An earlier paper [ref 6] described how modern frequency domain methods could be used for the design of a 2-wheeler systems. In particular, a procedure for calculating the probability of collisions between 2 parts is presented where a relative random response calculation is performed between two parts of a motorbike rear cowl in order to assess the possibility of collision (rattle) between the parts. Later on the concept of surrogate Load is introduced where a measured multi-channel loading on the front cowl of the motorbike is used to derive a simpler (surrogate) load that has a similar affect in terms of structural response (stress and fatigue), and so can be used more efficiently in a testing laboratory.

Page 1 of 7

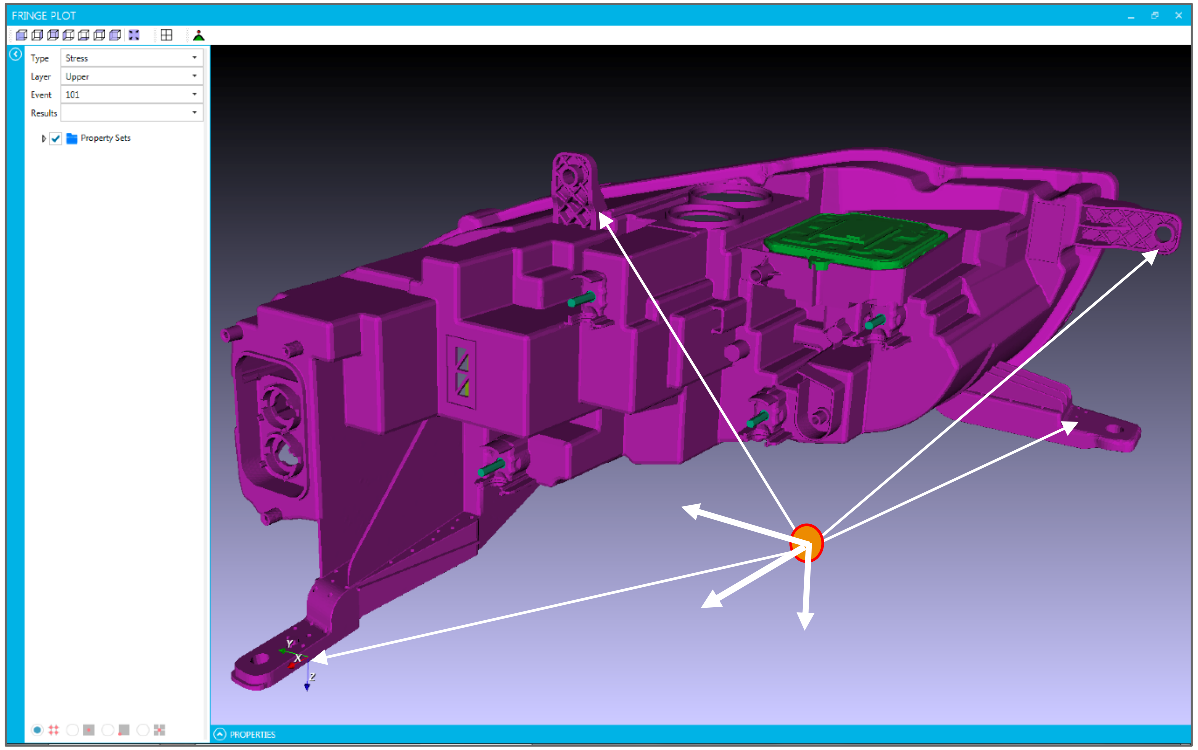
HEXAGON

*Key technologies covered: **Base Shake** and **Multi-input** analysis using **Loads Conditioning** and advanced **Random** results. **Loads Cascading** is used to generate additional input loads and **Surrogate Loads** is used to simplify those input loads into a single input PSD.*

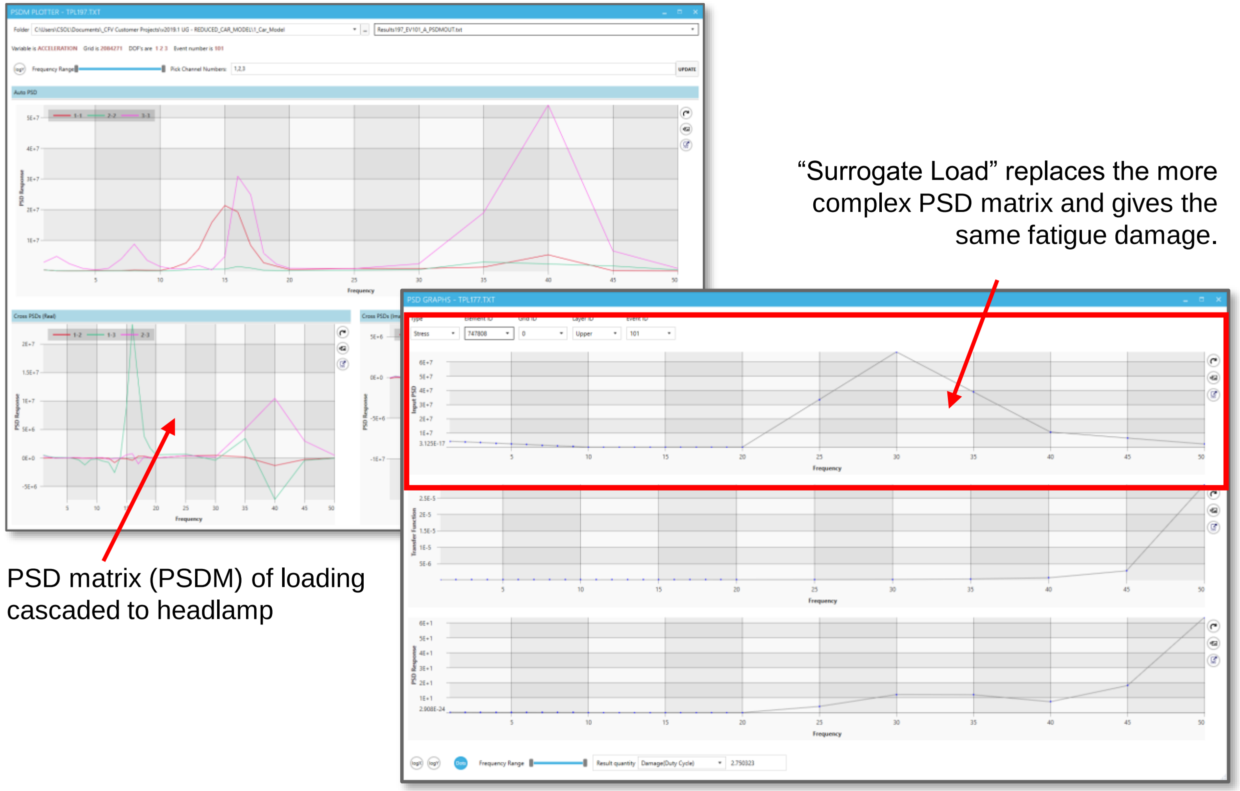
US Car Company

Headlamps

Advanced Surrogate Loads analysis for an automotive headlamp. Allows for the conversion of complicated multi-input loading into a simple single input PSD loading, that produces virtually the <u>same damage</u>.



Acceleration X, Y and Z loading cascaded (using CFR) from chassis point to headlamp



“Surrogate Load” replaces the more complex PSD matrix and gives the same fatigue damage.

PSD matrix (PSDM) of loading cascaded to headlamp

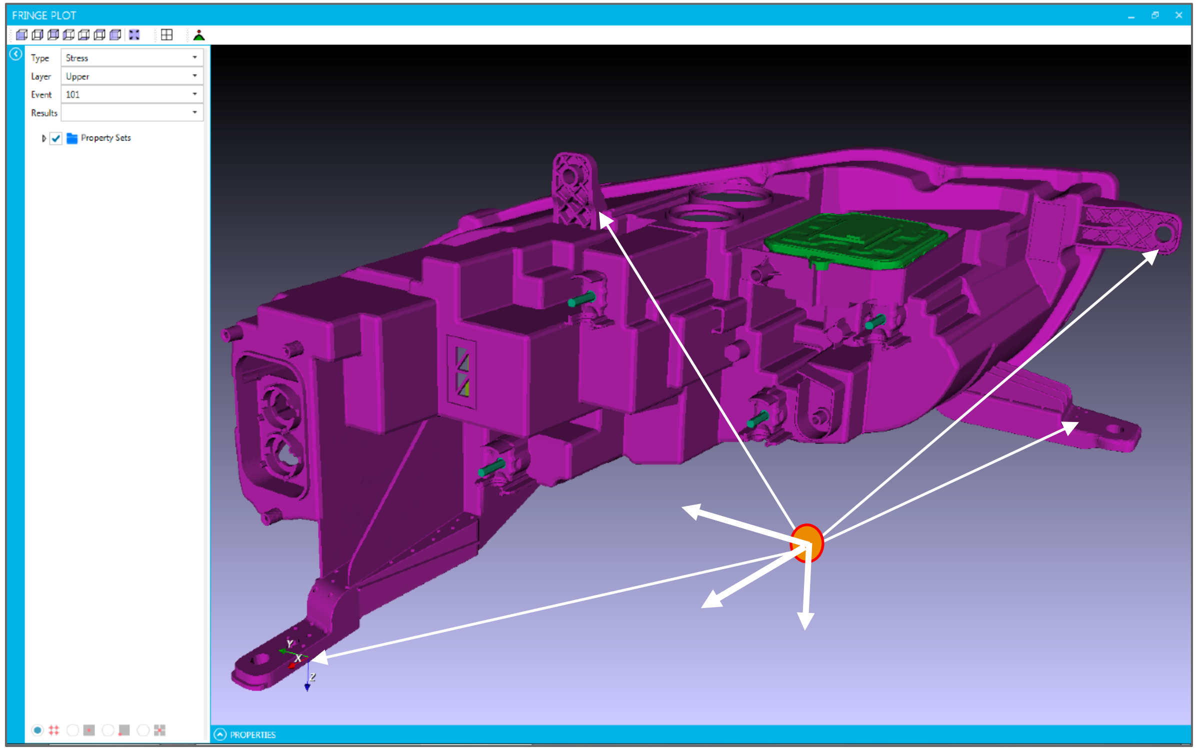
Example: Road Load Data can be cascaded from a chassis mounting point to a node(s) representing the headlamp mounting point(s) and those loads can be reduced to a single input PSD for simplified laboratory testing of a separate headlamp model. Excellent tool to reduce testing cost by permitting the use of less expensive test equipment.

HEXAGON

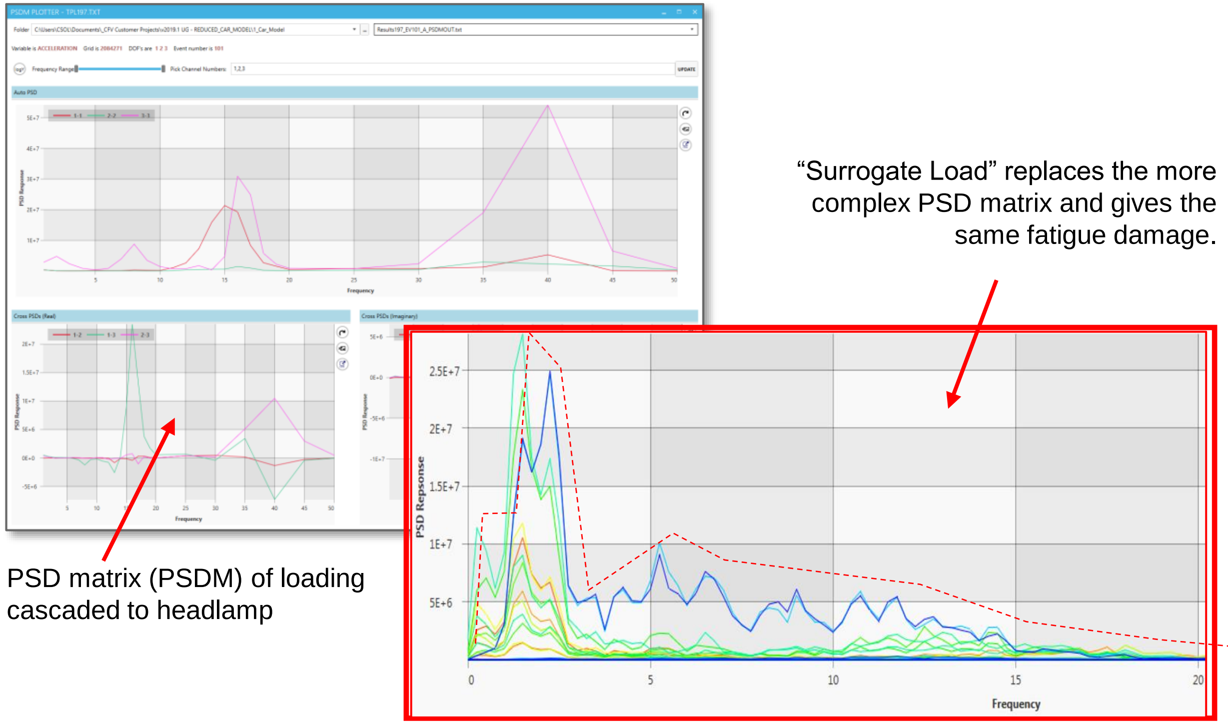
US Car Company

Headlamps

Advanced Surrogate Loads analysis for an automotive headlamp. Allows for the conversion of complicated multi-input loading into a simple single input a PSD Envelope, that produces virtually the same damage.



Acceleration X, Y and Z loading cascaded (using CFR) from chassis point to headlamp



PSD matrix (PSDM) of loading cascaded to headlamp

“Surrogate Load” replaces the more complex PSD matrix and gives the same fatigue damage.

Example: Road Load Data can be cascaded from a chassis mounting point to a node(s) representing the headlamp mounting point(s) and those loads can be reduced to a a PSD Envelope for simplified laboratory testing of a separate headlamp model. Note: PSD Envelopes are **VERY** conservative as they add RMS to the loading.

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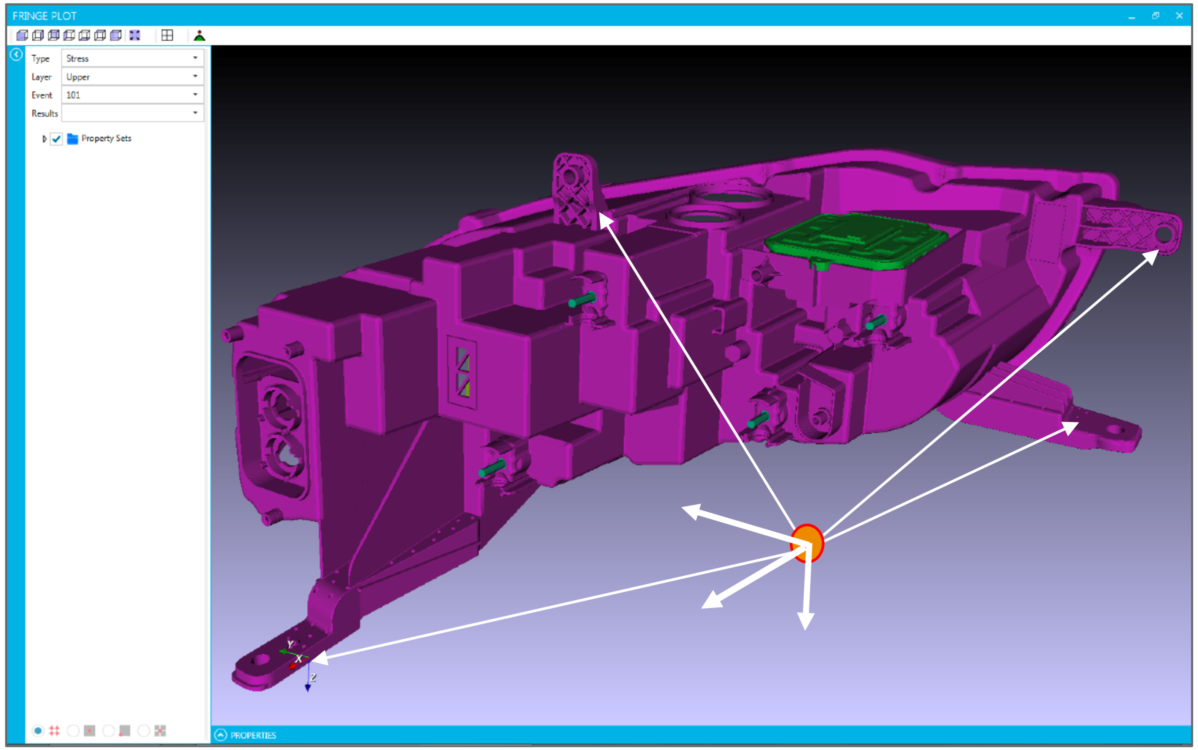
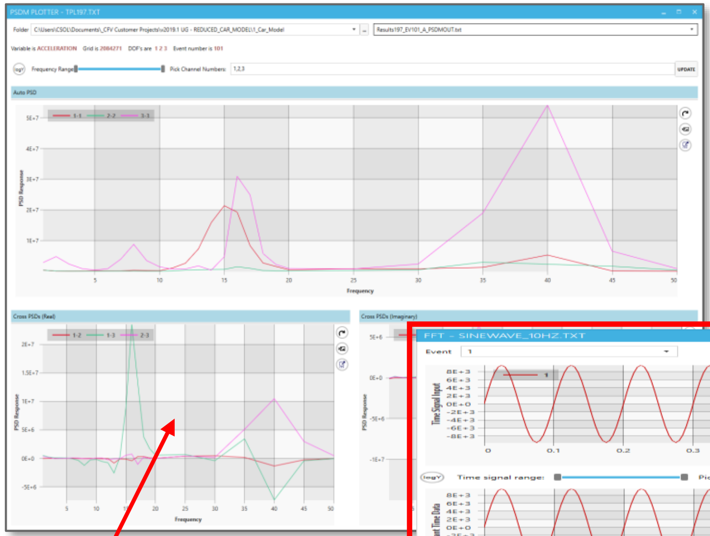
US Car Company

Headlamps

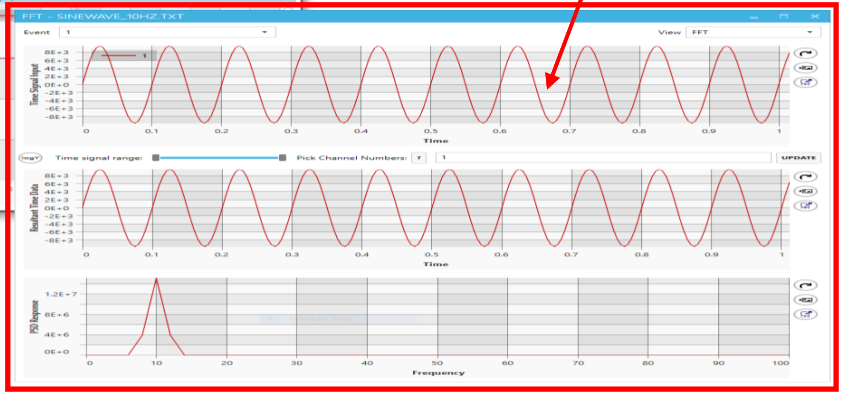
Advanced Surrogate Loads analysis for an automotive headlamp. Allows for the conversion of complicated multi-input loading into a simple single sine wave loading, that produces virtually the same damage.



Acceleration X, Y and Z loading cascaded (using CFR) from chassis point to headlamp



PSD matrix (PSDM) of loading cascaded to headlamp



“Surrogate Load” replaces the more complex PSD matrix and gives the same fatigue damage.

Example: Road Load Data can be cascaded from a chassis mounting point to a node(s) representing the headlamp mounting point(s) and those loads can be reduced to a sine wave for simplified laboratory testing of a separate headlamp model. Excellent tool to reduce testing cost by permitting the use of less expensive test equipment.

HEXAGON

Chinese Auto Company

Auto

Example of Surrogate Loads process

Preliminary Step: Conduct the CFV – Multi Input analysis on Hood with cascaded PSDM input loads.

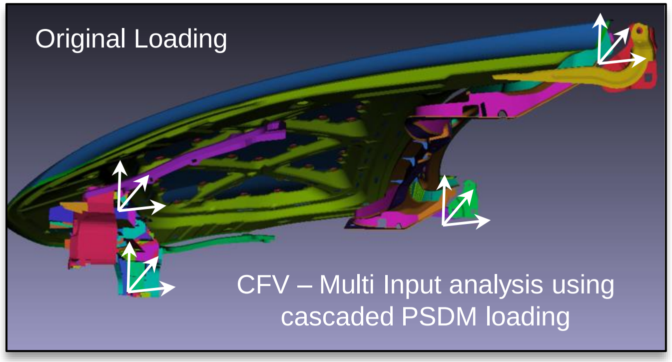
Run **CFV - MULTI** analysis on the hood with the cascaded PSDM inputs to generate a “TARGET” set of HOTSPOT elements with stress and fatigue target data.

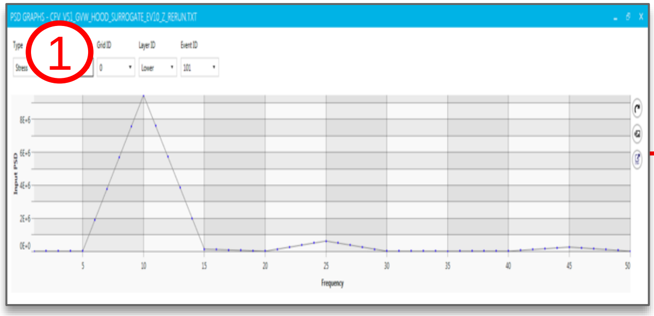
Now, reverse the process and use the target hotspot stress or damage fields to derive a NEW, simplified input loading function which creates (as near as possible) the same stress and damage identified in the Targets.

Start with step 1 and repeat steps 2 and 3 until a new simplified loading is obtained. The loading can be a single PSD, a PSD “envelope” or a single sine wave.

Targets

Element	Event	rms	rms_strain	Damage
144092	101	1.60	7.63E-06	6.29E-11
144092	102	2.08	9.93E-06	2.48E-11
144092	103	3.51	1.68E-05	3.07E-10
144092	104	[illegible]	[illegible]	1.06E-09
144092	[illegible]	[illegible]	[illegible]	1.42E-10
144092	[illegible]	[illegible]	[illegible]	3.22E-11
144092	[illegible]	[illegible]	[illegible]	1.71E-08
144092	108	[illegible]	[illegible]	1.66E-07
144092	109	10.39	4.98E-05	7.01E-07
144092	110	10.14	4.86E-05	1.01E-05
144092	111	5.21	2.49E-05	6.71E-08
144092	112	8.37	4.01E-05	8.40E-08
144092	113	0.49	2.34E-06	3.70E-14
144092	114	9.14	4.38E-05	6.74E-08
144092	115	5.01	2.39E-05	5.19E-09
144092	116	7.72	3.70E-05	1.85E-06
144092	117	9.88	4.74E-05	1.00E-18
144092	118	7.82	3.74E-05	1.00E-18
144092	119	3.27	1.56E-05	1.59E-09
144092	120	3.22	1.54E-05	5.78E-10
144092	121	5.54	2.65E-05	2.94E-08
144092	**SUM**	**10.39**	**4.98E-05**	**1.31E-05**
144624	101	5.18	2.48E-05	4.08E-08
144624	102	6.41	3.07E-05	1.04E-08
144624	103	11.15	5.35E-05	1.14E-07
144624	104	14.56	7.02E-05	4.13E-07





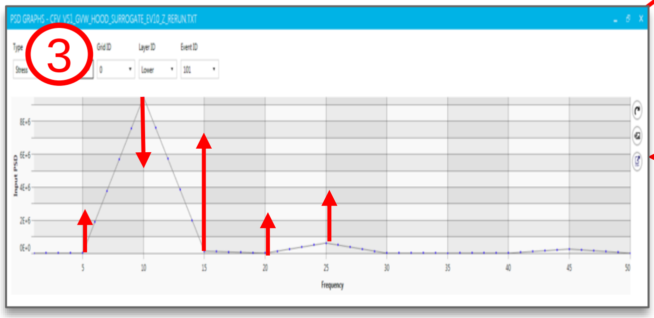
Define initial PSD input loading values per frequency for the X, Y or Z direction.

2

	TARGET		SURROGATE			
El[illegible]	rms	Damage	rms	Damage	Stress Error	Damage Error
74[illegible]	21.0	3.82	20.7	2.75	-1%	-28%
747606	20.0	1.46	19.6	1.03	-2%	-29%
747801	20.0	1.53	19.7	1.20	-2%	-22%
747499	18.0	0.21	17.7	0.15	-2%	-29%
747500	17.5	0.12	17.2	0.09	-2%	-25%
749940	17.2	0.08	16.9	0.06	-1%	-25%
748020	17.1	0.08	16.8	0.06	-2%	-25%
746782	17.3	0.09	17.0	0.07	-1%	-22%
746777	17.0	0.07	16.8	0.05	-1%	-29%
735637	16.6	0.04	16.3	0.03	-2%	-25%

Run **CFV – Base Shake** analysis for HOTSPOT elements only and compare results to TARGET

If results are not acceptable …



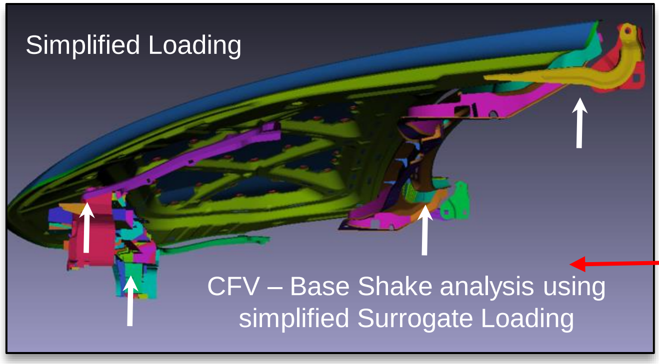
CFV will adjust PSD input loading values per frequency and repeat the analysis loop.

If results are acceptable …

Note: If we want to calculate a simple sine wave of loading, we simply apply the PSD input at <u>one</u> frequency for step (1) and continuously adjust that amplitude until the best match is achieved.

4 **Output PSD loading for each event and the envelope values.**

5 **Use PSD Surrogate Loads as input to a simplified component test.**



Ford

Automotive OEM

Advanced Surrogate Loading to produce a simplified envelope of loading

NAFEMS World Congress 2017, Stockholm June 11-14th 2017

Creating Accurate Surrogate and/or Accelerated Loads for Known Models or Structural Systems Types

Abstract

The specification of loads for durability is a very imp
used in an analysis, or in a laboratory based test proce
possible to the customers use of the product as possib
to this that can be achieved is to measure multi event,
correlation) from prototype vehicles (test tracks) and
or analysis environment. For analysis, this is feasible
approach although it doesn't easily deal with the topic
laboratory based simulation, especially of parts or sub
because of available test equipment) to simplify the lo
one at a time (eg, X, then Y, then Z) and this poses si
approaches are currently used.

First, an enveloping procedure can be applied to the l
combined into a single smoothed profile. A classical
involves no knowledge of the structural system so the
enveloped loads will cause the same damage values o

The second approach is to use the concept of a Fatigu
a simplified loading is created which does the same d
system (where the resonance is moved through the lo
account of any knowledge of the actual structural sys

This paper presents the details of a new approach whi
approach but using the actual system properties.

1. Introduction

Recently a new technological approach has been deve
VIBRATION (CFV) analysis package that allows the
creation of surrogate loads. Working entirely in the fr
approach allows the structural system (characterized
Abaqus as a system transfer function) to be further re

NAFEMS World Congress 2019 – NWC19-382

Paper Title: Loads Enveloping

Neil Bishop and Stuart Kerr
(CAEfatigue Limited, UK);
Edinilson Costa
(Ford Motor Company, Brasil);
Trenton Meehan
(Ford Motor Company, USA).

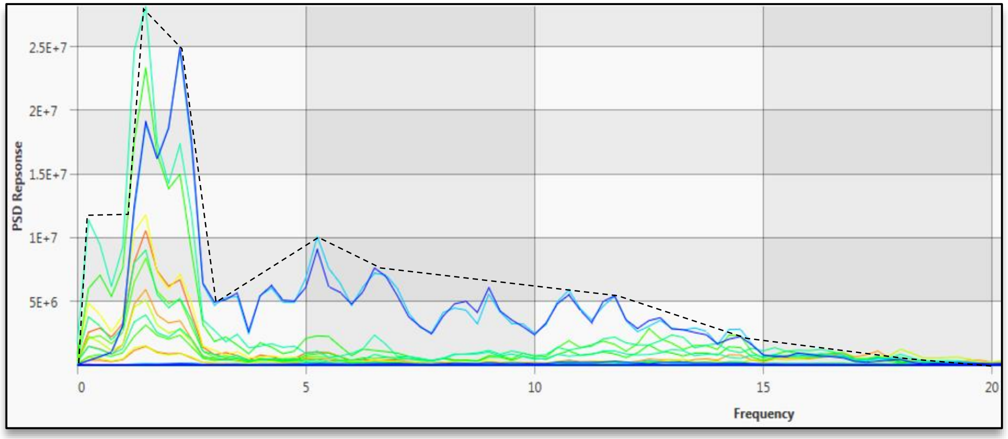
Abstract

Frequency domain methods for analysis and testing are experiencing a resurgence due to improved computer processing methods and newly developed algorithms. Recent SAE and NWC17 papers have demonstrated both the accuracy and benefits of these frequency domain methods. Time based methods will continue to be used for non-dynamic applications and for other specific types of dynamic calculations involving transients and non-linear systems. But, between the frequency domain and the time domain there is another kind of analysis, which uses sine sweeps.

Sometimes these sine sweeps are mixed with random loads (sine on random) and other times they are applied individually. The sweep is basically a sine input of specified magnitude and frequency where the frequency is varied in a prescribed manner. The sweep can typically be linear, logarithmic, or some kind of dwell around resonances. The dwell portions can also sometimes be perturbed in order to ensure that changing system properties (changing dynamic resonance frequencies) do not cause a "miss" in the maximum resonance. Typically, these test specifications will involve multiple "events" where each event represents a specific type of loading and a specific loading direction. Load specifications can be single channel (common) or multi-channel (less common in the frequency domain). All of the above procedures can be applied in a test laboratory, and there is now considerable interest in applying exactly the same procedures in a CAE analysis environment.

This paper will explore the relationship between the sine sweep test and an equivalent PSD loading and show that under certain conditions the two load types can be made equivalent in terms of fatigue damage potential. The details of this "equivalence" will be investigated in terms of not just fatigue damage but also the level of localized plasticity at hot spot locations.

Secondly, the consolidation of multiple events (and multiple channel data) into simplified (enveloped) test specifications will be explored. Specifically, the concept of reducing a multiple input loading scenario (with channel correlation)



	Original			Surrogate					
Element	rms	Damage	Plasticity Index	rms	Damage	Plasticity Index	Stress Error	Damage Error	PI Error
42949	185.38	8.81	3.35	187.54	9.47	3.38	1.01	1.1	1.01
25324	168.16	5.24	3.11	168.78	5.38	3.11	1.00	1.0	1.00
25714	163.92	4.54	3.05	164.53	4.69	3.05	1.00	1.0	1.00
40667	147.33	2.58	2.81	147.21	2.53	2.81	1.00	1.0	1.00
40648	144.08	2.28	2.76	143.92	2.24	2.76	1.00	1.0	1.00
24892	136.99	1.61	2.66	137.48	1.64	2.66	1.00	1.0	1.00
49124	130.90	1.31	2.57	131.08	1.30	2.57	1.00	1.0	1.00
25001	124.40	0.91	2.47	124.66	0.92	2.47	1.00	1.0	1.00
49571	123.86	0.93	2.46	123.76	0.92	2.46	1.00	1.0	1.00
49149	118.78	0.71	2.38	119.12	0.72	2.39	1.00	1.0	1.00

HEXAGON

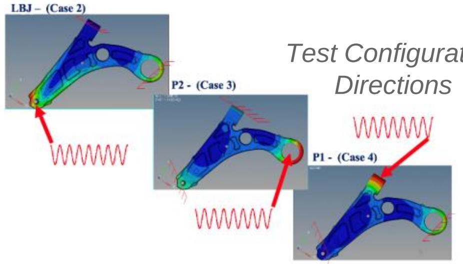
*Key technologies covered: **Base Shake** and **Multi-input** analysis using **Loads Conditioning** and advanced **Random** results. **Loads Cascading** is used to generate additional input loads and **Surrogate Loads** is used to create an envelope around those PSD loads.*

Customer Success Stories

Application of Surrogate Loading with Vector analysis to determine the best simplified sine wave loading and direction

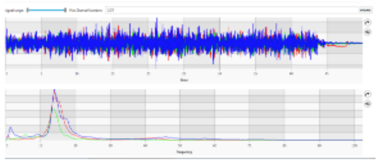
FCA

Automotive OEM



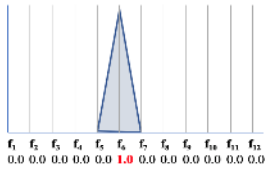
Test Configuration Directions

Complicated Input



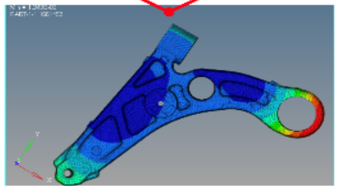
x, y and z (Test Track) multi input loading at the same time (also for multiple Events if needed)

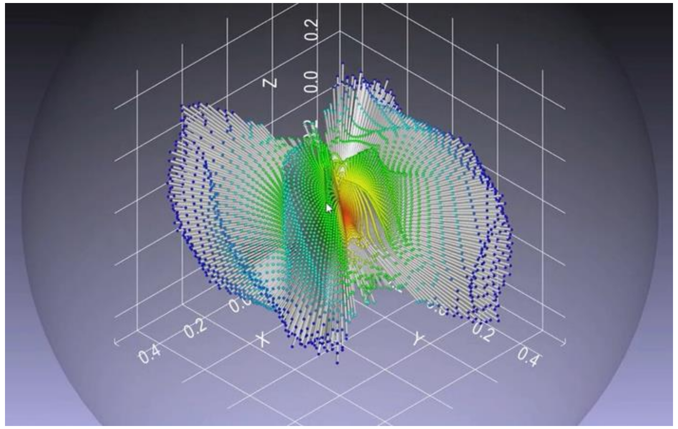
Simplified Input



Simple (Laboratory) PSD for one Event (plus other single input Events if needed)

Create a Surrogate Load that causes the same damage for the actual system

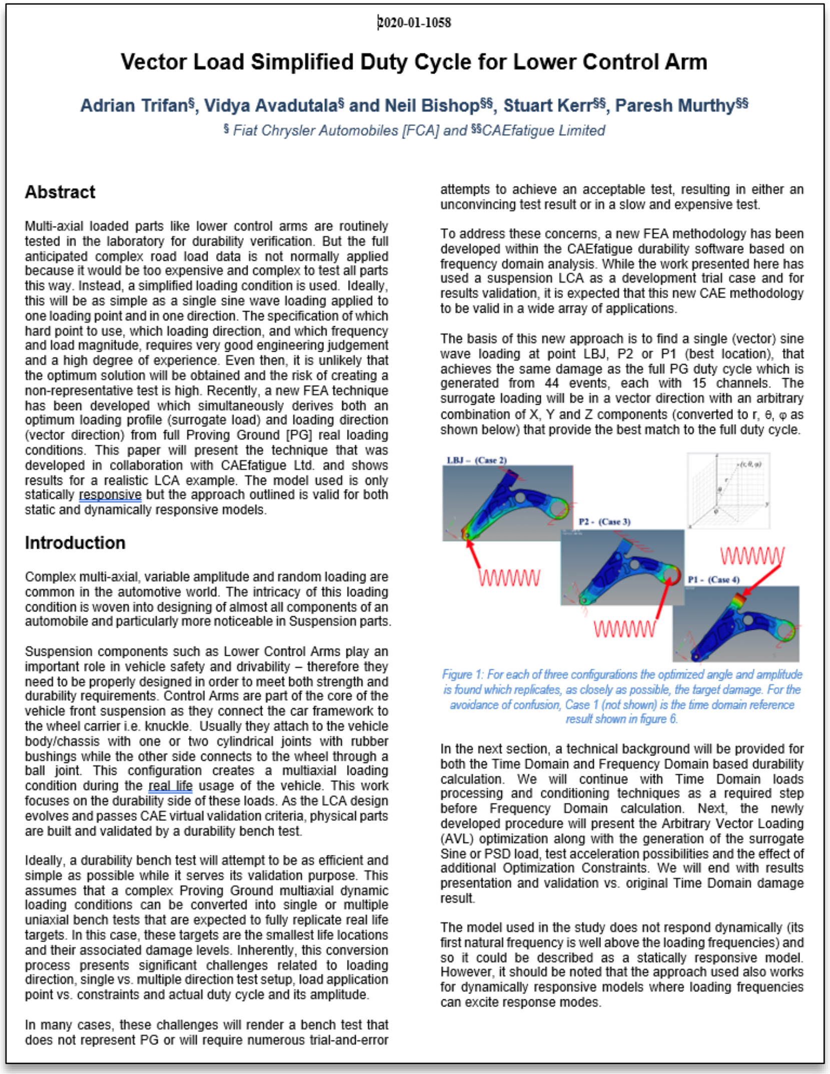




Determine the best direction to apply the simplified sine waves using VECTOR

Comparison of Damage Results from Simplified Sines vs. Original Time Domain

ElementID	Time Domain Damage	Damage Case 3 8Hz	FD/TD	Plasticity Index	ABS(FD-TD)/ MAX(FD, TD)
28691	1.000000	1.409900	1.41	1.000263	0.291
28095	0.646682	0.538407	0.83	1.000070	0.167
4308	0.162255	0.136187	0.84	1.000014	0.161
21913	0.165724	0.086438	0.52	1.000010	0.478
8675	0.049887	0.024850	0.50	1.000004	0.502
			T=30 Hours	average error	0.320
				QoF	0.680



2020-01-1058

Vector Load Simplified Duty Cycle for Lower Control Arm

Adrian Trifan[§], Vidya Avadutala[§] and Neil Bishop[§§], Stuart Kerr[§§], Paresh Murthy[§§]
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Abstract

Multi-axial loaded parts like lower control arms are routinely tested in the laboratory for durability verification. But the full anticipated complex road load data is not normally applied because it would be too expensive and complex to test all parts this way. Instead, a simplified loading condition is used. Ideally, this will be as simple as a single sine wave loading applied to one loading point and in one direction. The specification of which hard point to use, which loading direction, and which frequency and load magnitude, requires very good engineering judgement and a high degree of experience. Even then, it is unlikely that the optimum solution will be obtained and the risk of creating a non-representative test is high. Recently, a new FEA technique has been developed which simultaneously derives both an optimum loading profile (surrogate load) and loading direction (vector direction) from full Proving Ground [PG] real loading conditions. This paper will present the technique that was developed in collaboration with CAEfatigue Ltd. and shows results for a realistic LCA example. The model used is only statically responsive but the approach outlined is valid for both static and dynamically responsive models.

Introduction

Complex multi-axial, variable amplitude and random loading are common in the automotive world. The intricacy of this loading condition is woven into designing of almost all components of an automobile and particularly more noticeable in Suspension parts.

Suspension components such as Lower Control Arms play an important role in vehicle safety and drivability – therefore they need to be properly designed in order to meet both strength and durability requirements. Control Arms are part of the core of the vehicle front suspension as they connect the car framework to the wheel carrier i.e. knuckle. Usually they attach to the vehicle body/chassis with one or two cylindrical joints with rubber bushings while the other side connects to the wheel through a ball joint. This configuration creates a multiaxial loading condition during the real life usage of the vehicle. This work focuses on the durability side of these loads. As the LCA design evolves and passes CAE virtual validation criteria, physical parts are built and validated by a durability bench test.

Ideally, a durability bench test will attempt to be as efficient and simple as possible while it serves its validation purpose. This assumes that a complex Proving Ground multiaxial dynamic loading conditions can be converted into single or multiple uniaxial bench tests that are expected to fully replicate real life targets. In this case, these targets are the smallest life locations and their associated damage levels. Inherently, this conversion process presents significant challenges related to loading direction, single vs. multiple direction test setup, load application point vs. constraints and actual duty cycle and its amplitude.

In many cases, these challenges will render a bench test that does not represent PG or will require numerous trial-and-error attempts to achieve an acceptable test, resulting in either an unconvincing test result or in a slow and expensive test.

To address these concerns, a new FEA methodology has been developed within the CAEfatigue durability software based on frequency domain analysis. While the work presented here has used a suspension LCA as a development trial case and for results validation, it is expected that this new CAE methodology to be valid in a wide array of applications.

The basis of this new approach is to find a single (vector) sine wave loading at point LBJ, P2 or P1 (best location), that achieves the same damage as the full PG duty cycle which is generated from 44 events, each with 15 channels. The surrogate loading will be in a vector direction with an arbitrary combination of X, Y and Z components (converted to r, θ, φ as shown below) that provide the best match to the full duty cycle.

Figure 1: For each of three configurations the optimized angle and amplitude is found which replicates, as closely as possible, the target damage. For the avoidance of confusion, Case 1 (not shown) is the time domain reference result shown in figure 6.

In the next section, a technical background will be provided for both the Time Domain and Frequency Domain based durability calculation. We will continue with Time Domain loads processing and conditioning techniques as a required step before Frequency Domain calculation. Next, the newly developed procedure will present the Arbitrary Vector Loading (AVL) optimization along with the generation of the surrogate Sine or PSD load, test acceleration possibilities and the effect of additional Optimization Constraints. We will end with results presentation and validation vs. original Time Domain damage result.

The model used in the study does not respond dynamically (its first natural frequency is well above the loading frequencies) and so it could be described as a statically responsive model. However, it should be noted that the approach used also works for dynamically responsive models where loading frequencies can excite response modes.

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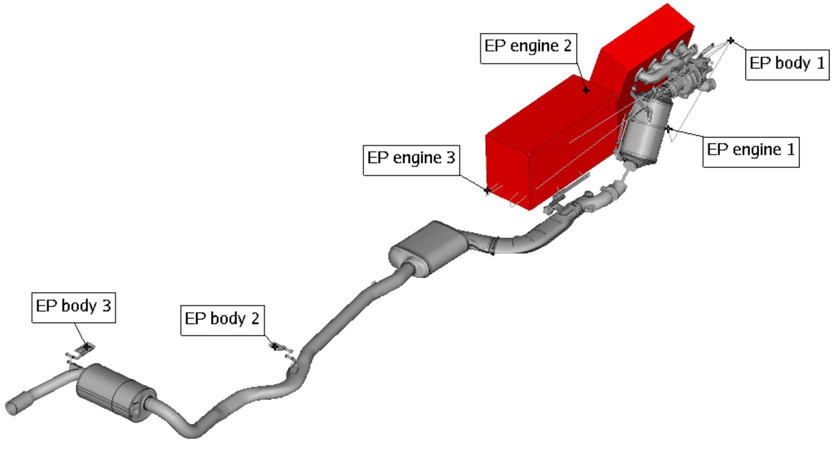
*Key technologies covered: **Base Shake** and **Multi-input** analysis using **Loads Conditioning** and advanced **Random** results. **Loads Cascading** is used to generate additional input loads and **Surrogate Loads** is used to simplify those input loads to sine waves. **Time Domain** analysis is used to compare results.*

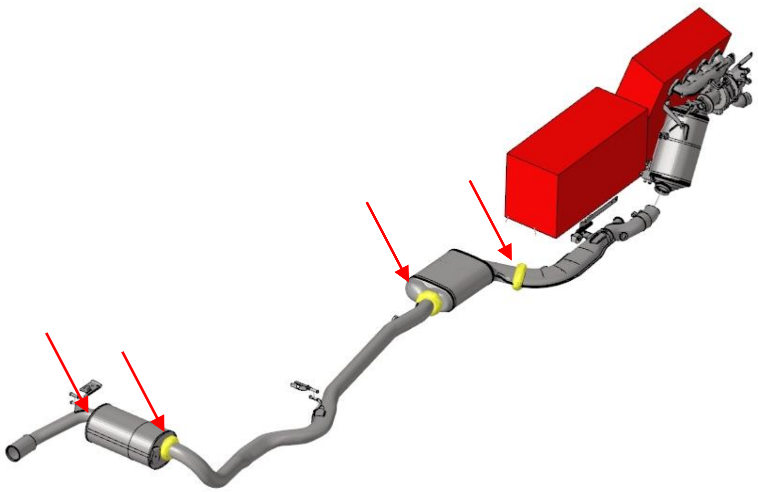
Eberspaecher

Exhausts

Advanced Surrogate Load analysis used to generate cutting plane loads for exhausts



Measured Responses



Derived Cutting Plane Loads

2020-01-0180

Frequency Domain Loads Processing for Exhaust Systems

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Abstract

A previous SAE paper (2018-01-1396), (see ref 5) introduced the concept of using the frequency domain for exhaust system analysis under road excitation as a fast and efficient approach. During the intervening period further benchmarks have confirmed the validity of the approach for several exhaust systems comparing the results of simulations performed in the time domain and frequency domain for different applications.

This paper will present that data and also introduce a new technique which has been developed to create, from the originating road load data (RLD), the cutting plane loads at any desired section of the exhaust system. Those loads can be used directly to determine safety factors against load capacities based, for example, on part SN curves from current or former hardware testing or even derived from statistics.

The advantage lies in the early availability of virtual models in the development process, providing information about the loads acting on the system before any hardware can be built or measured.

Another big benefit relates to the high variability and flexibility of post-processing using CAE tools which stands in opposition to the fixed strain gage positions in measurement which makes sensitivity studies or additional evaluations impossible.

Last but not least, this method can also be applied to provide better information about load targets to be achieved by, for example, supplier parts, like flex pipes or flaps. This can therefore support early application-tailored designs. Or it can be used for the extraction of loads for shaker testing and replace generic loads that may lead to over engineering.

Introduction

The development of exhaust systems follows, as for many industrial goods, a well-known and standardized procedure, defined in Advanced Product Quality Planning (APQP). Any new developed product will have to go through this procedure, which is separated in different maturity phases where various functions of the systems to be developed have to be analyzed and compared to targets usually defined in a specification book at project start.

In the development of automotive parts, we usually speak of planning phase, concept phase, design verification phase and a product validation phase. Design & Development (D&D) teams are typically active in the concept, design verification and product validation phases. Figure 1 illustrates the phase model also used in development of exhaust systems.

Figure 1: phase model in development following APQP

In the past decade, we have observed a gradual increase in the usage of CAE methods, mostly supporting the development process in those phases. Their main focus was to confirm designs in concept phase and avoid unnecessary loops in physical testing during design verification and product validation phases.

Today, we see an increased need for lower cost and shorter time to market coming directly from the Customer. Car manufactures are, today even more than in the past, in the dilemma of on the one hand meeting the legal and economic regulations and on the other hand to be competitive in the market with a wide range of products.

The consequence is that a large variety of products has to be developed in a shorter time and with lower D&D costs.

In the described situation, it is crucial to have a maximum of information about the system available in earlier phases in the project to enable effective risk assessment right in the concept phase. That is where simulation methods play an ever increasing role in solving the challenges ahead.

Physical tests use prototypes of a certain maturity to validate certain functions in an experimental way. Very often, those tests are one of the drivers of D&D cost, due to the high cost of prototype production. Additionally, those tests and prototype usually represent the critical path during the development phase, thus defining the timeline.

CAE methods will, in future, play an even more central role in the development, because of the possibility to solve these issues. First, their application is possible at the start of the concept phase without the presence of physical prototypes and secondly, the cost is relatively low in comparison to experimental methods.

One of the functions to be analyzed during the development phase is mechanical resistance of the parts, which is a discipline usually described as durability. In this paper we concentrate on the durability aspect, although similar observations are true for other disciplines (e.g. acoustics and thermodynamics).

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01/28/2020

*Key technologies covered: **Base Shake** (single input) analysis and **Surrogate Loads** to simply the input loads at cutting planes.*

Advanced Surrogate Loading to produce simplified sine waves of loading

GM

Automotive OEM

2020-01-1056

Loads simplification on multi input axle systems

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Abstract

The time domain is currently the most widely chosen option in fatigue testing to fully represent random events occurring in multiple simultaneous input channels. In vehicles for example, time domain tests can represent the same conditions of the road, by applying the same loads at the hard points of the vehicle along a time history. The main drawback of this methodology is the extensive testing duration and hardware cost. Time domain based fatigue tests are composed of a complex hardware, which requires servo motors to work, in order to induce the specific amount of load at a specific time window. These tests are time consuming, since they require the same length duration of the event they are reproducing, times the required repetitions. The frequency domain method for fatigue testing, on the other hand, requires simpler hardware, since there are no need for servomotors and the test length is reduced, since there is no need to run the full event times the required repetitions. The drawback in this case is the ability to represent random events with multiple simultaneous input channels. For this reason, frequency domain tests are mainly applied for simple round-robin tests, unable to represent random events like a road condition. In this work, we will present a new methodology that uses the fatigue result as input, reversing it in order to find the loads that would be required to originate it in frequency domain. This new methodology can help future developments in fatigue test, replacing time domain tests without losing accuracy and the ability to represent random events occurring in multiple simultaneous input channels.

Introduction

Durability is an extremely important aspect to be considered during the development of a vehicle. It guarantees the vehicle will keep its structural integrity during use, not only preserving performance aspects of the vehicle itself, but also the safety of its occupants and its environment.

A durability test can be executed in two different ways. The first one is using the vehicle itself on a road and the second is to bench test components or mechanisms in a laboratory. Bench tests are usually more preferable, since they do not require an entire drivable vehicle and a test road. However, a bench test that fully represents the same fatigue performance of a driving vehicle on a road requires extremely complex and expensive hardware that is able to reproduce the same loads as the road using servo actuators in time domain.

Although the complexity, and cost, of these "full complex loading" bench tests might not be an issue for the largest automakers, who see this approach as a huge evolution over traditional durability methods using full vehicles on the road, the approach can be prohibitive for smaller companies, such as suppliers. These smaller companies cannot afford the financial investment required. These suppliers are also unable to do full vehicle testing on a test track because the vehicle is usually not accessible to them. Therefore, the typical approach available to suppliers is only the simple round-robin testing approach.

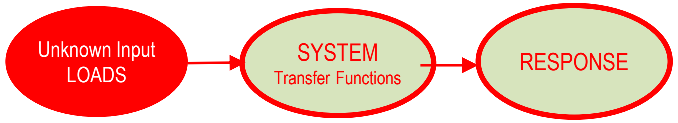
Round-robin tests are conducted using simple actuators (instead of servo actuators) and are used in a process without feedback control. Due to this, they are inexpensive to purchase, extremely easy to set-up and do not require the full vehicle. The main drawback is that the approach does not represent the road loads in a complete way. A common workaround, used in the industry, is to combine several different round-robin tests, with different sets of actuators, applying forces in different vectors in attempt to generate an approximation of the road loads as close as possible. However, significant differences will often still exist.

These differences are such that, after suppliers finish their simplified in-house durability tests, the automaker also performs in-house durability tests, but using the servo based time-domain machine to make sure the component is able to stand vehicle condition. If failures occur, the component returns to the supplier to start the process again and this development-test loop between the automaker and supplier continues until the component is able to withstand the automakers final testing.

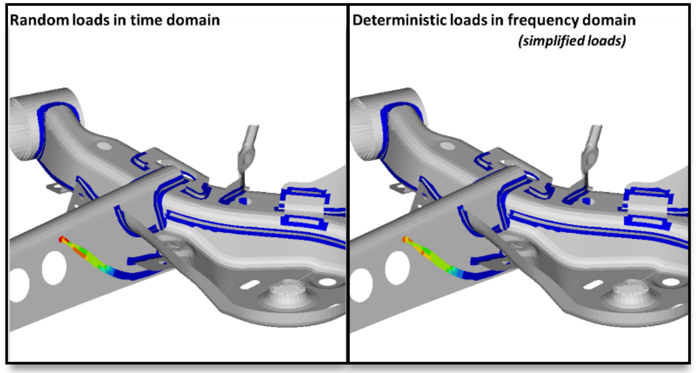
The work presented in this paper proposes a new method to evaluate and test for fatigue damage using a more robust frequency domain approach. The core idea is to determine a vector of forces, which will not necessarily exist inside all the road loading history but, when acting at given frequencies, will reproduce the same fatigue behavior in the full structure, as would be generated by the full road loading in the time domain. For this new method, simple actuators can be used allowing even smaller companies to substitute their inaccurate round-robin testing for more robust testing that may reduce or even remove development-test loops with the automakers.

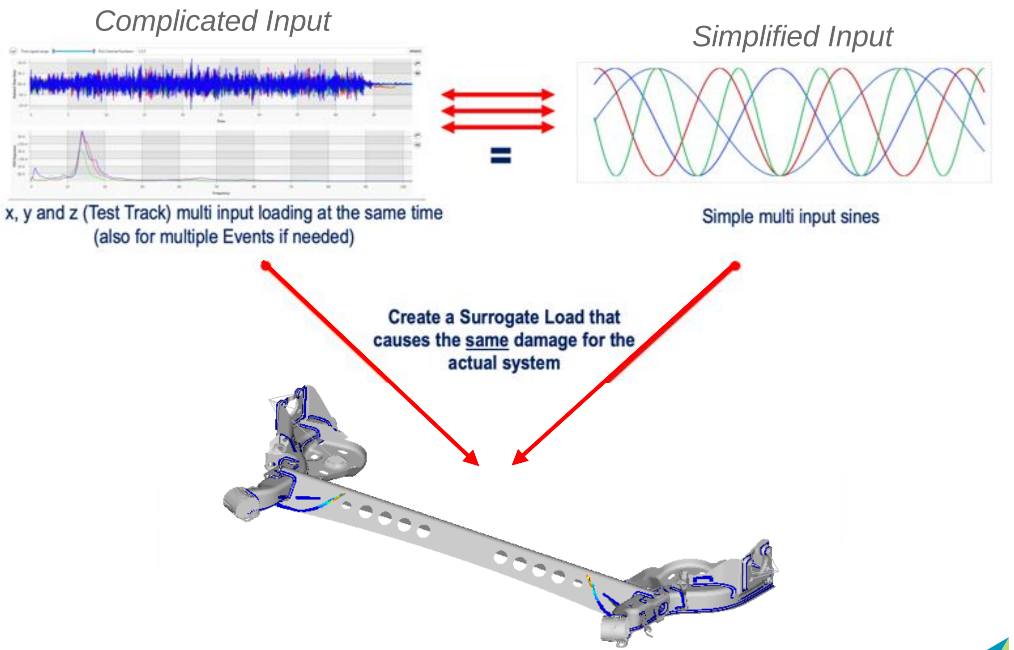
Methods and Materials

For this work, two methods will be evaluated using CAE tools [8]. This new CAE technology works predominantly in the frequency domain and has been radically extended over the past 5 years [10-22]. The use of simulation for this evaluation is desirable due to its simplicity and



Simplify the Input … get the same response.





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*Key technologies covered: **Base Shake** and **Multi-input** analysis using **Loads Conditioning** and advanced **Random** results. **Loads Cascading** is used to generate additional input loads and **Surrogate Loads** is used to simplify those input loads into simple sine waves.*

IDIADA

Proving Grounds

IDIADA

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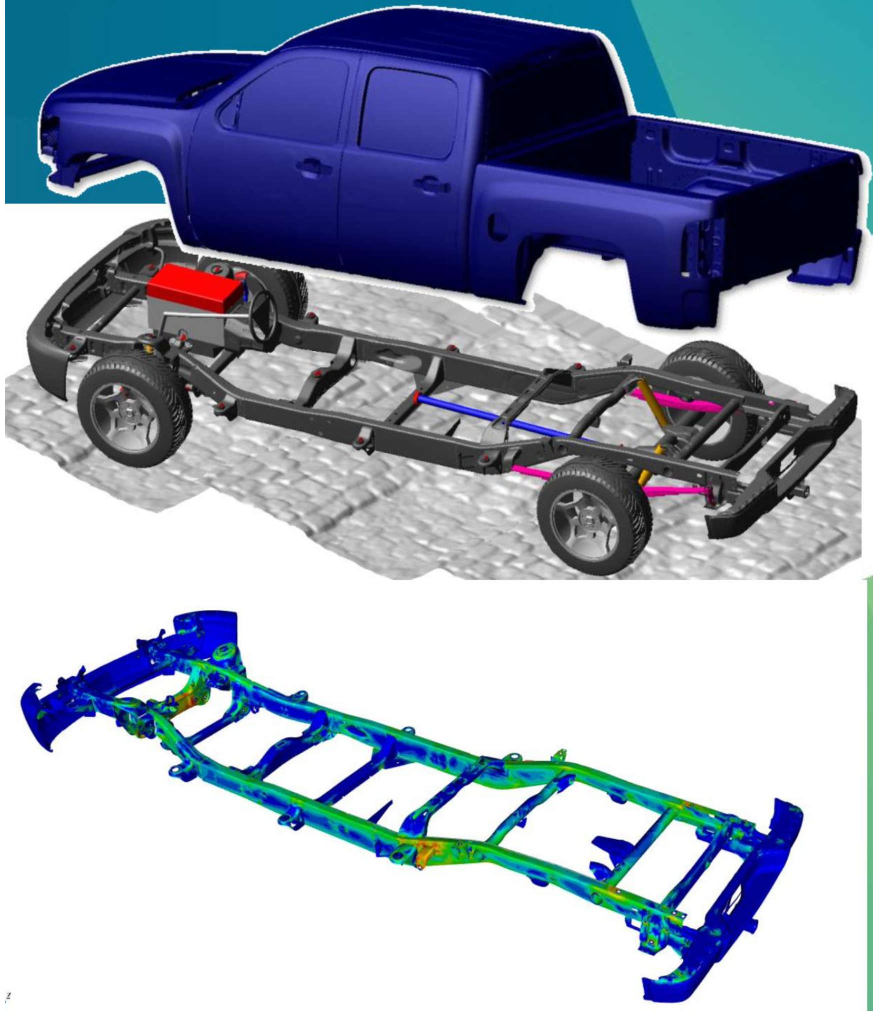


Adams-CAEfatigue based Process Flow, from Virtual Proving Ground to Full Vehicle Durability Duty Cycle Analysis

Presenting

Fredrik Sjögren and Marco Veltri

Hexagon / MSC Software

Adams Durability Symposium – February 24 - 2021

Nauti-Craft: from MBD time to frequency domain

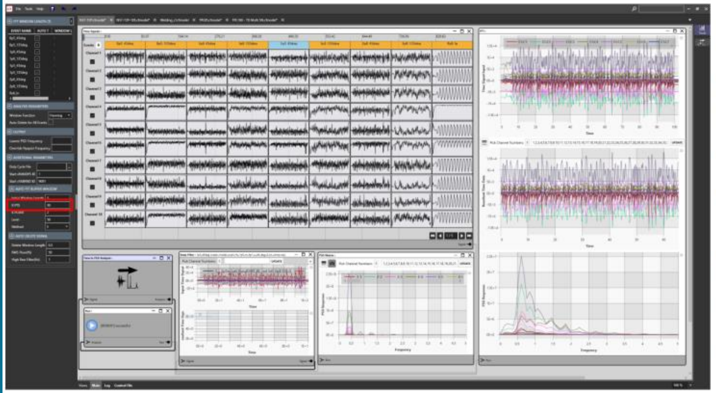
Nauti-Craft

Boats

NWC 2021 NAFEMS WORLD CONGRESS
A WORLD OF ENGINEERING SIMULATION
INCORPORATING
spdm INTERNATIONAL CONFERENCE Simulation Process & Data Management

Frequency Domain Fatigue Methodology Applied to Boat Design

Liam MacFarlane, Laurie Walker, Nauti-Craft
Stuart Kerr, Neil Bishop, MSC Software



Conditioning and Conversion done with **CAEfatigue** (an MSC product).

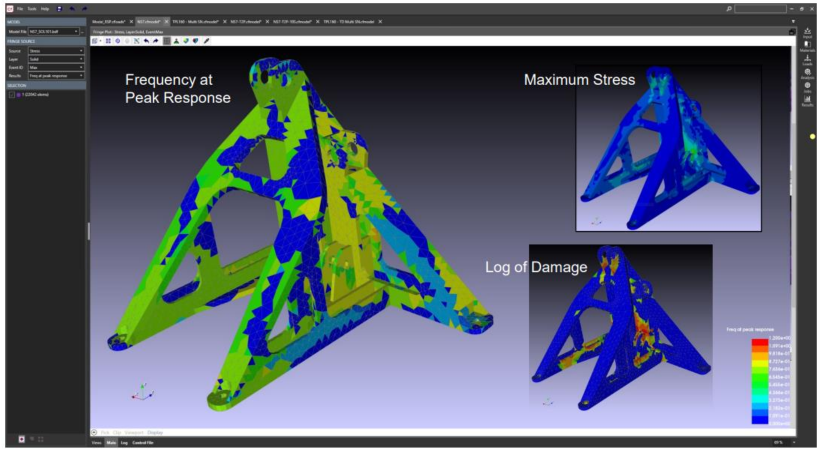
Time signals are automatically conditioned to remove non-stationary, low RMS sections (AutoD filter).

Time signals are automatically converted to PSD matrix files primarily using a KPTS setting. (AutoT filter).

PSD Matrix file contains an array of PSDs that represent the direct responses and cross PSD responses (phasing between channels).

One Event of time signals equals one PSD Matrix file.

Outputs from Frequency Domain Analysis Fringe Plots



The Frequency Domain fatigue analysis provides **fringe plots** at any location for numerous output types.

At the left is a fringe plot showing several available outputs including max stress, log of damage and frequency at peak response; a plot only available in the Frequency Domain.

Freq at Peak Response shows at what frequencies the model is responding. Although some areas are responding at different frequencies, no part of the model is responding at a higher frequency than 1.2 Hz. This can be valuable information to help resolve issues.

NAFEMS World Congress 2021 – NWC21-340

FREQUENCY DOMAIN FATIGUE METHODOLOGY APPLIED TO BOAT DESIGN

Liam MacFarlane, Dr. Laurie Walker
(Nauti-Craft, Australia)

Dr. Stuart Kerr, Dr. Neil Bishop
(MSC Software, United States)

water surface is defined as twenty superimposed sinusoidal ocean waves propagating from a single source in distance. The period (T1) of the sea state is defined for a selected significant wave height (Hs) according to data obtained from the Naturaliste region in the south-west of Australia. The amplitude of each sinusoidal wave component is determined according to a JONSWAP spectrum.

Thrust model

This group defines the thrust force exerted by each outboard motor. The thrust force controlled using a proportional controller to maintain the vessel velocity at the setpoint.

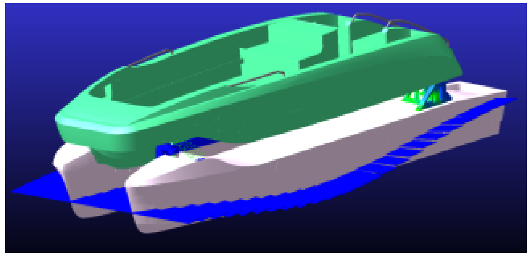


Figure 3: MBD model of the NS7 constructed in MSC Adams

7. Duty Cycle Definition

The NS7 has been designed to a fatigue life of 1500 hours of operation as a recreational vessel. Using MSC Adams, an operational profile was defined for the vessel at which a peak vertical acceleration of 1g would be experienced at the helm position on the deck with the suspension system operational. This operational profile is shown in the figure below.

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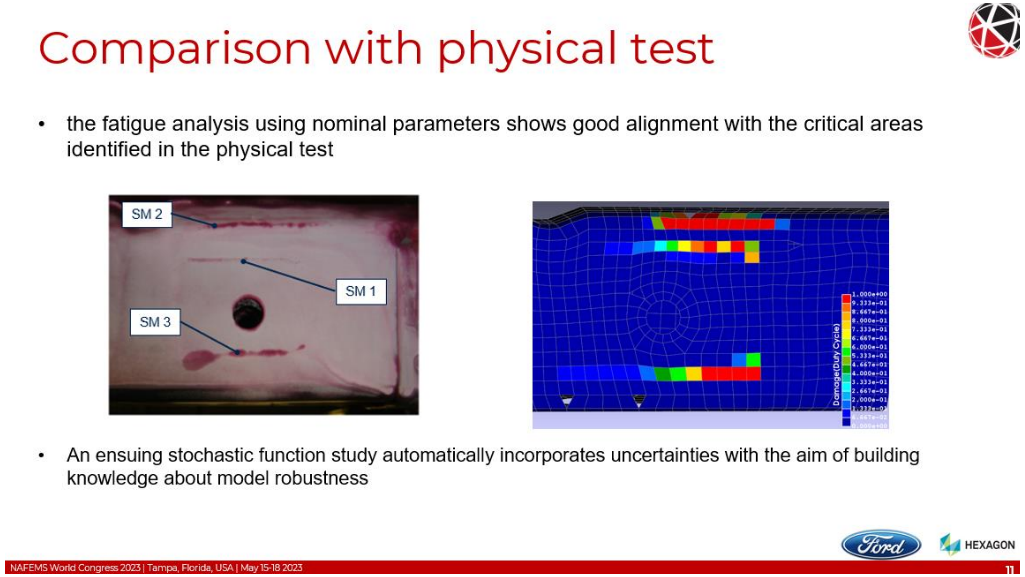
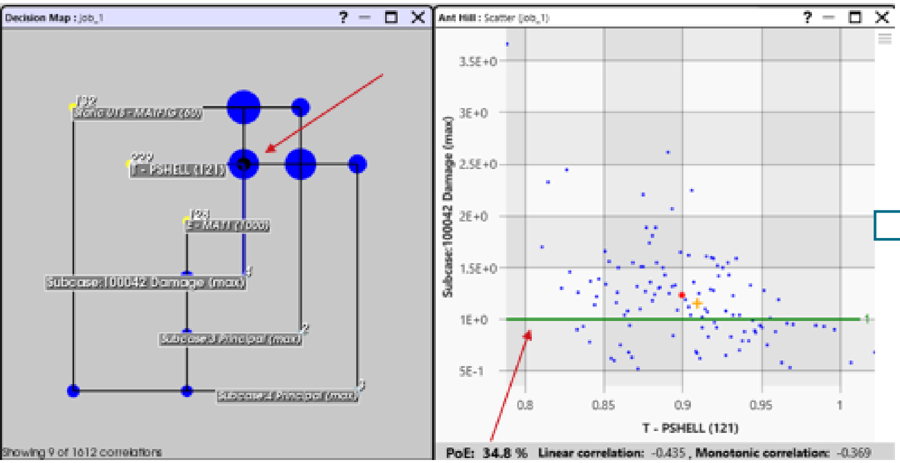
Application example Robust Design to automoti

Ford

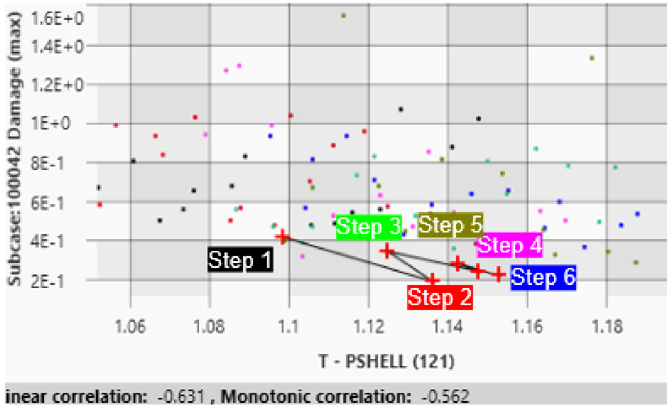
Auto

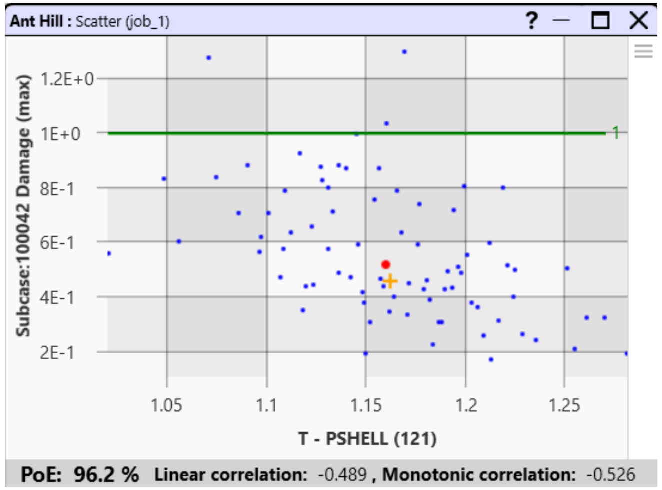


From stochastic Design function study



To stochastic Design Improvement cycles





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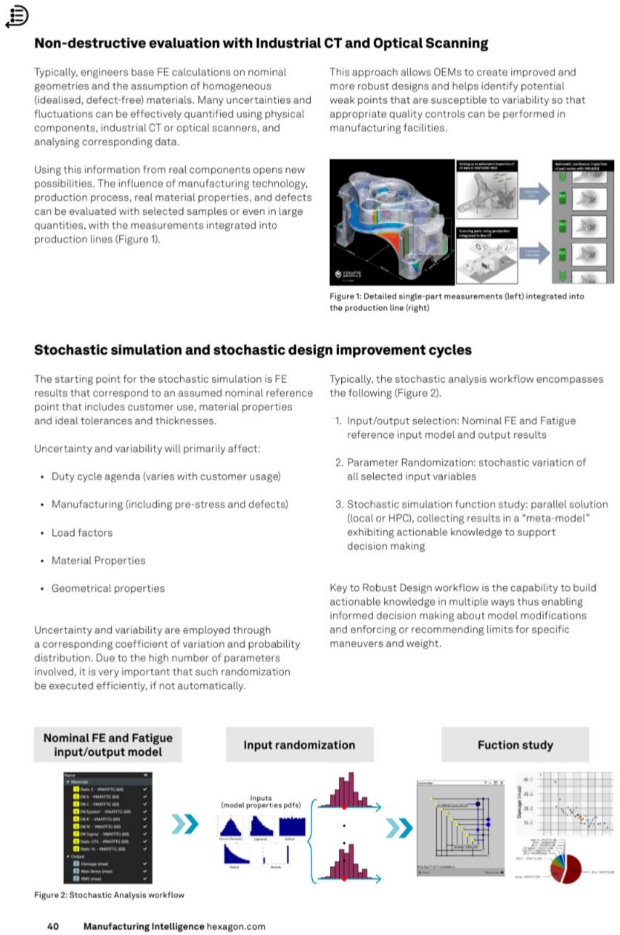
Developing robust structures in the auto industry

Breaking Barriers, e-book by Engineering Reality



Non-destructive evaluation with Industrial CT and Optical Scanning

Typically, engineers base FE calculations on nominal geometries and the assumption of homogeneous (idealised, defect-free) materials. Many uncertainties and fluctuations can be effectively quantified using physical components, industrial CT or optical scanners, and analysing corresponding data.

Using this information from real components opens new possibilities. The influence of manufacturing technology, production process, real material properties, and defects can be evaluated with selected samples or even in large quantities, with the measurements integrated into production lines (Figure 1).

This approach allows OEMs to create improved and more robust designs and helps identify potential weak points that are susceptible to variability so that appropriate quality controls can be performed in manufacturing facilities.

Figure 1: Detailed single-part measurements (left) integrated into the production line (right)

Stochastic simulation and stochastic design improvement cycles

The starting point for the stochastic simulation is FE results that correspond to an assumed nominal reference point that includes customer use, material properties and ideal tolerances and thicknesses.

Uncertainty and variability will primarily affect:

- Duty cycle agenda (varies with customer usage)
- Manufacturing (including pre-stress and defects)
- Load factors
- Material Properties
- Geometrical properties

Uncertainty and variability are employed through a corresponding coefficient of variation and probability distribution. Due to the high number of parameters involved, it is very important that such randomization be executed efficiently, if not automatically.

Typically, the stochastic analysis workflow encompasses the following (Figure 2).

1. Input/output selection: Nominal FE and Fatigue reference input model and output results
2. Parameter Randomization: stochastic variation of all selected input variables
3. Stochastic simulation function study: parallel solution (local or HPC), collecting results in a "meta-model" exhibiting actionable knowledge to support decision making

Key to Robust Design workflow is the capability to build actionable knowledge in multiple ways thus enabling informed decision making about model modifications and enforcing or recommending limits for specific maneuvers and weight.

Nominal FE and Fatigue input/output model | Input randomization | Fuction study

Inputs (model properties pdfs)

Figure 2: Stochastic Analysis workflow

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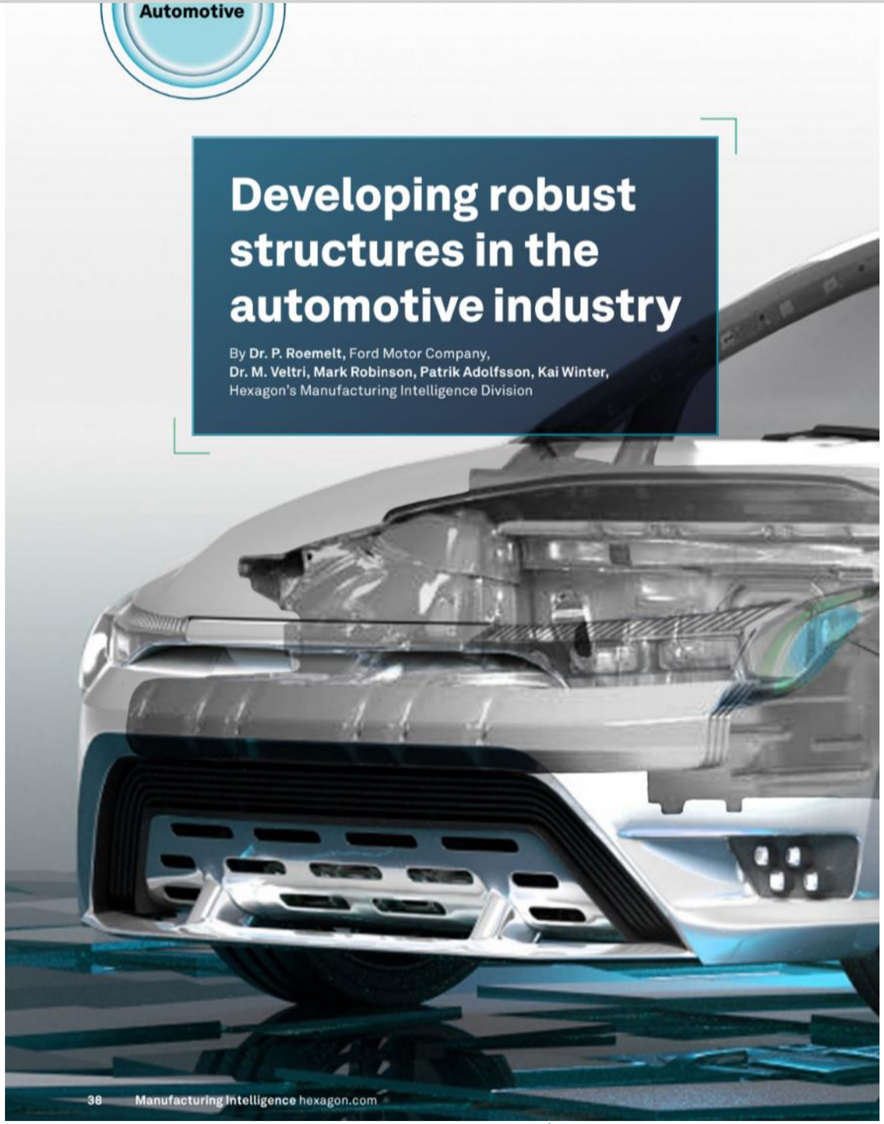

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[Access full e-book here]

Dana Incorporated

Ground Vehicles

Incorporating uncertainty in durability assessments

Challenge

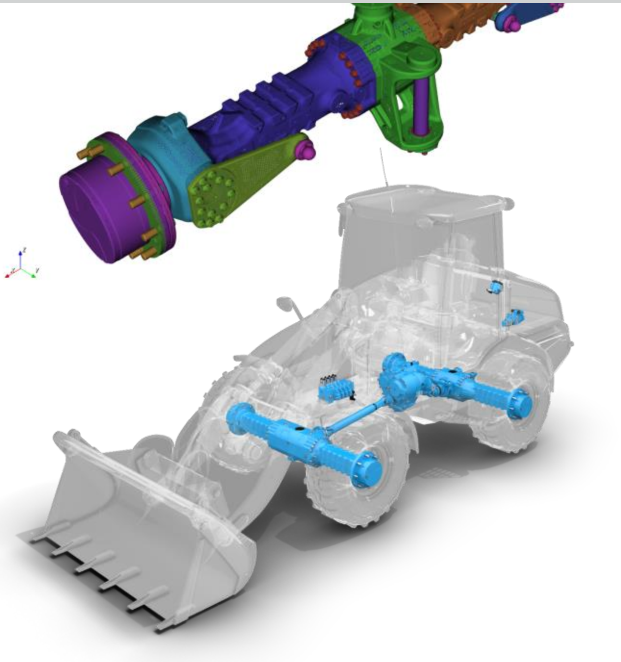
- Incorporate parameters and loads variability in durability design
- Identify loads and locations contributing the most to the damage of the axle structure.
- Determine the best combination of loads and event representative of customer operations

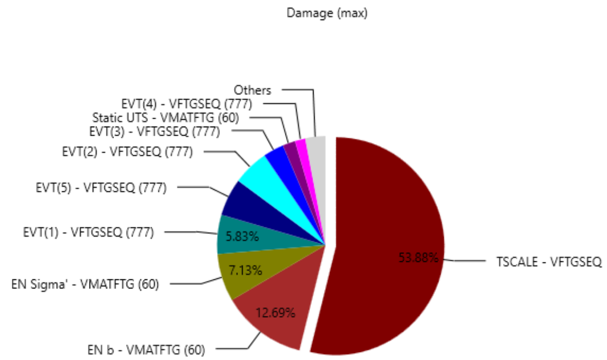
Solution

- Robust Design tool provides actionable outputs that quantify the effect of each variable and loading (e.g., y-cycle loading is the most destructive) while ALL input variables vary.
- Surrogate loads through SDI for a most realistic and robust component certification physical test

Benefits

- Gain insight into the underlying interrelations of variables to understand which are the most influential.
- Reduce and improve physical testing.
- Reduce liability costs by exploring the whole design space of loading conditions upfront.





Access full article here

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Dana Improves off-road durability with Robust Design

Article in Engineering REALITY by Hexagon

Figure 1: Comparison between Max Neuber Stress results related to every maneuver.

Figure 2. Identification of critical fatigue areas.

Figure 3. Correlation between parameter variations.

Figure 4. Clouds of input-output pairs obtained with the stochastic study.

Phase 2. Stochastic analysis after randomisation of loads and fatigue parameters to determine the effects of variation

After performing the fatigue simulation in phase 1, the teams deployed CAEfatigue Robust Design to automatically enforce a stochastic variation on all the input variables and perform Monte Carlo simulation. All parameters follow a Gauss Normal distribution based on three pieces of information: the mean value, the coefficient of variation, and the cutoff (the number of standard deviations that determines the accuracy percentage). The only exception is customer usage or the number of repetitions for every different maneuver.

Embracing such probabilistic approach, the teams could evaluate the effects of the uncertainties on the structure's robustness, identifying the input variations that led to the highest potential for damage and other undesirable conditions, thus empowering them to create more robust designs capable of withstanding the uncertainties with in-field conditions.

Hexagon's CAEfatigue Robust Design provides two graphic tools (Figure 3) — the decision map and pie chart — that help engineers understand how different variables influence the axle component performance. The decision map shows the relationship between the input parameters (yellow squares) and requested outputs (light blue squares), with the size of the blue circles showing the relative input-output influence. The pie chart displays all relative parameter influence on a given output— for example, their impact on max damage value.

CAEfatigue Robust Design can also generate a point-cloud visualization of the input-output pairs distribution. This visual representation makes it easy for engineers to identify the combinations of inputs most likely to lead to an unacceptable output (e.g., damage exceedance probability) or undesirable behaviors (e.g., outliers or bifurcations).

Phase 3. Loads synthesis through stochastic design improvement, aimed at designing a representative physical test

Once the engineers reached their robust design objectives, they began the loads synthesis phase, searching for combinations of test loads conditions causing results closest to the target in-field damage. For this endeavor, the teams deployed the stochastic design improvement workflow inside CAEfatigue Robust Design. Through this probabilistic approach, a set number of stochastic simulation steps are undertaken in sequence, each including small variations of key design variable parameters affecting the test bench set-up.

"The third phase is where we identify the procedure that determined the best set of loads in the physical tests that mimicked the real damage accumulated during the simulation," said Bavaresco. "We use a simplification of the maneuvers, which is represented in different kinds of loads. The whole purpose is to correlate the damage predicted with the damage we actually get in the test setup."

"We'll take the initial results and refine them, taking variability into account to perform the stochastic design improvement. To accomplish that, we start from an initial assumption, then change that initial condition —for example, we may tweak the number of repetitions in the scaling factor, modulating them until we get a damage that is as close as possible to what we'd calculated with the actual maneuvers. It isn't a typical simulation validation exercise; it's that we're trying to get our results to emulate as best as possible what we've calculated in simulation," Bavaresco said.

Dedicated CAEfatigue Robust Design graphic tools allow engineers to quickly interpret a vast number of results, letting them follow design variables trends step-by-step.

All design parameter combinations have the same probability. For every simulation step, CAEfatigue Robust Design identifies the best solution (red cross in the point cloud) as the starting point for the next step (Figure 6). At the steps sequence conclusion, CAEfatigue displays the test parameters combination causing damage closest to the damage calculated with the real in-field load events. Assuming that the conclusion reached meets the target requirements, engineers can perform a final robustness check to validate and confirm their findings, using the improved test parameter combination as input for a new stochastic study.

Figure 5. Design parameters variation and damage objectives trend according to the step number.

Figure 6. Improved configurations for every step.

Results: Randomness and variability

Dana has been pleased with the results that Hexagon and the CAEfatigue and Robust Design solutions have been able to provide. "Knowing the variability of the damage allows us to reduce the margin because we've already taken the variability of the inputs into account. That means we can optimise the weight and close the variability gap," said Bavaresco.

"With Hexagon allowing us to introduce randomness and variability into the equation, we can right-size our components — which helps us ensure reliability in the field and remain competitive. In the end, working with Hexagon creates not just a massive value for us, but also our customers," he shared.

The company plans to continue its stochastic design improvement in the future, further evaluating how various input parameters impact structure damage and determining a simplified, representative robust surrogate load set for test validation. Ultimately, the company will be able to develop a fully-repeatable process that can be easily and consistently applied to multiple parts of its production line.

"Hexagon's simulations allow us to virtually create and predict the same damage we see in real life. Those simulations will serve as the basis for future optimization because if we can optimise in simulation, we protect against failure on the field," Bavaresco concluded.

[Access full article here]

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Technical Papers for further reading / review

1988, N Bishop, **The use of frequency domain parameters to predict structural fatigue*, Ph.D. Thesis***, *University of Warwick*. Original coding, testing, of Dirlik formula.

March 1989, N Bishop and F Sherratt, **Fatigue life prediction from power spectral density data. Part 1, traditional approaches**. Jnl. Environmental Engineering, Vol. 2, No. 1. Mechanical Engineering Publications, pp 11-29. Background material on frequency domain approach.

June 1989, N Bishop and F Sherratt, **Fatigue life prediction from power spectral density data. Part 2, recent developments.** Jnl. Environmental Engineering, Vol. 2, No.2. Mechanical Engineering Publications, pp 11-19. Background material on frequency domain approach.

June 1990, N Bishop and F Sherratt, **A theoretical solution for the estimation of `rainflow' ranges from power spectral density data.** Jnl. Fatigue and Fracture of Engineering Materials and Structures, Vol. 13, no 4. Pergamon Press, pp 311-326. The only theoretical verification ever done on the process of calculating rainflow ranges from a PSD of stress.

Oct 2000, N Bishop and F Sherratt, **Finite Element Based Fatigue Calculations**, a book published by *NAFEMS Ltd*, Whitworth Building, Scottish Enterprise Technology Park, East Kilbride, Glasgow. A “How To” guide commissioned by NAFEMS dealing with FE Based fatigue Calculations.

June 2013, N Bishop et al, **Practical Fatigue Calculations for Combined Random and Deterministic Loads:** *2013 Spacecraft and Launch Vehicle Dynamic Environments Workshop, The Aerospace Corporation in El Segundo, Calif.* Detailed investigation of mixed loadings (sine on random).

April 2014, N Bishop et al, ***Advances Relating to Fatigue Calculations for Combined Random and Deterministic Loads***, SAE Paper 2014-01-0725. Work done with Booz Allen Hamilton on fatigue of systems with mixed loadings (sine on random).

May 2014, N Bishop et al, ***Improved methods of (frequency domain) fatigue calculations for automotive systems***, NAFEMS event, New Frontiers in Product Modeling and Simulation, Colorado Springs, CO, USA. Work done with Ford (Dearborn) on fatigue of AC lines within full body car systems.

April 2015, N Bishop et al, ***Time v Frequency Domain Analysis for Large Automotive Systems***, SAE Paper 2015-01-0535. Pioneering work done with Booz Allen Hamilton on fatigue of large automotive system (truck cab) done in both the time and frequency domain.

April 2016, N Bishop et al, ***A Comparative Study of Automotive System Fatigue Models Processed in the Time and Frequency Domain,*** SAE Paper 2016-01-0377. Very important benchmarks study done with Ford, Dearborn and Ford Brazil comparing time and frequency domain results for large automotive systems including truck trailer frame with 24 load inputs.

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Technical Papers for further reading / review

April 2016, N Bishop et al, ***Modern Methods for Random Fatigue of Automotive Parts,*** SAE Paper 2016-01-0372. Work done with Hella, Germany on frequency domain fatigue analysis of headlamps.

April 2017, N Bishop et al, ***Frequency FE-Based Weld Fatigue Life Prediction of Dynamic Systems,*** SAE Paper 2017-01-0355. Work done with Ford, Brazil on fatigue of seam welds in the frequency domain.

April 2017, N Bishop et al, ***Simultaneous Durability Assessment and Relative Random Analysis Under Base Shake Loading Conditions,*** SAE Paper 2017-01-0339. Work done with FCA, Michigan, on fatigue and random response (collision detection) for exhaust systems.

June 2017, N Bishop et al, ***Creating Accurate Surrogate and/or Accelerated Loads for Known Models or Structural Systems Types,*** NAFEMS World Congress, Stockholm 11-14th June 2017. Work done with Ford, UK, on surrogate load analysis for exhaust systems.

June 2017, N Bishop et al, ***Simultaneous Durability Assessment and Relative Random Analysis Under Base Shake Loading Conditions,*** NAFEMS World Congress, Stockholm 11-14th June 2017. Work done with OHB, Germany (Satellite company) on random response and fatigue analysis of satellite parts.

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April 2018, N Bishop, K Sweitzer. et al, ***Loads Cascading in the Frequency Domain***, SAE Technical Paper 2018-01-0138. Work done with Ford, Brazil and Booz Allen Hamilton on loads cascading (from road load data to bracket location) for a truck trailer.

April 2018, P Leisten et al, ***Frequency Domain Fatigue Analysis of Exhaust Systems***, SAE Technical Paper 2018-01-0396. Work done with Eberspaecher, Germany, on frequency domain fatigue analysis of exhausts.

June 2019, M Sethi, A Sharma, et al, ***Two-Wheeler Fatigue and Random Response,*** NAFEMS World Congress 2019, Paper NWC19-375. Work done with HeroMotoCorp, India, on fatigue and random response (collision detection) for w wheeler motor bikes.

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Technical Papers for further reading / review

June 2019, N Bishop, P Murthy, et al. ***Frequency Domain Spot Weld and Seam Weld Analysis:*** NAFEMS World Congress 2019, NWC19-380. Work done with Ford, Germany on spot weld fatigue of large models and comparisons with test.

June 2019, N Bishop, S Kerr, et al, ***Loads Enveloping:*** NAFEMS World Congress 2019, Paper NWC19-382. Work done with Ford, Brazil and Ford, US on loads simplification and surrogate loads analysis.

April 2020, T Thesing et al, ***User Defined FE Based Connector Joints For Plastics*.** SAE World Congress paper 2020-01-0186. Work done with Hella Germany, on a new user connector.

April 2020, P Roemelt et al, ***Loads Cascading for Full Vehicle Component Design.*** SAE World Congress paper 2020-01-0762. Work done with Ford, Germany, on loads cascading.

April 2020, A Trifan et al, ***Vector Load Simplified Duty Cycle For Lower Control Arm.*** SAE World Congress paper 2020-01-1058. Work done with FCA, Canada, on test design simplification and surrogate loads analysis.

April 2020, r goncalves et al, ***Loads simplification on multi input axle systems.*** SAE World Congress paper 2020-01-1056. Work done with GM Brazil, and GM US, on loads simplification and surrogate loads analysis.

April 2020, N Bishop, S Kerr, et al, ***Full Body Car Analysis in the Time and Frequency Domains - Sheet, Spot and Seam Weld Fatigue Benchmark Studies.*** SAE World Congress paper 2020-01-0195. Work done with Ford, US, on loads simplification and surrogate loads analysis.

April 2020, N Bishop, S Kerr, et al, ***Frequency Domain Analysis of 2-Wheeler Systems.*** SAE World Congress paper 2020-01-0476. Work done with Ford, Brazil and Ford, US on loads simplification and surrogate loads analysis.

April 2020, N Bishop, S Kerr, et al, ***Frequency Domain Loads Processing For Exhaust Systems.*** SAE World Congress paper 2020-01-0180. Work done with Ford, Brazil and Ford, US on loads simplification and surrogate loads analysis.

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Technical Papers for further reading / review

October 2021Liam MacFarlane, Dr. Laurie Walker et al. **FREQUENCY DOMAIN FATIGUE METHODOLOGY APPLIED TO BOAT DESIGN**, NAFEMS World Congress 2021, NWC21-340. Collaboration with Nauti-Craft rom time analysis in Adams to automatic conversion to frequency domain and multi events vibration fatigue.

May 2023, Dr. P. Roemelt, Dr Marco Veltri et al. **A ROBUST DESIGN APPROACH INCORPORATING UNCERTAINTIES IN AN AUTOMOTIVE DURABILITY ASSESSMENT**, NAFEMS World Congress 2023, NWC23-0499, Collaboration with Ford Werke, Germany, deploying Robust Design and Nastran

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